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(54) **DATA TRANSMISSION METHOD AND DATA TRANSMISSION APPARATUS**

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(52) **U.S. Cl.**
CPC **H04B 10/541** (2013.01)

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(57) **ABSTRACT**

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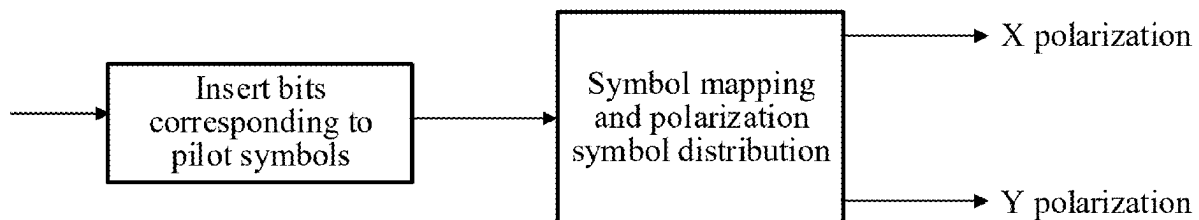
Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/
116034, filed on Aug. 31, 2023.

A generated data frame includes N symbols in a polarization direction. Every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, where $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. Q pilot symbols are generated by using the target polynomial and a seed. Each pilot symbol is one of $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, where A is a real number. The Q pilot symbols are direct current balanced. A degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8.

(30) **Foreign Application Priority Data**

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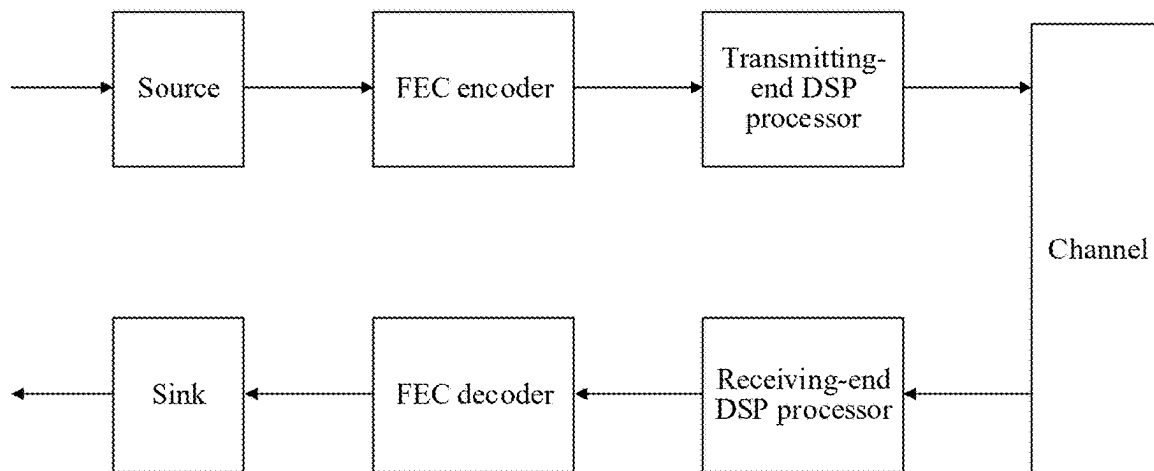


FIG. 1

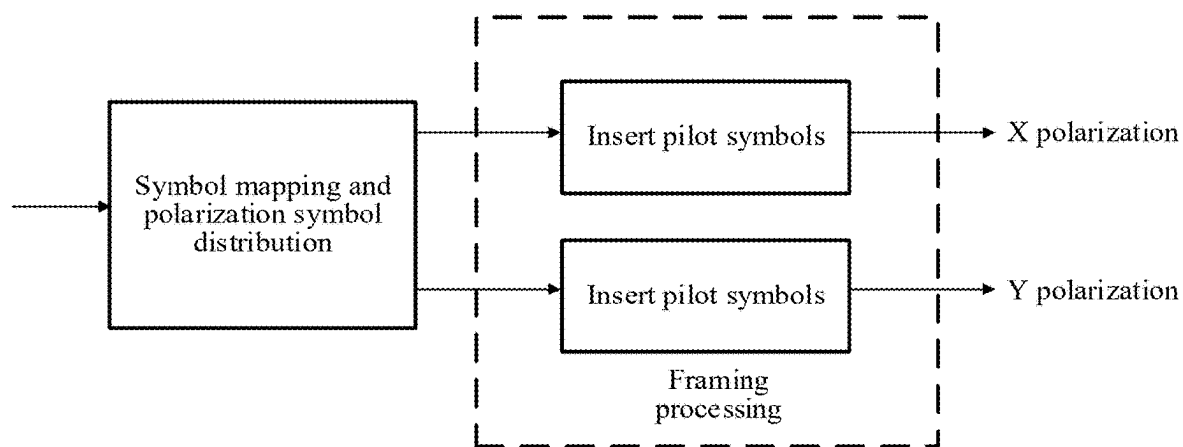


FIG. 2

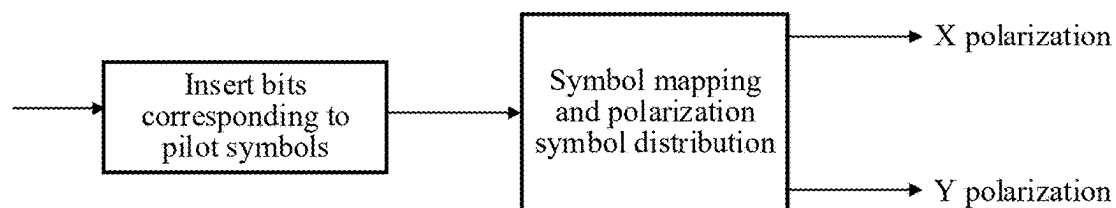


FIG. 3

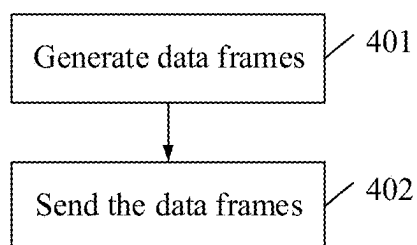


FIG. 4

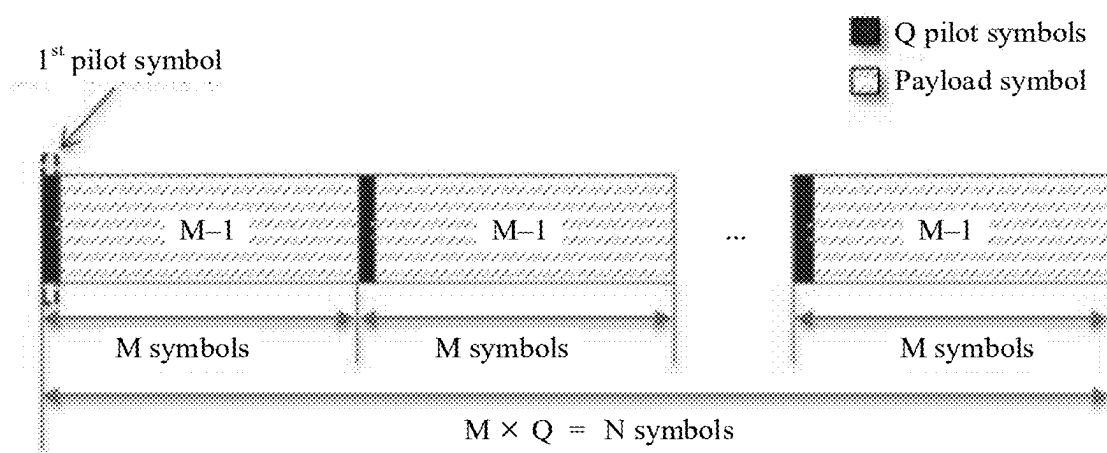


FIG. 5

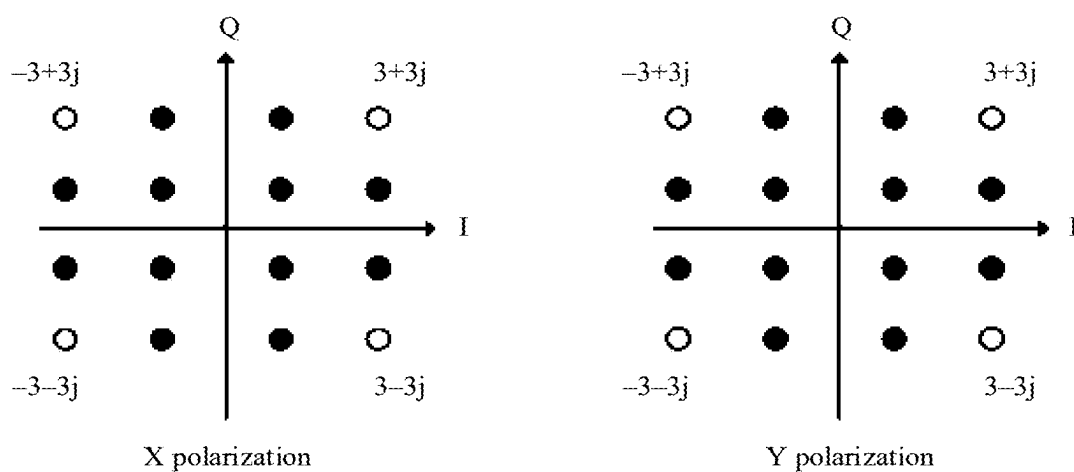


FIG. 6

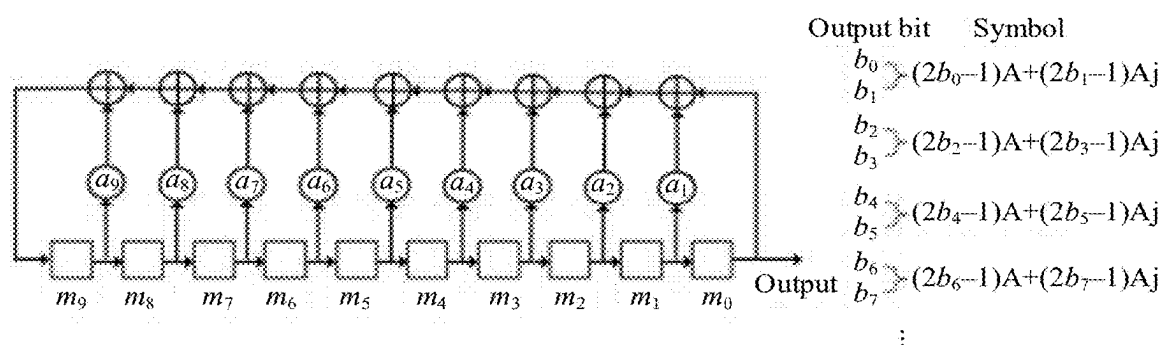


FIG. 7

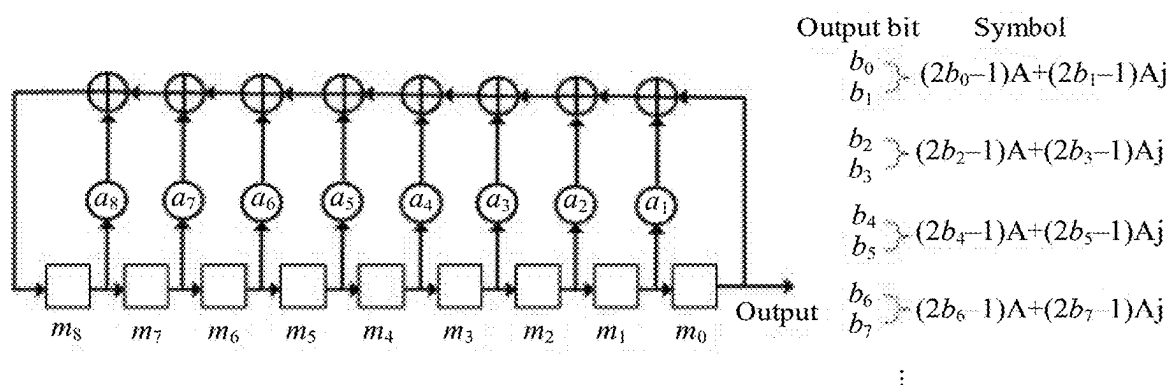


FIG. 8

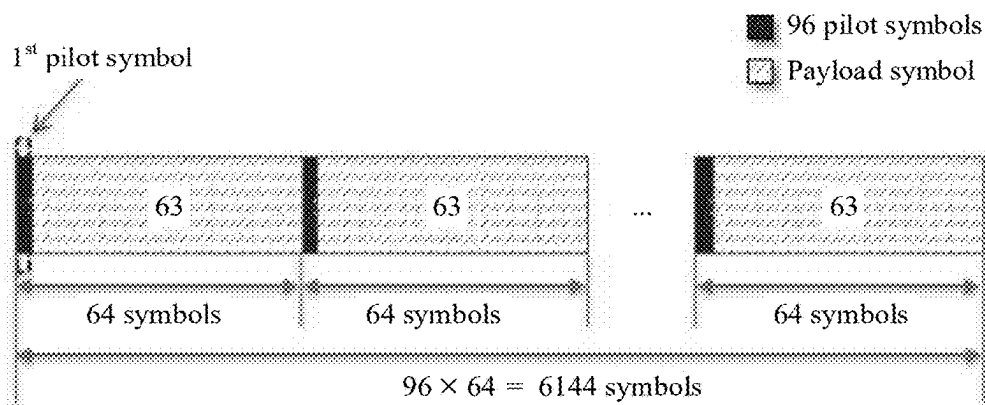
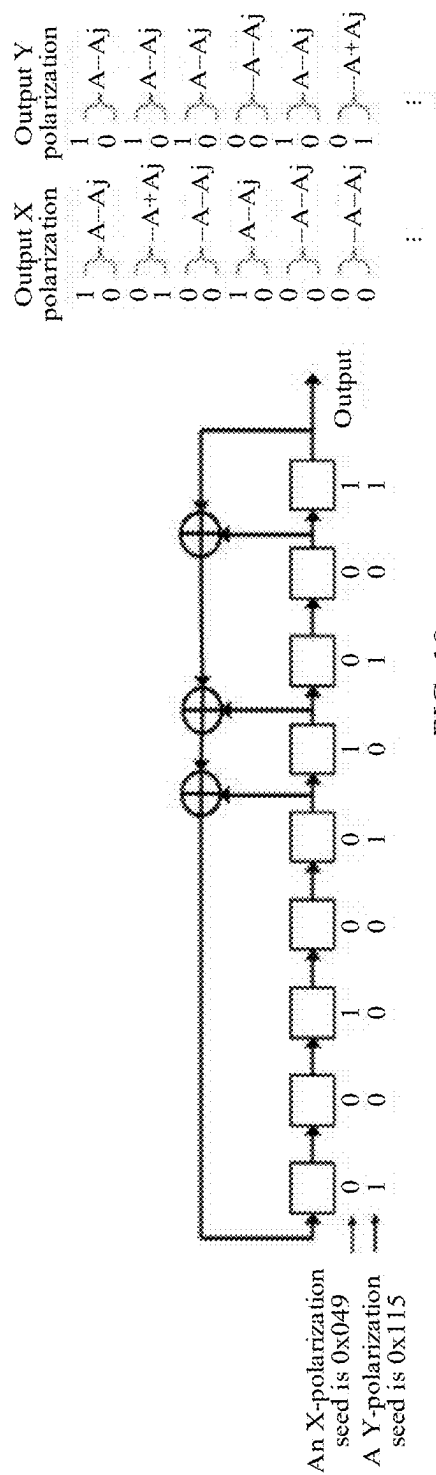


FIG. 9



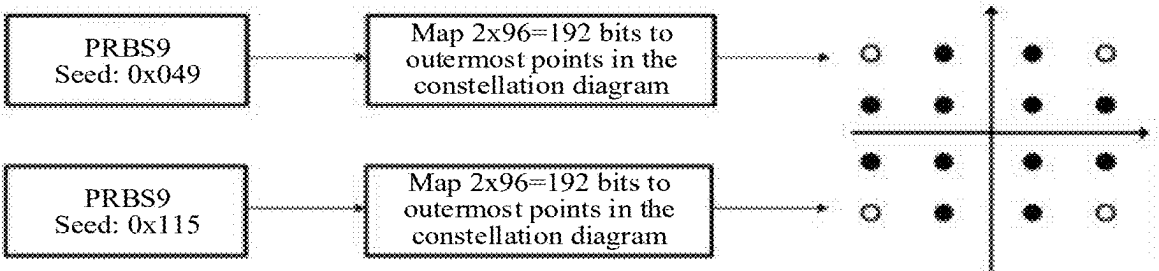


FIG. 11

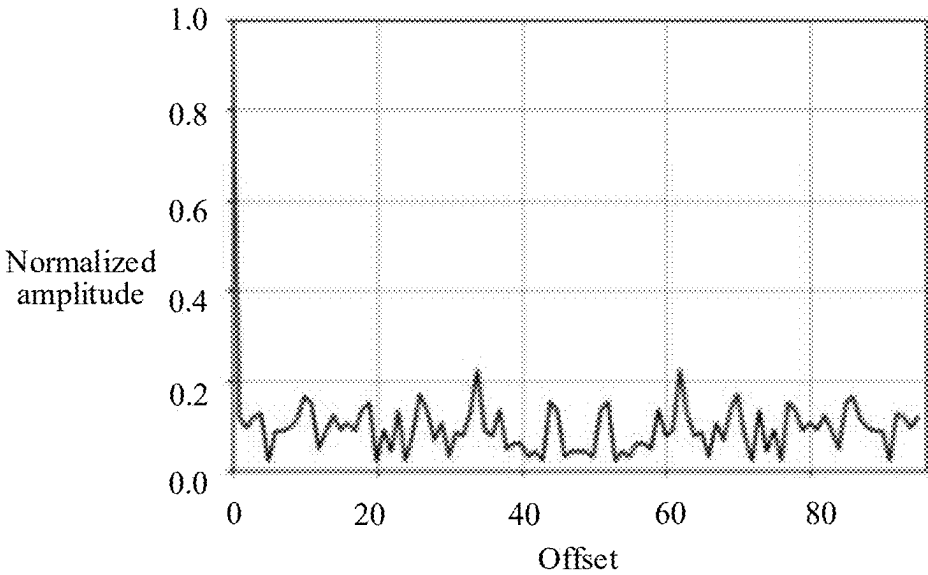


FIG. 12A

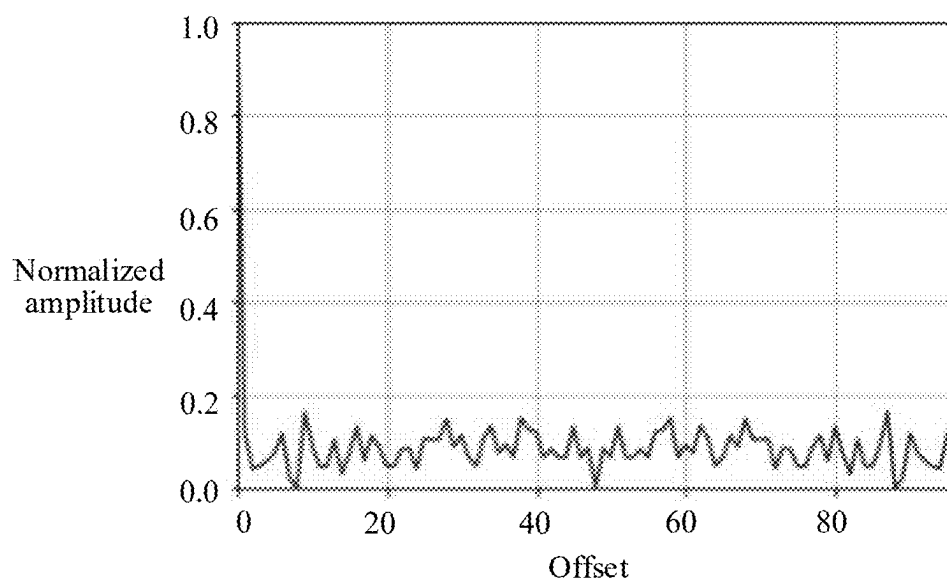


FIG. 12B

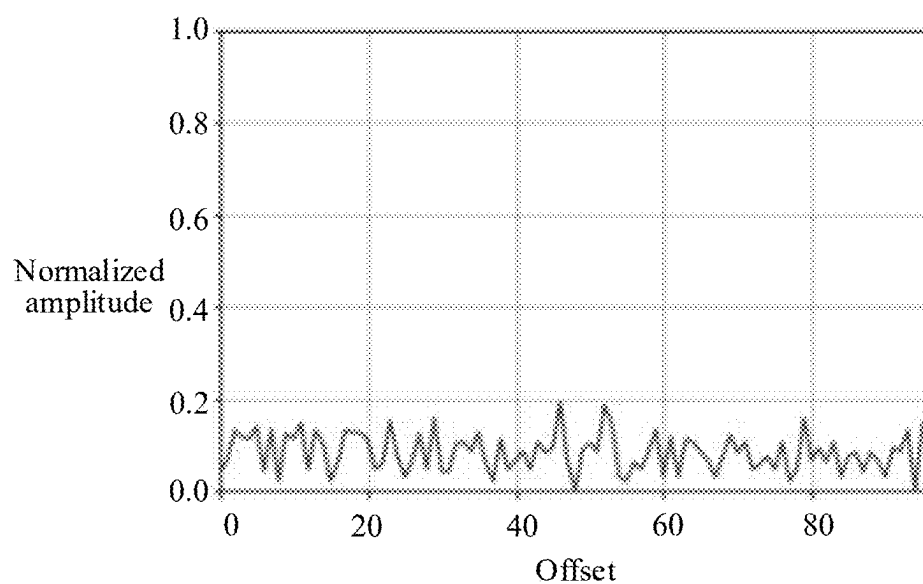


FIG. 12C

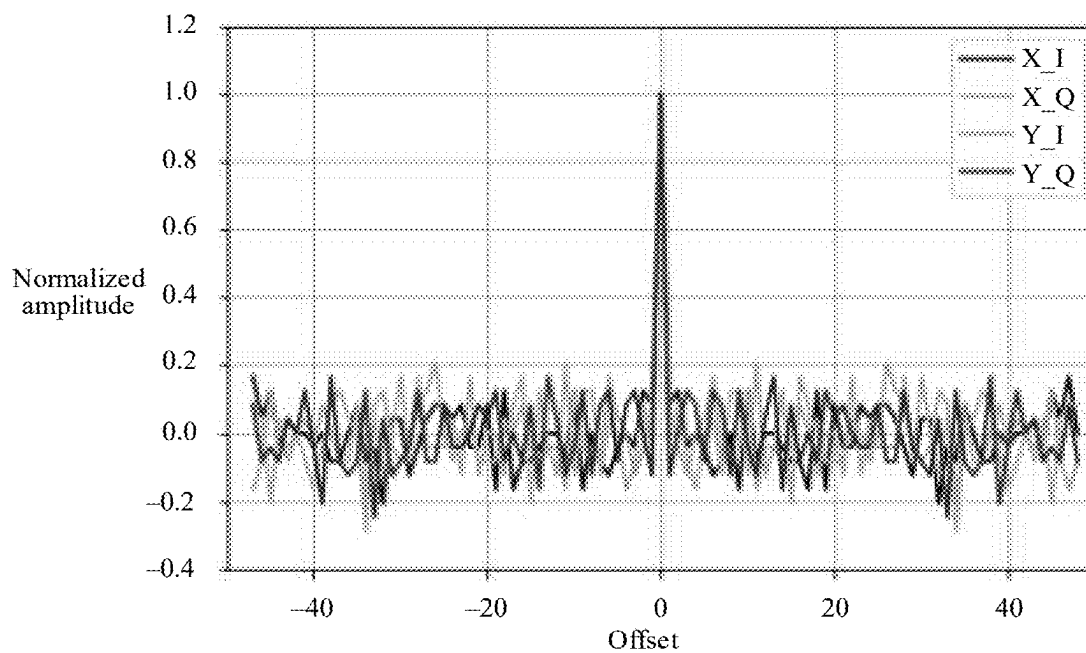


FIG. 13A

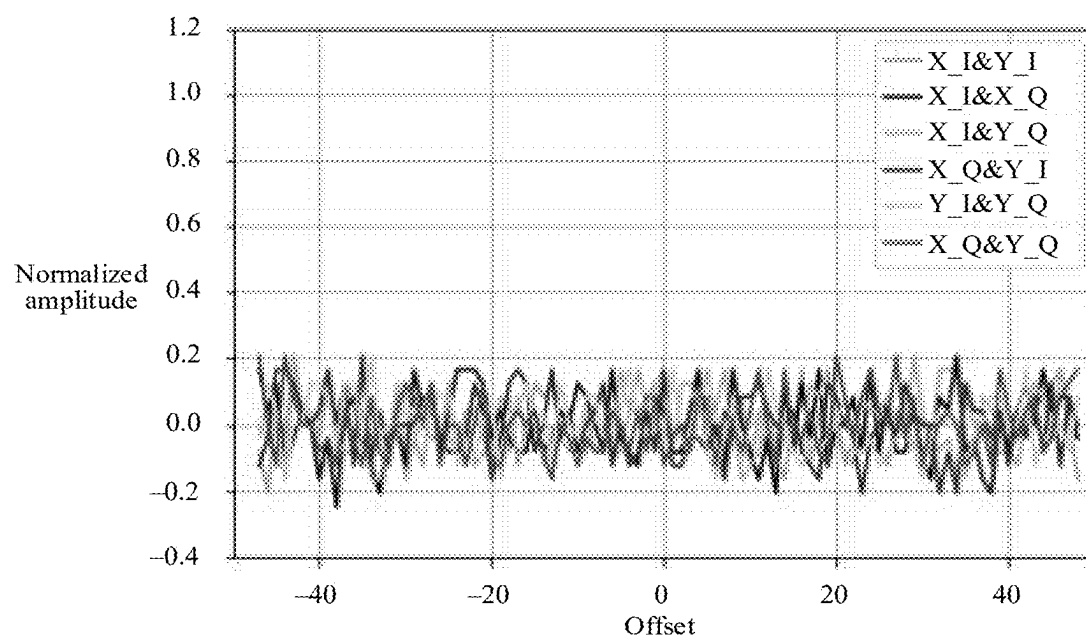
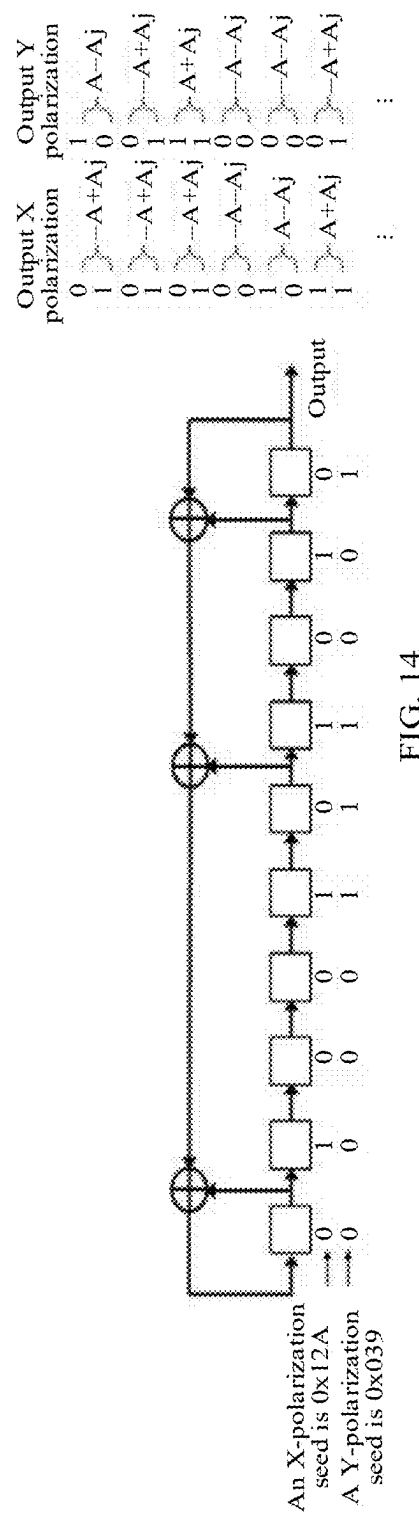


FIG. 13B



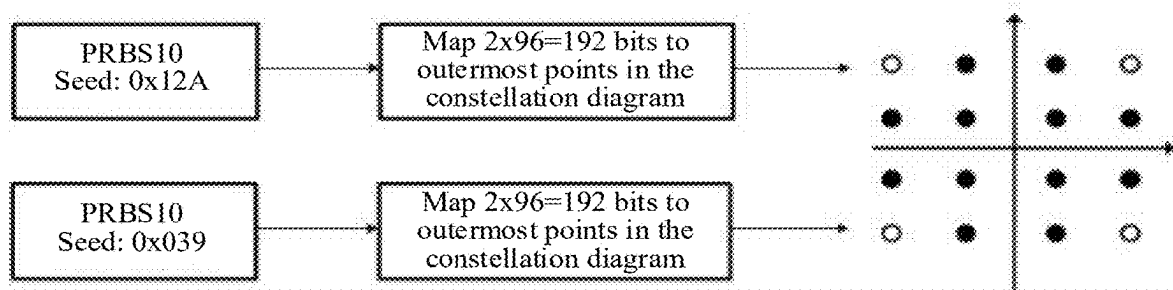


FIG. 15

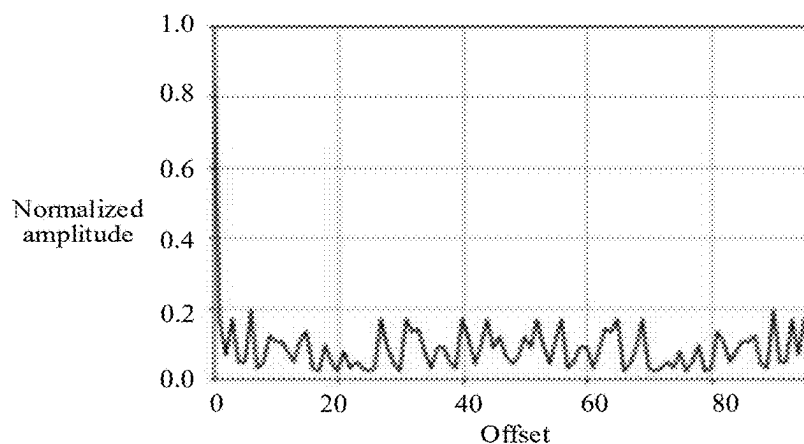


FIG. 16A

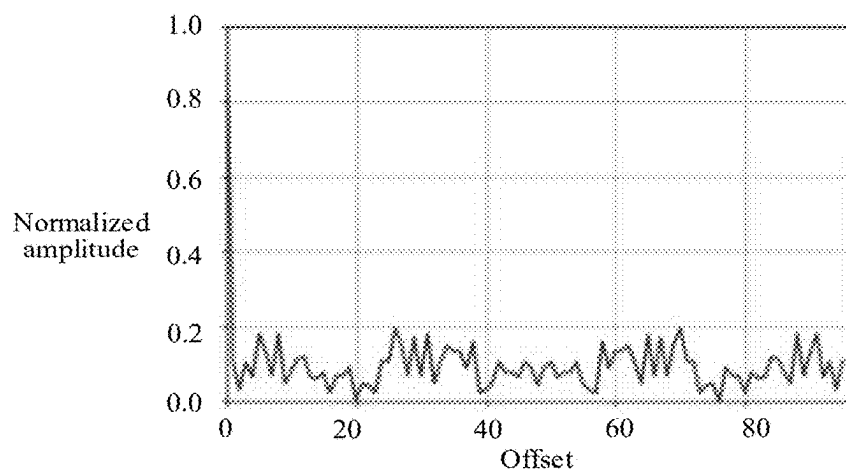


FIG. 16B

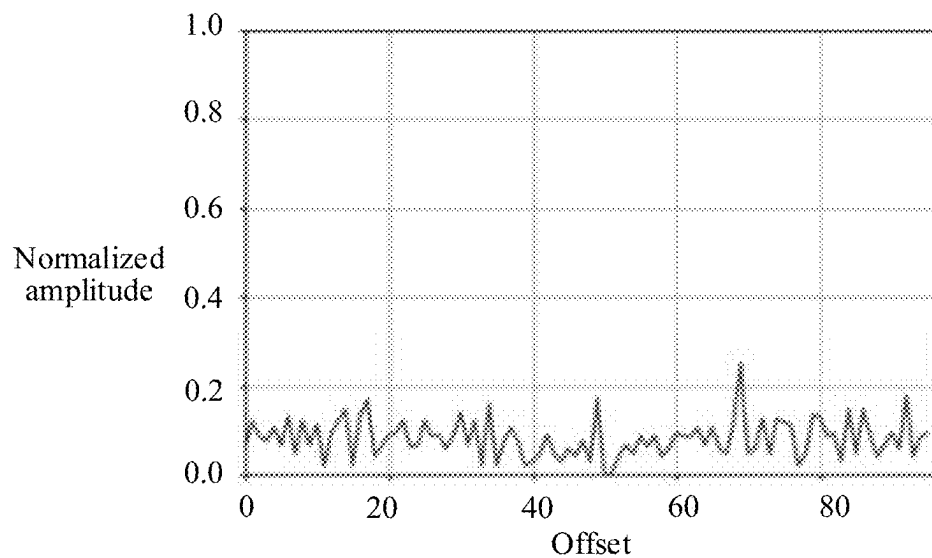


FIG. 16C

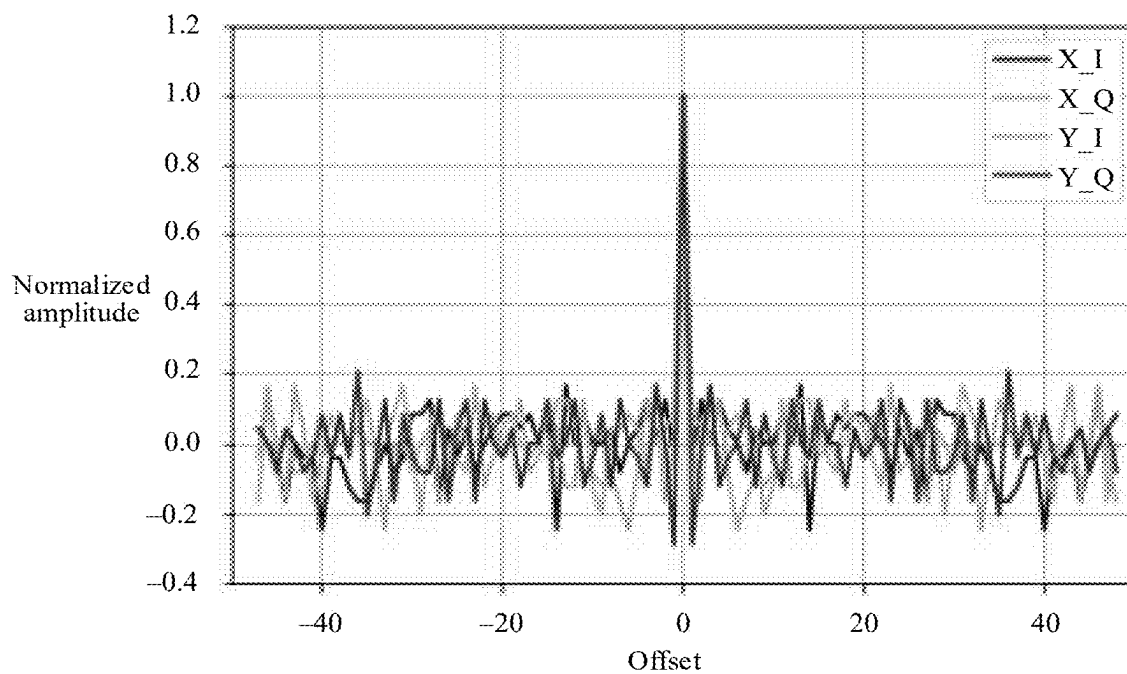


FIG. 17A

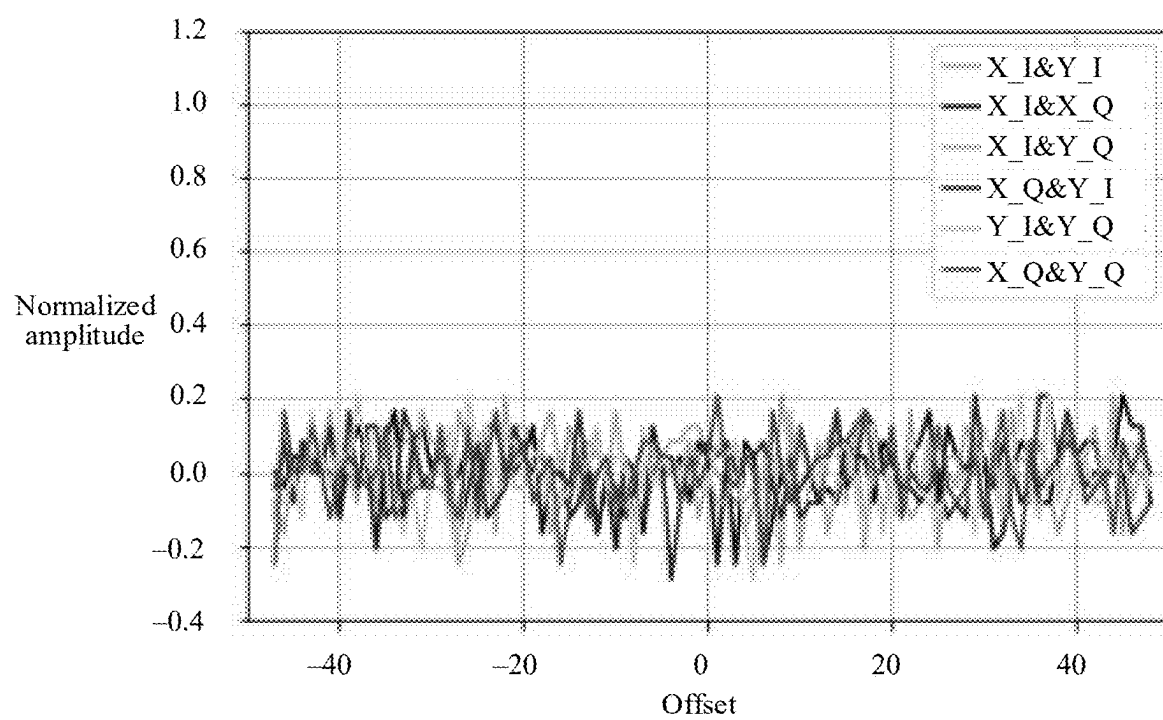


FIG. 17B

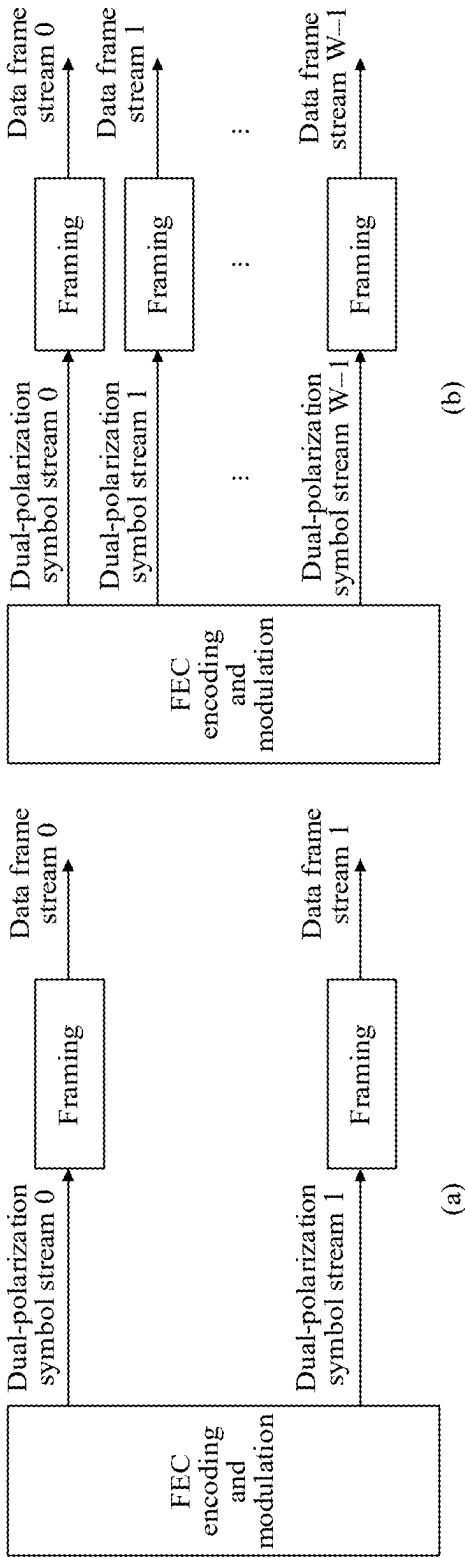
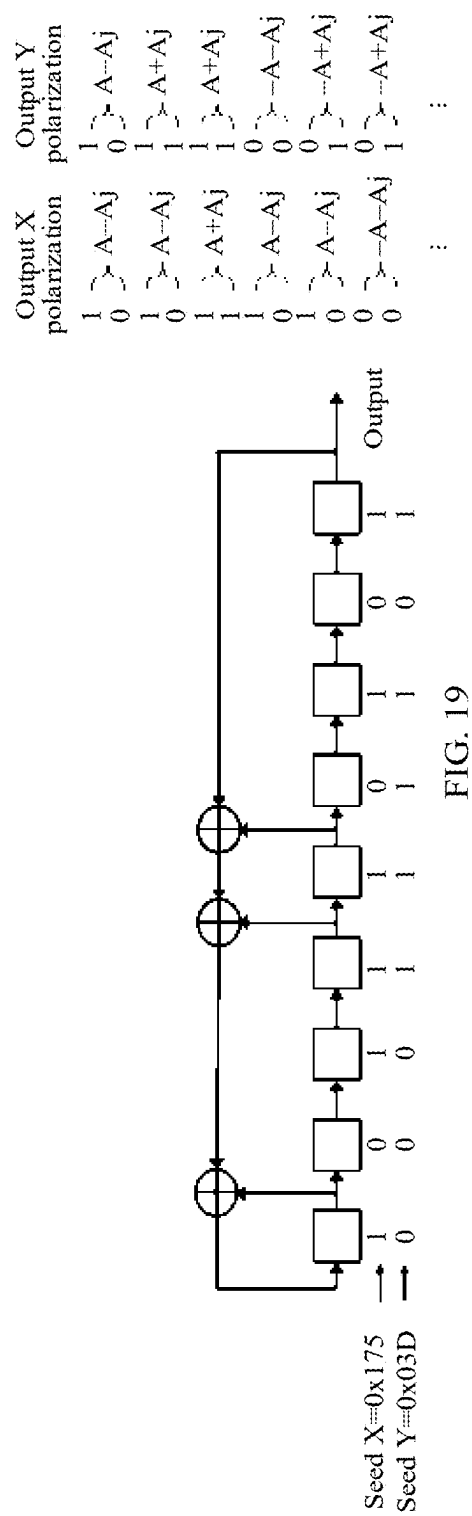


FIG. 18



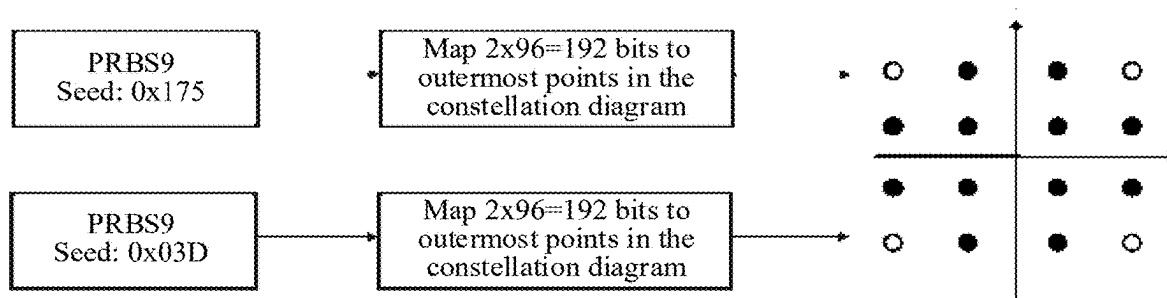


FIG. 20

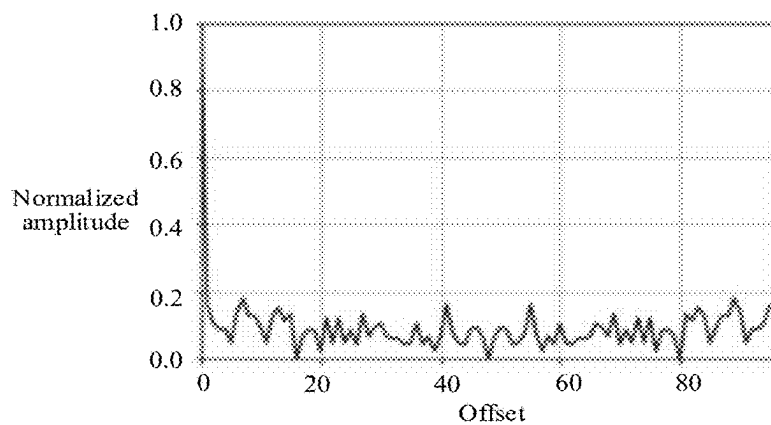


FIG. 21A

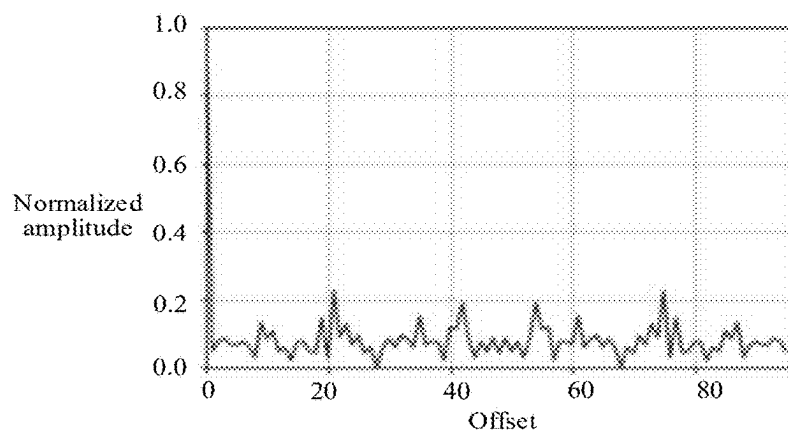


FIG. 21B

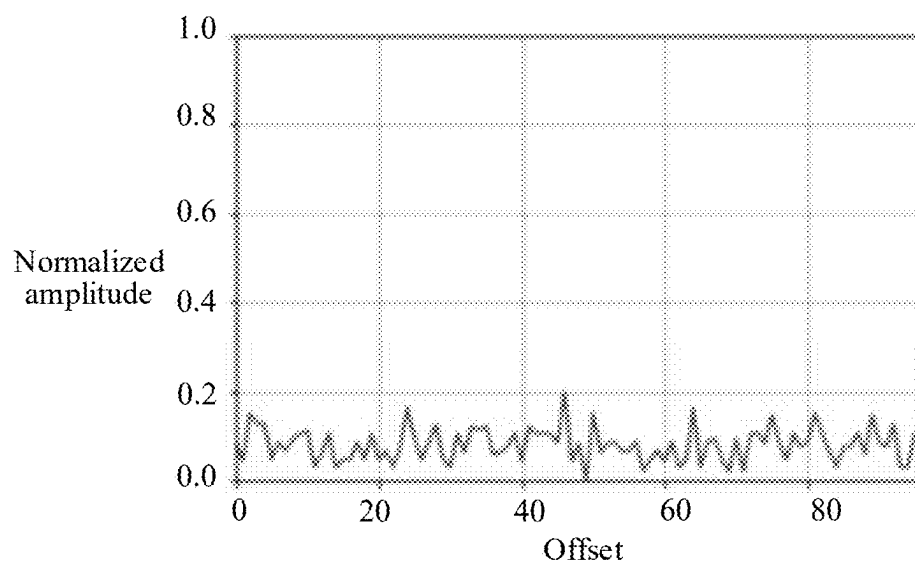


FIG. 21C

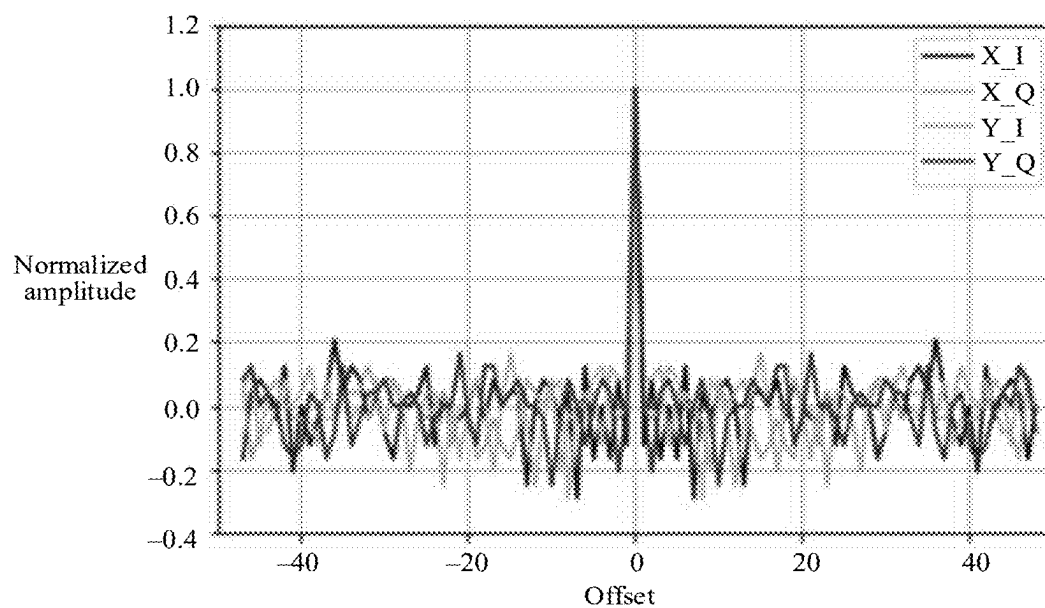


FIG. 22A

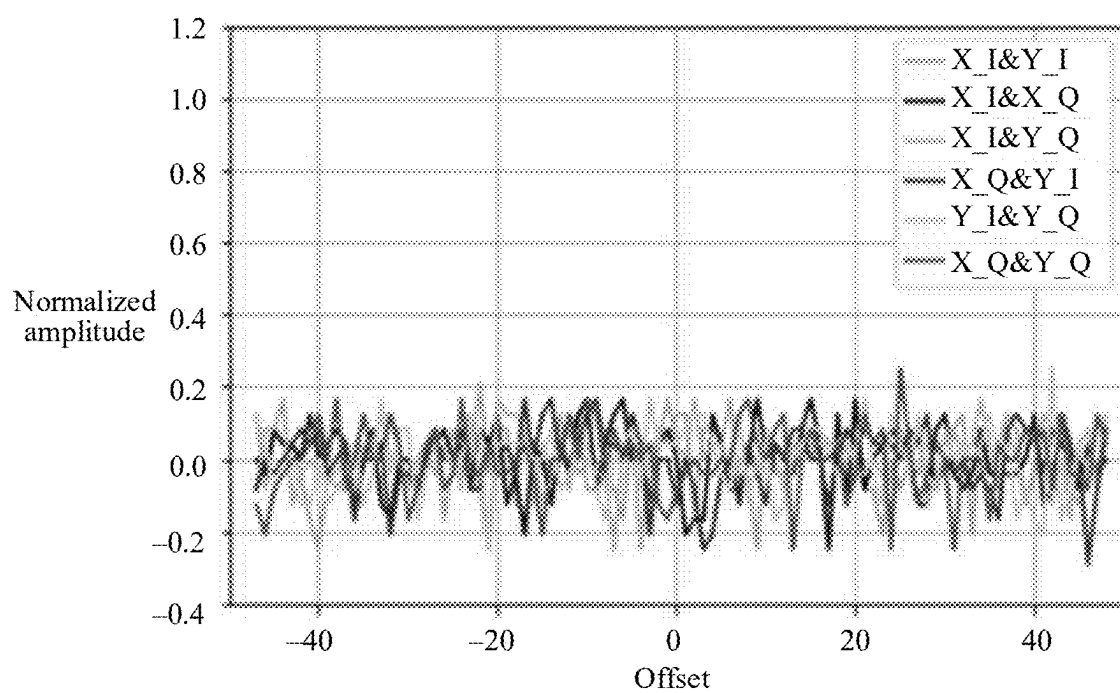


FIG. 22B

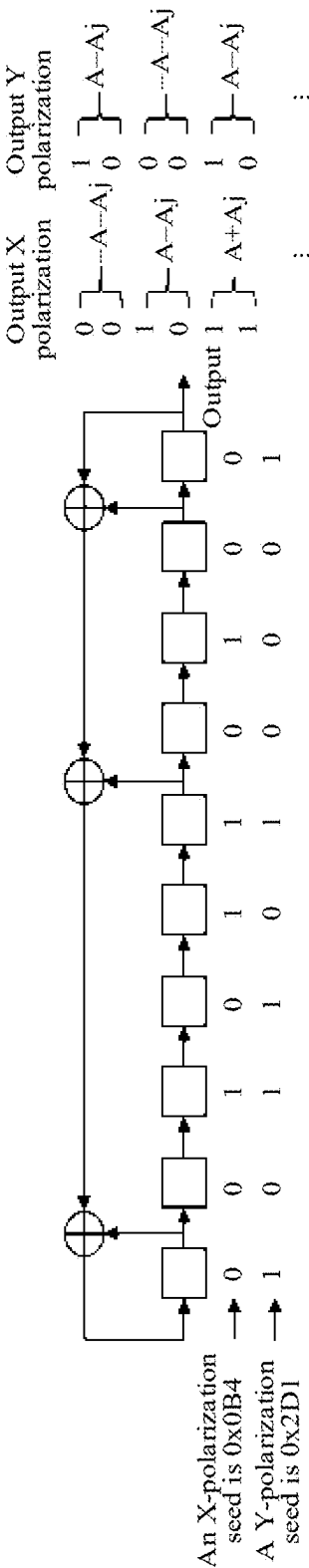


FIG. 23

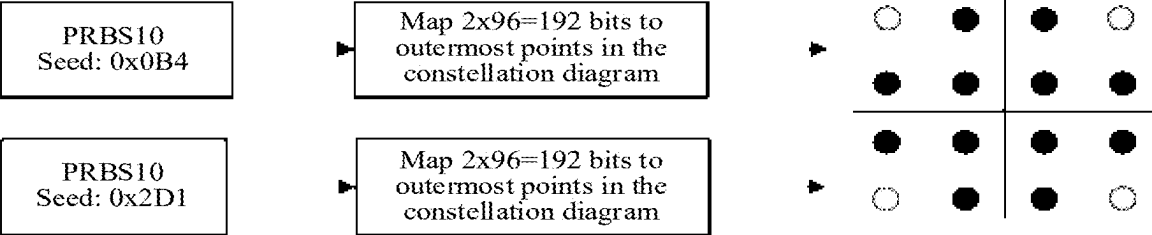


FIG. 24

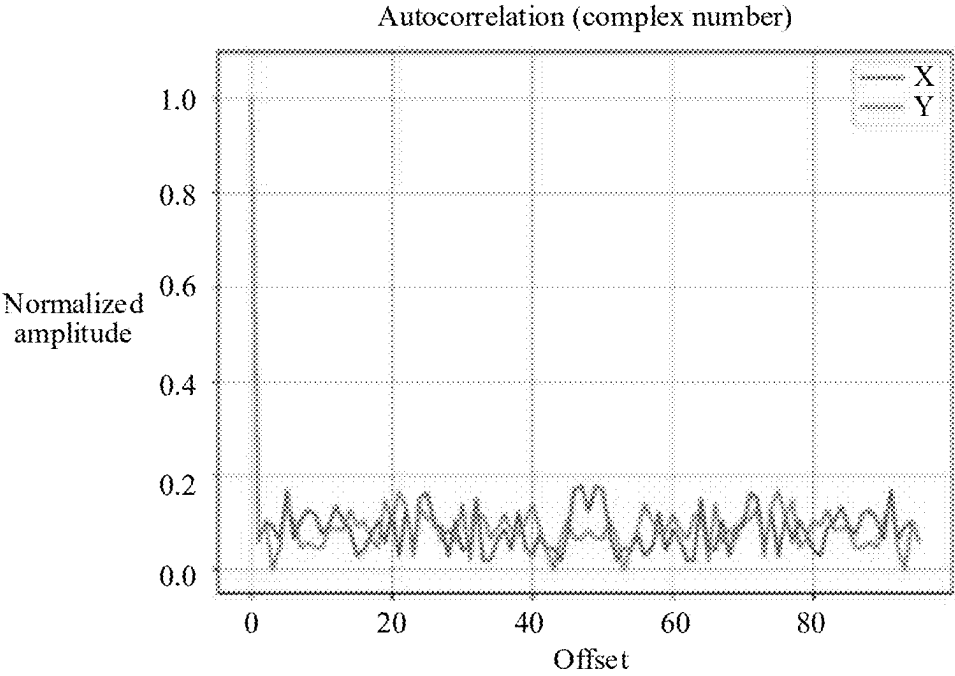


FIG. 25A

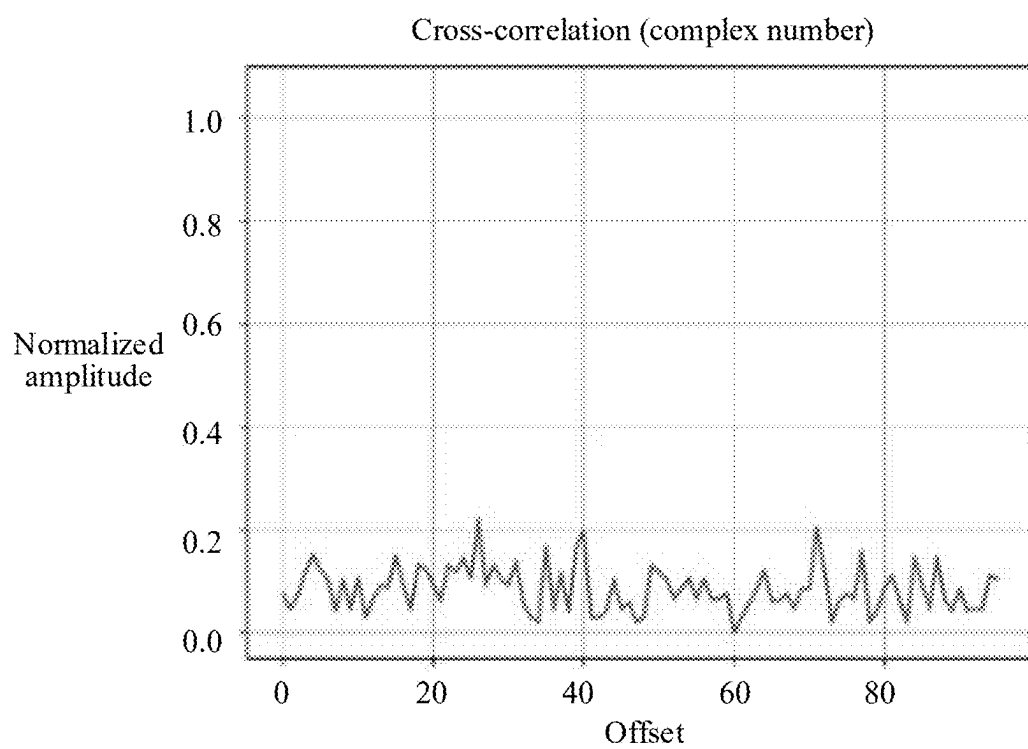


FIG. 25B

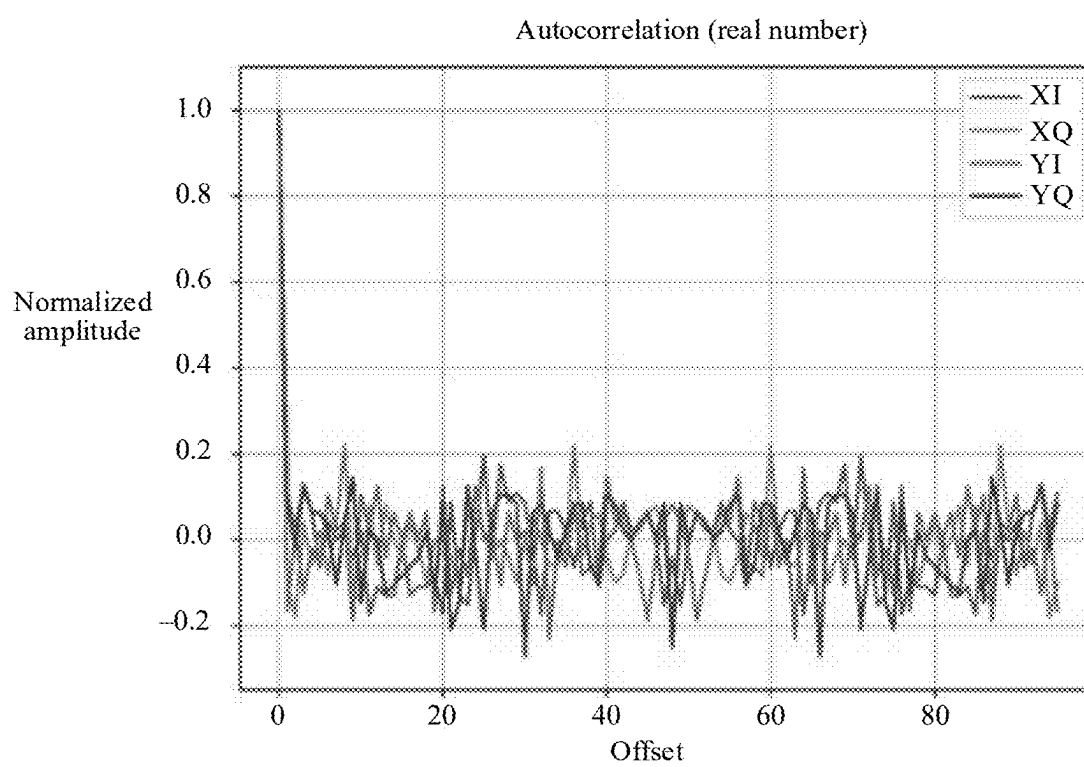


FIG. 26A

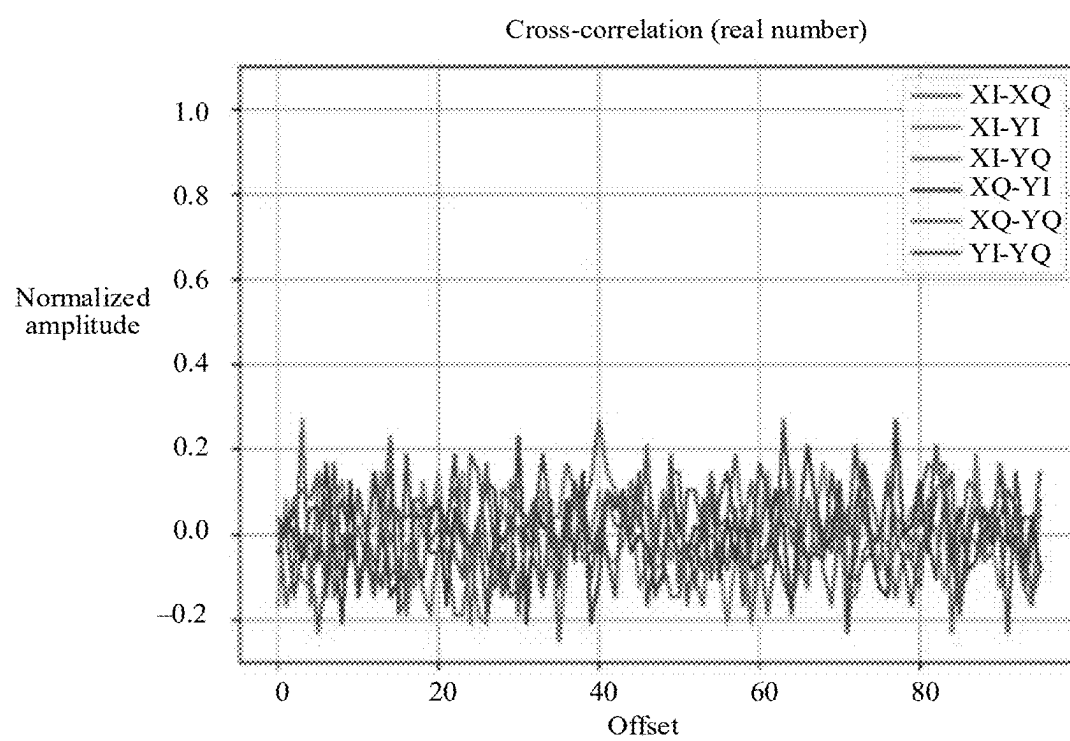
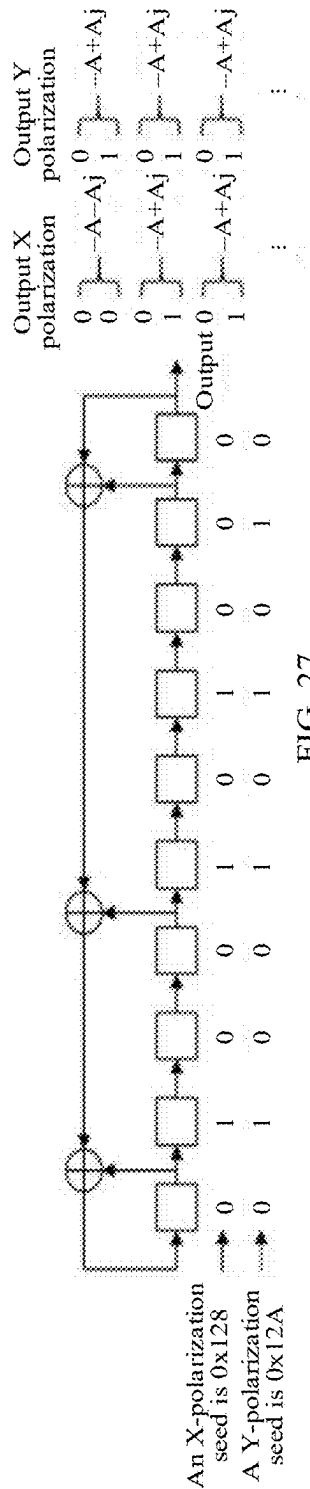


FIG. 26B



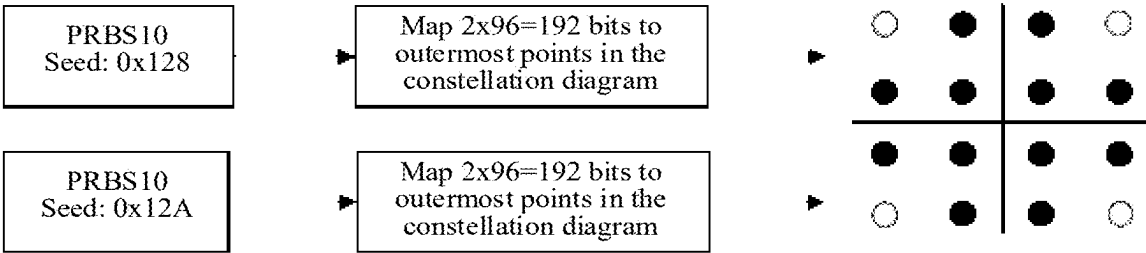


FIG. 28

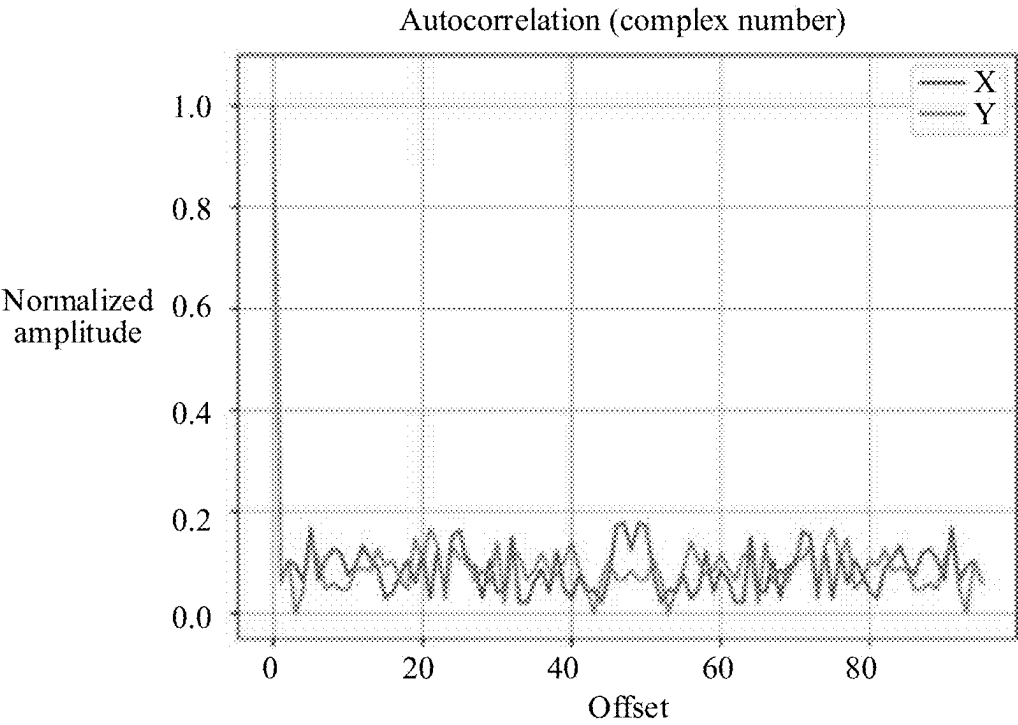


FIG. 29A

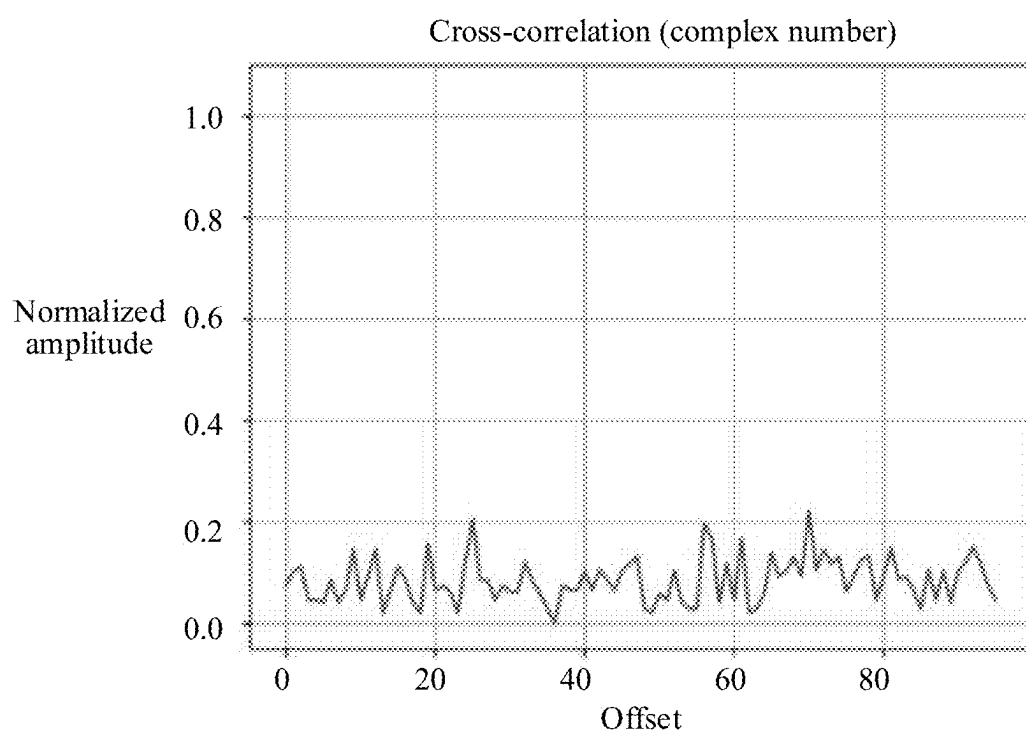


FIG. 29B

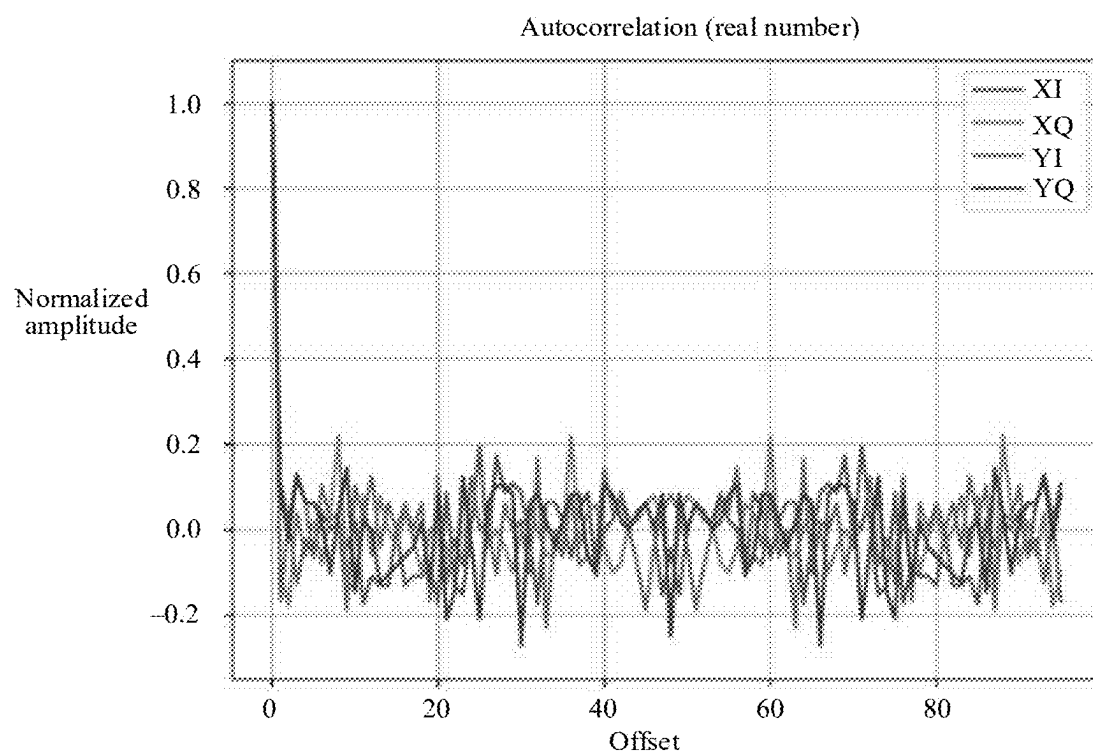


FIG. 30A

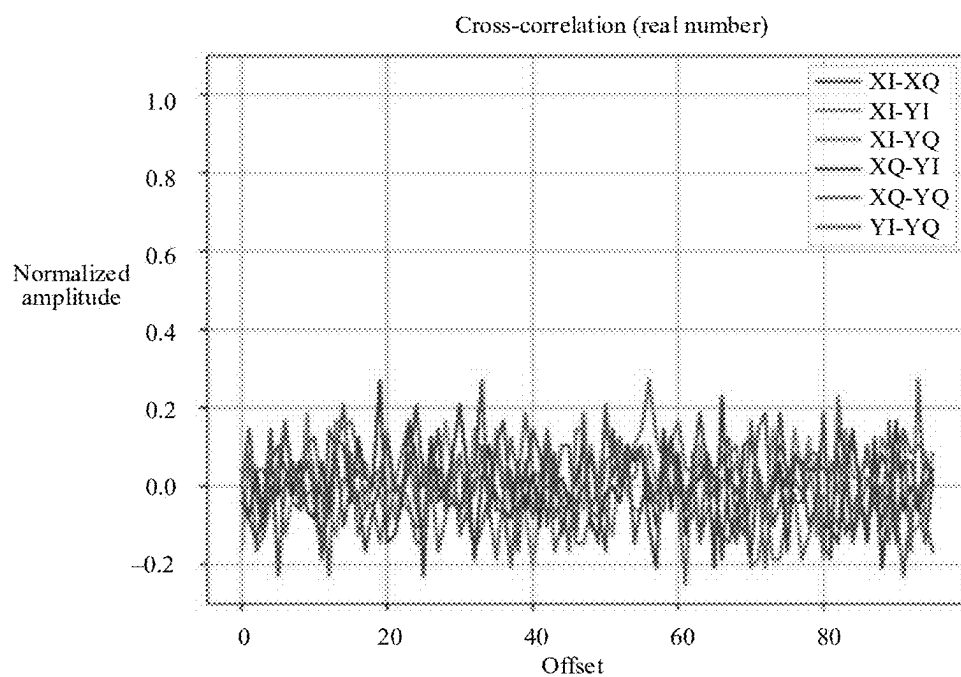


FIG. 30B

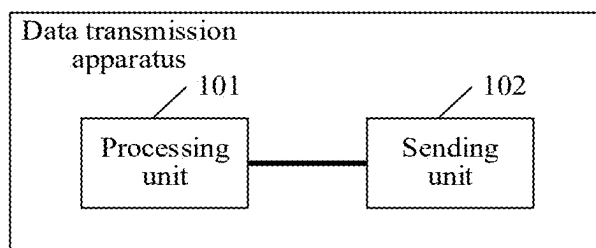


FIG. 31

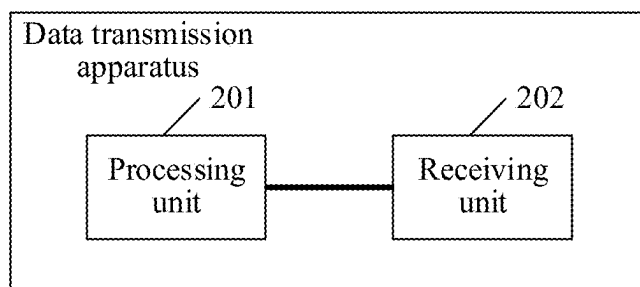


FIG. 32

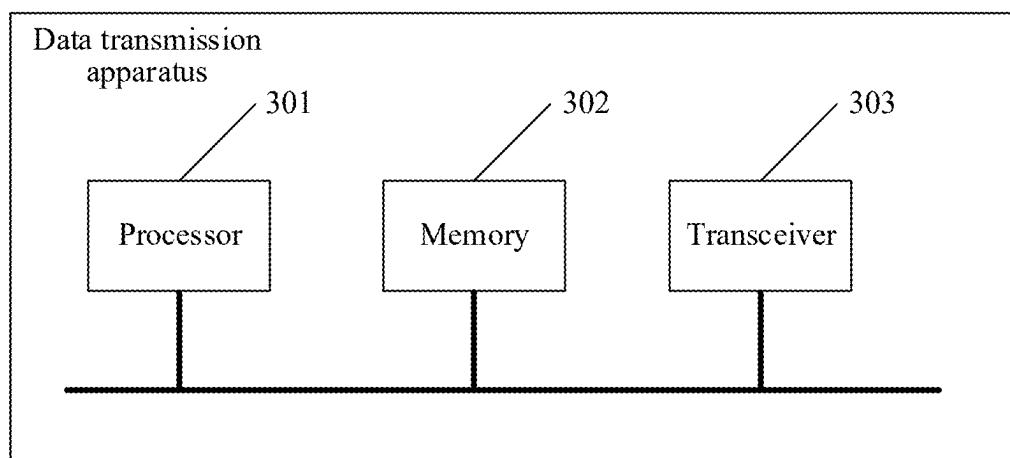


FIG. 33

DATA TRANSMISSION METHOD AND DATA TRANSMISSION APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation of International Patent Application No. PCT/CN2023/116034 filed on Aug. 31, 2023, which claims priority to Chinese Patent Application No. 202211495134.6 filed on Nov. 26, 2022 and Chinese Patent Application No. 202310668432.9 filed on Jun. 6, 2023. All of the aforementioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates to the communication field, and in particular, to a data transmission method and a data transmission apparatus.

BACKGROUND

[0003] With continuous promotion of 5G, cloud computing, big data, and artificial intelligence, high-speed optical transport networks are developing toward high capacity, packetization, and intelligence. A coherent optical communication system uses amplitudes, phases, polarization, and frequencies of optical waves to carry information. To resist optical signal distortion caused by dispersion, polarization-dependent impairment, noise, non-linear effects, and other factors in a transmission process and maintain long-distance transmission, the coherent optical communication system usually adds some designed fixed symbol sequences to a transmission symbol sequence, so that a receiving end can restore a sent symbol.

[0004] An existing transmission symbol sequence is mainly used in a scenario of 400 gigabits per second (Gbps), and cannot adapt to future scenarios over 400 Gbps (including 600 Gbps, 800 Gbps, and the like). In addition, solutions in another technology have a problem that hardware implementation complexity is high, which needs to be resolved urgently in the future.

SUMMARY

[0005] Embodiments of this disclosure provide a data transmission method and a data transmission apparatus. A simple target polynomial is designed to generate pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the generated pilot symbols have good autocorrelation and cross-correlation characteristics, and are direct current balanced, which helps a receiving end restore signal quality.

[0006] According to a first aspect, an embodiment of this disclosure provides a data transmission method. The method applied to a transmitting end includes the following steps. First, data frames are generated. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, A is a real number, the Q pilot symbols are direct current balanced, a degree of the target polynomial is less than or equal to 10, and a quantity

of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. Then, the data frames are sent.

[0007] In the implementation, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N=M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0008] In some possible implementations, the target polynomial is one of the following:

$$[0009] \quad x^9 + x^4 + x^3 + x + 1;$$

$$[0010] \quad x^9 + x^5 + x^4 + x + 1;$$

$$[0011] \quad x^9 + x^8 + x^5 + x^4 + 1;$$

$$[0012] \quad x^9 + x^8 + x^6 + x^5 + 1;$$

$$[0013] \quad x^{10} + x^4 + x^3 + x + 1;$$

$$[0014] \quad x^{10} + x^5 + x^2 + x + 1;$$

$$[0015] \quad x^{10} + x^8 + x^5 + x + 1;$$

$$[0016] \quad x^{10} + x^9 + x^5 + x^2 + 1;$$

$$[0017] \quad x^{10} + x^9 + x^6 + x + 1;$$

$$[0018] \quad x^{10} + x^9 + x^7 + x^6 + 1;$$

$$[0019] \quad x^{10} + x^9 + x^8 + x^5 + 1;$$

$$[0020] \quad x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1;$$

$$[0021] \quad x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1;$$

$$[0022] \quad x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1;$$

$$[0023] \quad x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1; \text{ or}$$

$$[0024] \quad x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1.$$

[0025] In some possible implementations, the target polynomial is one of the following:

$$[0026] \quad x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1; \text{ or}$$

$$[0027] \quad x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1.$$

[0028] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at the transmitting end and a laser at the receiving end.

[0029] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0030] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0031] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following:

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- [0175] the target polynomial: $x^{10}+x^8+x^7+x^3+x^2+x+1$, the seed in the first polarization direction: 0x0D4, and the seed in the second polarization direction: 0x0ED;
- [0176] the target polynomial: $x^{10}+x^8+x^7+x^3+x^2+x+1$, the seed in the first polarization direction: 0x01A, and the seed in the second polarization direction: 0x0ED;
- [0177] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x186;
- [0178] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x261;
- [0179] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x1A3;
- [0180] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x0C7;
- [0181] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x346;
- [0182] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x1C5;
- [0183] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x063;
- [0184] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x1C5, and the seed in the second polarization direction: 0x327;
- [0185] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x240, and the seed in the second polarization direction: 0x3C4;
- [0186] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x3C4, and the seed in the second polarization direction: 0x103;
- [0187] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x226;
- [0188] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x165;
- [0189] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x165, and the seed in the second polarization direction: 0x0C7;
- [0190] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x266, and the seed in the second polarization direction: 0x0C7;

- [0191]** the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x3E2;

- [0192] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x0C4, and the seed in the second polarization direction: 0x226; or the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x1D8, and the seed in the second polarization direction: 0x2D9.

[0193] It should be understood that the foregoing polynomials and seeds have good differential autocorrelation and differential cross-correlation characteristics, and the good correlation effects can still be achieved when there is the frequency difference between the laser at the transmitting end and the laser at the receiving end.

[0194] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^9+x^4+x^3+x+1$, the seed in the first polarization direction is 0x049, the seed in the second polarization direction is 0x115, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0195] 96 pilot symbols in the first polarization direction are sequentially:

[0196] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, A+Aj, -A+Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, A-Aj, A+Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, and A+Aj.

[0197] 96 pilot symbols in the second polarization direction are sequentially:

[0198] A-Aj, A-Aj, A-Aj, -A-Aj, A-Aj, -A+Aj,
A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A-Aj,
-A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A-Aj,
A-Aj, A-Aj, -A+Aj, A-Aj, -A-Aj, -A+Aj, A+Aj,
A-Aj, A+Aj, A+Aj, A+Aj, -A+Aj, A-Aj, -A+Aj,
A-Aj, -A+Aj, A+Aj, A+Aj, A+Aj, -A+Aj, -A-Aj,
-A+Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj,
-A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A-Aj, A+Aj,
-A-Aj, A-Aj, -A+Aj, A+Aj, A+Aj, A+Aj, -A-Aj,
-A-Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, A-Aj,
-A+Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, A-Aj,
-A-Aj, A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj,
-A-Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, and A+Aj.

[0199] In some possible implementations, N=6144, M=64, Q=96, the target polynomial is $x^{10}+x^9+x^4+1$, the seed in the first polarization direction is 0x12A, the seed in the second polarization direction is 0x039, and the first polarization direction and the second polarization direction are orthogonal to each other.

are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor+1$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor+1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$. $\lfloor a \rfloor$ represents rounding down a positive real number a to the nearest integer.

[0218] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^9 + x^8 + x^5 + x^4 + 1$, a seed in a first polarization direction is 0x175, a seed in a second polarization direction is 0x03D, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0219] In some possible implementations, 96 pilot symbols in the first polarization direction are sequentially:

[0220] A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, -A-Aj, A-Aj, A-Aj, A+Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj, -A-Aj, A-Aj, -A-Aj, A+Aj, A-Aj, -A+Aj, A-Aj, A-Aj, A-Aj, -A+Aj, A-Aj, -A+Aj, A-Aj, A-Aj, A+Aj, -A-Aj, A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, and -A-Aj.

[0221] 96 pilot symbols in the second polarization direction are sequentially:

[0222] A-Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj,
-A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj,
-A-Aj, A+Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, -A+Aj,
A+Aj, A+Aj, A-Aj, A+Aj, -A-Aj, -A+Aj, A-Aj,
-A-Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, -A+Aj, A+Aj,
A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj,
A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj,
-A+Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj,
-A-Aj, A+Aj, -A+Aj, A+Aj, -A+Aj, A-Aj, -A+Aj,
-A-Aj, A-Aj, -A+Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj,
-A-Aj, -A+Aj, -A-Aj, -A+Aj, -A+Aj, A+Aj, A+Aj,
A+Aj, A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, -A+Aj, A+Aj,
-A-Aj, A-Aj, A+Aj, A+Aj, and A+Aj.

[0223] In some possible implementations, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$; and in a second polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$. Alternatively, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor+1$; and in a second polarization direction, in the data frame, the quantity

of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$. $\lfloor a \rfloor$ represents rounding down the positive real number a to the nearest integer, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0224] In the implementation, in the two polarization directions, a total quantity of pilot symbols that are $-A-A_j$ is $Q/2$, a total quantity of pilot symbols that are $-A+A_j$ is $Q/2$, a total quantity of pilot symbols that are $A-A_j$ is $Q/2$, and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[0225] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0226] 96 pilot symbols in the first polarization direction are sequentially:

[0227] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj,
A+Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, -A+Aj,
-A-Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A-Aj,
A-Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj, A+Aj, A+Aj,
-A-Aj, A+Aj, A+Aj, -A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj,
-A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj, -A-Aj,
A+Aj, A+Aj, A+Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj,
-A-Aj, A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj,
-A-Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, -A-Aj, A+Aj,
-A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, A+Aj,
A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, A+Aj, A+Aj,
-A+Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, and A+Aj.

[0228] 96 pilot symbols in the second polarization direction are sequentially:

[0229] A-Aj, A-Aj, A-Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A-Aj, A-Aj, A-Aj, -A+Aj, A-Aj, -A-Aj, -A+Aj, A+Aj, A+Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A+Aj, -A-Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, and A+Aj.

[0230] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0231] 96 pilot symbols in the first polarization direction are sequentially:

[0232] -A+Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj,
-A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj,
-A+Aj, A+Aj, -A+Aj, -A-Aj, -A-Aj, A-Aj, -A+Aj,
-A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A+Aj,
-A+Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj,
A+Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj,
-A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A-Aj,
A+Aj, A-Aj, A+Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj.

[0264] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at a transmitting end and a laser at the receiving end.

[0265] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0266] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0267] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following:

[0268] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x115;

[0269] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x02B;

[0270] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x115;

[0271] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x02B;

[0272] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x115, and the seed in the second polarization direction: 0x08D;

[0273] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x08D, and the seed in the second polarization direction: 0x02B;

[0274] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0FE;

[0275] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0BF;

[0276] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x17F;

[0277] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0FE;

[0278] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0BF;

[0279] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x17F;

[0280] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0FE;

[0281] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0BF;

[0282] the target polynomial: $x^9+x^5+x^4+x+1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x17F;

[0283] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x11E;

[0284] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x03D;

[0285] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x08F;

[0286] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x11E;

[0287] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x03D;

[0288] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x08F;

[0289] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x11E, and the seed in the second polarization direction: 0x175;

[0290] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x175, and the seed in the second polarization direction: 0x03D;

[0291] the target polynomial: $x^9+x^8+x^5+x^4+1$, the seed in the first polarization direction: 0x175, and the seed in the second polarization direction: 0x08F;

[0292] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x16A, and the seed in the second polarization direction: 0x1E1;

[0293] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x16A, and the seed in the second polarization direction: 0x1C3;

[0294] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x1E1, and the seed in the second polarization direction: 0x069;

[0295] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x1E1, and the seed in the second polarization direction: 0x113;

[0296] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x069, and the seed in the second polarization direction: 0x1C3;

[0297] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x1C3, and the seed in the second polarization direction: 0x113;

[0298] the target polynomial: $x^{10}+x^4+x^3+x+1$, the seed in the first polarization direction: 0x0E6, and the seed in the second polarization direction: 0x36E;

[0299] the target polynomial: $x^{10}+x^5+x^2+x+1$, the seed in the first polarization direction: 0x3DC, and the seed in the second polarization direction: 0x36A;

[0300] the target polynomial: $x^{10}+x^5+x^2+x+1$, the seed in the first polarization direction: 0x36A, and the seed in the second polarization direction: 0x35E;

[0301] the target polynomial: $x^{10}+x^5+x^2+x+1$, the seed in the first polarization direction: 0x36A, and the seed in the second polarization direction: 0x1AF;

[0302] the target polynomial: $x^{10}+x^8+x^5+x+1$, the seed in the first polarization direction: 0x1FD, and the seed in the second polarization direction: 0x3A7;

- [illegible]

- [0370] the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x014, and the seed in the second polarization direction: 0x34C;
- [0371] the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x014, and the seed in the second polarization direction: 0x261;
- [0372] the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x34C, and the seed in the second polarization direction: 0x282; or the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x282, and the seed in the second polarization direction: 0x261.
- [0373] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following:
- [0374] the target polynomial: $x^9+x^8+x^6+x^5+1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x1E1;
- [0375] the target polynomial: $x^9+x^4+x^3+x+1$, the seed in the first polarization direction: 0x094, and the seed in the second polarization direction: 0x02B;
- [0376] the target polynomial: $x^{10}+x^9+x^7+x^6+1$, the seed in the first polarization direction: 0x058, and the seed in the second polarization direction: 0x3CC;
- [0377] the target polynomial: $x^{10}+x^9+x^7+x^6+1$, the seed in the first polarization direction: 0x3CC, and the seed in the second polarization direction: 0x316;
- [0378] the target polynomial: $x^{10}+x^9+x^7+x^6+1$, the seed in the first polarization direction: 0x3CC, and the seed in the second polarization direction: 0x18E;
- [0379] the target polynomial: $x^{10}+x^9+x^7+x^6+1$, the seed in the first polarization direction: 0x3CC, and the seed in the second polarization direction: 0x163;
- [0380] the target polynomial: $x^{10}+x^9+x^8+x^5+1$, the seed in the first polarization direction: 0x152, and the seed in the second polarization direction: 0x14D;
- [0381] the target polynomial: $x^{10}+x^9+x^8+x^5+1$, the seed in the first polarization direction: 0x2A4, and the seed in the second polarization direction: 0x316;
- [0382] the target polynomial: $x^{10}+x^9+x^6+x+1$, the seed in the first polarization direction: 0x0B4, and the seed in the second polarization direction: 0x2D1;
- [0383] the target polynomial: $x^{10}+x^9+x^6+x+1$, the seed in the first polarization direction: 0x1A2, and the seed in the second polarization direction: 0x05A;
- [0384] the target polynomial: $x^{10}+x^9+x^6+x+1$, the seed in the first polarization direction: 0x05A, and the seed in the second polarization direction: 0x2D1;
- [0385] the target polynomial: $x^{10}+x^9+x^4+x+1$, the seed in the first polarization direction: 0x128, and the seed in the second polarization direction: 0x12A;
- [0386] the target polynomial: $x^{10}+x^9+x^4+x+1$, the seed in the first polarization direction: 0x250, and the seed in the second polarization direction: 0x12A;
- [0387] the target polynomial: $x^{10}+x^9+x^4+x+1$, the seed in the first polarization direction: 0x250, and the seed in the second polarization direction: 0x295;
- [0388] the target polynomial: $x^{10}+x^4+x^3+x+1$, the seed in the first polarization direction: 0x39C, and the seed in the second polarization direction: 0x36E;
- [0389] the target polynomial: $x^{10}+x^4+x^3+x+1$, the seed in the first polarization direction: 0x18E, and the seed in the second polarization direction: 0x36E;
- [0390] the target polynomial: $x^{10}+x^4+x^3+x+1$, the seed in the first polarization direction: 0x36E, and the seed in the second polarization direction: 0x239;
- [0391] the target polynomial: $x^{10}+x^4+x^3+x+1$, the seed in the first polarization direction: 0x36E, and the seed in the second polarization direction: 0x0E7;
- [0392] the target polynomial: $x^{10}+x^5+x^2+x+1$, the seed in the first polarization direction: 0x232, and the seed in the second polarization direction: 0x36A;
- [0393] the target polynomial: $x^{10}+x^5+x^2+x+1$, the seed in the first polarization direction: 0x36A, and the seed in the second polarization direction: 0x119;
- [0394] the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x282, and the seed in the second polarization direction: 0x01F;
- [0395] the target polynomial: $x^{10}+x^9+x^8+x^7+x^3+x^2+1$, the seed in the first polarization direction: 0x014, and the seed in the second polarization direction: 0x01F;
- [0396] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x1BA, and the seed in the second polarization direction: 0x2CE;
- [0397] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x2A6, and the seed in the second polarization direction: 0x2CE;
- [0398] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x196, and the seed in the second polarization direction: 0x2CE;
- [0399] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x2CE, and the seed in the second polarization direction: 0x259;
- [0400] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x380, and the seed in the second polarization direction: 0x2CE;
- [0401] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x380, and the seed in the second polarization direction: 0x081;
- [0402] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x2CE, and the seed in the second polarization direction: 0x375;
- [0403] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x2CE, and the seed in the second polarization direction: 0x353;
- [0404] the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x2CE, and the seed in the second polarization direction: 0x2FF;

- [illegible]

[0453] In some possible implementations, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor+1$. Alternatively, in a polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor+1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$. [a] represents rounding down a positive real number a to the nearest integer.

[0454] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^9+x^8+x^5+x^4+1$, a seed in a first polarization direction is 0x175, a seed in a second polarization direction is 0x03D, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0455] In some possible implementations, 96 pilot symbols in the first polarization direction are sequentially:

[0456] A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, A-Aj, and -A-Aj.

[0457] 96 pilot symbols in the second polarization direction are sequentially:

[0458] A-Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj,
-A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj,
-A-Aj, A+Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, -A+Aj,
A+Aj, A+Aj, A-Aj, A+Aj, -A-Aj, -A+Aj, A-Aj,
-A-Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, -A+Aj, A+Aj.

A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, A+Aj, -A+Aj, A-Aj, -A+Aj, A-Aj, A+Aj, A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A+Aj, A+Aj, and A+Aj.

[0459] In some possible implementations, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$; and in a second polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$. Alternatively, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor+1$; and in a second polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor+1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$. [a] represents rounding down the positive real number a to the nearest integer, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0460] In the implementation, in the two polarization directions, a total quantity of pilot symbols that are $-A-A_j$ is $Q/2$, a total quantity of pilot symbols that are $-A+A_j$ is $Q/2$, a total quantity of pilot symbols that are $A-A_j$ is $Q/2$, and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[0461] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0462] 96 pilot symbols in the first polarization direction are sequentially:

[0463] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj,
A+Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj,
-A+Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, -A-Aj,
A-Aj, A-Aj, A-Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, A-Aj,
A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj, -A+Aj,
A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, -A+Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj,
-A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj, -A-Aj, A-Aj,
A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, -A-Aj, -A+Aj, -A-Aj, -A-Aj, and A+Aj.

and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0484] In some possible implementations, the target polynomial is one of the following:

- [0485] $x^9 + x^4 + x^3 + x + 1$;
- [0486] $x^9 + x^5 + x^4 + x + 1$;
- [0487] $x^9 + x^8 + x^5 + x^4 + 1$;
- [0488] $x^9 + x^8 + x^6 + x^5 + 1$;
- [0489] $x^{10} + x^4 + x^3 + x + 1$;
- [0490] $x^{10} + x^5 + x^2 + x + 1$;
- [0491] $x^{10} + x^8 + x^5 + x + 1$;
- [0492] $x^{10} + x^9 + x^5 + x^2 + 1$;
- [0493] $x^{10} + x^9 + x^6 + x + 1$;
- [0494] $x^{10} + x^9 + x^7 + x^6 + 1$;
- [0495] $x^{10} + x^9 + x^8 + x^5 + 1$;
- [0496] $x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$;
- [0497] $x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$;
- [0498] $x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$;
- [0499] $x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$; or

[0500] In some possible implementations, the target polynomial is one of the following:

[0501] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at a transmitting end and a laser at the receiving end.

[0502] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0503] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0504] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following:

- [0505] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x115;
- [0506] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x002, and the seed in the second polarization direction: 0x02B;
- [0507] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x115;
- [0508] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x049, and the seed in the second polarization direction: 0x02B;

[0509] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x115, and the seed in the second polarization direction: 0x08D;

[0510] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: 0x08D, and the seed in the second polarization direction: 0x02B;

[0511] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0FE;

[0512] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x0BF;

[0513] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x098, and the seed in the second polarization direction: 0x17F;

[0514] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0FE;

[0515] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x0BF;

[0516] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x14C, and the seed in the second polarization direction: 0x17F;

[0517] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0FE;

[0518] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x0BF;

[0519] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: 0x0A6, and the seed in the second polarization direction: 0x17F;

[0520] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x11E;

[0521] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x03D;

[0522] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x1D4, and the seed in the second polarization direction: 0x08F;

[0523] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x11E;

[0524] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x03D;

[0525] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x0EA, and the seed in the second polarization direction: 0x08F;

[0526] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x11E, and the seed in the second polarization direction: 0x175;

[0527] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x175, and the seed in the second polarization direction: 0x03D;

[0528] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: 0x175, and the seed in the second polarization direction: 0x08F;

[0529] the target polynomial: $x^9 + x^8 + x^6 + x^5 + 1$, the seed in the first polarization direction: 0x16A, and the seed in the second polarization direction: 0x1E1;

- [illegible]

- [illegible]

- [illegible]

- [illegible]

and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[0698] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0699] 96 pilot symbols in the first polarization direction are sequentially:

[0701] 96 pilot symbols in the second polarization direction are sequentially:

[0703] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0704] 96 pilot symbols in the first polarization direction are sequentially:

[0705] $-A+A_j, -A+A_j, -A+A_j, -A-A_j, A-A_j, A+A_j,$
 $-A+A_j, A-A_j, A+A_j, -A-A_j, A-A_j, -A+A_j, -A+A_j,$
 $-A+A_j, A+A_j, -A+A_j, -A-A_j, -A-A_j, A-A_j, -A+A_j,$
 $-A+A_j, -A-A_j, -A-A_j, A-A_j, A-A_j, A-A_j, A+A_j,$
 $-A+A_j, A-A_j, -A-A_j, A+A_j, -A+A_j, -A-A_j, -A+A_j,$
 $A+A_j, A-A_j, A+A_j, -A+A_j, A+A_j, -A-A_j, -A-A_j,$
 $A+A_j, -A+A_j, A-A_j, -A-A_j, -A-A_j, A-A_j, A+A_j,$
 $-A-A_j, A+A_j, -A-A_j, A-A_j, A+A_j, -A-A_j, A+A_j,$
 $-A-A_j, A-A_j, A+A_j, A+A_j, -A-A_j, A+A_j, A+A_j,$
 $A+A_j, -A-A_j, A+A_j, -A+A_j, A+A_j, -A-A_j, -A-A_j,$
 $A+A_j, -A-A_j, -A+A_j, -A+A_j, A-A_j, -A-A_j, -A+A_j,$
 $A-A_j, A+A_j, -A+A_j, -A-A_j, \text{ and } A+A_j.$

[0706] 96 pilot symbols in the second polarization direction are sequentially:

[0707] A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj, A-Aj, -A+Aj, -A-Aj, A-Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,

A-Aj, -A-Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, A-Aj, A-Aj, -A+Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, and A+Aj.

[0708] In some possible implementations, a quantity of data frames is W, W is an integer greater than 1, and the W data frames are carried in a plurality of optical signals for transmission.

[0709] In some possible implementations, the W data frames are respectively carried in W optical signals. Any two of the W optical signals have different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0710] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the first target polynomial and a second seed. In the first polarization direction, Q pilot symbols in the second data frame are generated by using a second target polynomial and a third seed; and in the second polarization direction, the Q pilot symbols in the second data frame are generated by using the second target polynomial and a fourth seed.

[0711] In some possible implementations, a first pilot symbol sequence in the first data frame is the same as a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0712] In some possible implementations, the first target polynomial is the same as the second target polynomial, the first seed is the same as the third seed, the second seed is the same as the fourth seed.

[0713] In some possible implementations, a first pilot symbol sequence in the first data frame is different from a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0714] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0715] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0716] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0717] In some possible implementations, in a polarization direction, a modulation format of the symbol in the data frame is 16QAM, and $A = -1, 1, -3, 3, -\sqrt{5}$, or $\sqrt{5}$.

[0718] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A = -1$ or 1 .

[0719] According to a seventh aspect, an embodiment of this disclosure provides a data transmission apparatus. The data transmission apparatus includes a receiving unit and a processing unit. The receiving unit is configured to receive data frames. In a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and $M-1$ payload symbols, $N = M \times Q$, Q is an even number, M is an integer greater than or equal to 1 , Q pilot symbols are generated by using a target polynomial and a seed, each pilot symbol is one of four complex numbers: $-A - Aj$, $-A + Aj$, $A - Aj$, and $A + Aj$, A is a real number, the Q pilot symbols are direct current balanced, a degree of the target polynomial is less than or equal to 10 , and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8 . The processing unit is configured to process the data frames.

[0720] In the implementation, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the $M-1$ payload symbols, and $N = M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10 , and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8 . It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0721] In some possible implementations, the target polynomial is one of the following: $x^9 + x^4 + x^3 + x + 1$;

[0722] $x^9 + x^5 + x^4 + x + 1$;

[0723] $x^9 + x^8 + x^5 + x^4 + 1$;

[0724] $x^9 + x^8 + x^6 + x^5 + 1$;

[0725] $x^{10} + x^4 + x^3 + x + 1$;

[0726] $x^{10} + x^5 + x^2 + x + 1$;

[0727] $x^{10} + x^8 + x^5 + x + 1$;

[0728] $x^{10} + x^9 + x^4 + x + 1$;

[0729] $x^{10} + x^9 + x^5 + x^2 + 1$;

[0730] $x^{10} + x^9 + x^6 + x + 1$;

[0731] $x^{10} + x^9 + x^7 + x^6 + 1$;

[0732] $x^{10} + x^9 + x^8 + x^5 + 1$;

[0733] $x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$;

[0734] $x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$;

[0735] $x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$;

[0736] $x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$; or

[0737] $x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$.

[0738] In some possible implementations, the target polynomial is one of the following:

[0739] $x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$; or

[0740] $x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$.

[0741] It should be understood that the foregoing polynomials have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is a frequency difference between a laser at a transmitting end and a laser at the receiving end.

[0742] In some possible implementations, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located.

[0743] In some possible implementations, a sequence including Q pilot symbols in a first polarization direction is different from a sequence including Q pilot symbols in a second polarization direction, and the first polarization direction and the second polarization direction are orthogonal to each other. This avoids a problem that the receiving end cannot distinguish between two polarization directions in actual transmission.

[0744] In some possible implementations, $N = 6144$, $M = 64$, $Q = 96$, and a correspondence between the target polynomial, a seed in the first polarization direction, and a seed in the second polarization direction is one of the following:

[0745] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x002$, and the seed in the second polarization direction: $0x115$;

[0746] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x002$, and the seed in the second polarization direction: $0x02B$;

[0747] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x049$, and the seed in the second polarization direction: $0x115$;

[0748] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x049$, and the seed in the second polarization direction: $0x02B$;

[0749] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x115$, and the seed in the second polarization direction: $0x08D$;

[0750] the target polynomial: $x^9 + x^4 + x^3 + x + 1$, the seed in the first polarization direction: $0x08D$, and the seed in the second polarization direction: $0x02B$;

[0751] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x098$, and the seed in the second polarization direction: $0x0FE$;

[0752] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x098$, and the seed in the second polarization direction: $0x0BF$;

[0753] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x098$, and the seed in the second polarization direction: $0x17F$;

[0754] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x14C$, and the seed in the second polarization direction: $0x0FE$;

[0755] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x14C$, and the seed in the second polarization direction: $0x0BF$;

[0756] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x14C$, and the seed in the second polarization direction: $0x17F$;

[0757] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x0A6$, and the seed in the second polarization direction: $0x0FE$;

[0758] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x0A6$, and the seed in the second polarization direction: $0x0BF$;

[0759] the target polynomial: $x^9 + x^5 + x^4 + x + 1$, the seed in the first polarization direction: $0x0A6$, and the seed in the second polarization direction: $0x17F$;

[0760] the target polynomial: $x^9 + x^8 + x^5 + x^4 + 1$, the seed in the first polarization direction: $0x1D4$, and the seed in the second polarization direction: $0x11E$;

- [illegible]

- [illegible]

- [illegible]

- [illegible]

[0900] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x226;

[0901] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x165;

[0902] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x165, and the seed in the second polarization direction: 0x0C7;

[0903] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x266, and the seed in the second polarization direction: 0x0C7;

[0904] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x2E2, and the seed in the second polarization direction: 0x3E2;

[0905] the target polynomial: $x^{10}+x^9+x^6+x^3+x^2+x+1$, the seed in the first polarization direction: 0x0C4, and the seed in the second polarization direction: 0x226; or the target polynomial: $x^{10}+x^9+x^8+x^7+x^4+x+1$, the seed in the first polarization direction: 0x1D8, and the seed in the second polarization direction: 0x2D9.

[0906] It should be understood that the foregoing polynomials and seeds have good differential autocorrelation and differential cross-correlation characteristics, and the good correlation effects can still be achieved when there is the frequency difference between the laser at the transmitting end and the laser at the receiving end.

[0907] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^9+x^4+x^3+x+1$, the seed in the first polarization direction is 0x049, the seed in the second polarization direction is 0x115, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0908] 96 pilot symbols in the first polarization direction are sequentially:

[0909] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj,
A+Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj,
-A+Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, -A-Aj,
A-Aj, A-Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj,
A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj, -A+Aj,
A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj,
-A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj, -A-Aj, -A-Aj,
A-Aj, -A+Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, -A+Aj,
-A-Aj, -A+Aj, -A+Aj, A+Aj, -A+Aj, A+Aj, -A+Aj,
-A+Aj, -A-Aj, -A+Aj, -A-Aj, -A-Aj, and A+Aj.

[0910] 96 pilot symbols in the second polarization direction are sequentially:

[0911] A-Aj, A-Aj, A-Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A-Aj, A-Aj, A-Aj, -A+Aj, A-Aj, -A+Aj, -A+Aj, A+Aj, A-Aj, A+Aj, A+Aj, A+Aj, -A+Aj, A-Aj, -A+Aj, A-Aj, -A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj.

$-A+A_j$, $A-A_j$, $A+A_j$, $-A-A_j$, $A-A_j$, $-A-A_j$, $A+A_j$,
 $-A-A_j$, $A-A_j$, $-A+A_j$, $A+A_j$, $-A-A_j$, $-A+A_j$, $-A-A_j$,
 $-A-A_j$, $-A+A_j$, $-A+A_j$, $A-A_j$, $A+A_j$, $A+A_j$, $-A-A_j$,
 $-A+A_j$, $A+A_j$, $-A-A_j$, $-A-A_j$, $-A-A_j$, $-A-A_j$, $A-A_j$,
 $-A-A_j$, $-A+A_j$,

[0912] A-Aj, A+Aj, A-Aj, A-Aj, -A+Aj, -A+Aj, A+Aj, -A-Aj, A+Aj, A-Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, and A+Aj.

[0913] In some possible implementations, $N=6144$, $M=64$, $Q=96$, the target polynomial is $x^{10}+x^9+x^4+x+1$, the seed in the first polarization direction is 0x12A, the seed in the second polarization direction is 0x039, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0914] 96 pilot symbols in the first polarization direction are sequentially:

[0915] -A+Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj,
-A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj,
-A+Aj, A-Aj, -A+Aj, -A-Aj, -A-Aj, A-Aj, -A+Aj,
-A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A+Aj,
-A+Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj,
A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj,
-A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, A+Aj, A-Aj, A-Aj,
A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj,
A+Aj, -A-Aj, A+Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj,
A+Aj, -A-Aj, -A+Aj, A+Aj, A-Aj, A-Aj, -A+Aj,
A-Aj, A+Aj, -A+Aj, -A-Aj, and -A+Aj.

[0916] 96 pilot symbols in the second polarization direction are sequentially:

[0917] A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj,
-A-Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj,
A-Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,
A-Aj, -A-Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, A-Aj,
-A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, A-Aj, A+Aj,
-A-Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, -A-Aj, A-Aj,
A+Aj, -A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
-A+Aj, A-Aj, A+Aj, A-Aj, A+Aj, A-Aj, A-Aj, A-Aj,
-A+Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, -A-Aj, A-Aj,
-A+Aj, A+Aj, A-Aj, A+Aj, A-Aj, -A+Aj, -A+Aj,
-A-Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, -A-Aj, -A+Aj,
-A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, and A+Aj.

[0918] In some possible implementations, a quantity of data frames is W, W is an integer greater than 1, and the W data frames are carried in a plurality of optical signals for transmission.

[0919] In some possible implementations, the W data frames are respectively carried in W optical signals. Any two of the W optical signals have different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0920] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the

[0936] In some possible implementations, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$; and in a second polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor$. Alternatively, in a first polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor+1$; and in a second polarization direction, in the data frame, the quantity of pilot symbols that are $-A-A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor+1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor+1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2-\lfloor Q/4 \rfloor-1$. $\lfloor a \rfloor$ represents rounding down the positive real number a to the nearest integer, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0937] In the implementation, in the two polarization directions, a total quantity of pilot symbols that are $-A-A_j$ is $Q/2$, a total quantity of pilot symbols that are $-A+A_j$ is $Q/2$, a total quantity of pilot symbols that are $A-A_j$ is $Q/2$, and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[0938] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0939] 96 pilot symbols in the first polarization direction are sequentially:

[0940] A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, -A+Aj,
A+Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A+Aj,
-A+Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, -A-Aj,
A-Aj, A-Aj, A-Aj, A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj,
A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj, -A+Aj,
A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, -A+Aj, -A-Aj, -A-Aj, A+Aj, A+Aj, A+Aj,
-A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, -A+Aj,
A-Aj, -A+Aj, -A+Aj, A+Aj, -A+Aj, A+Aj, A-Aj,
A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, A-Aj, A+Aj, -A+Aj,
-A+Aj, -A-Aj, -A+Aj, -A-Aj, and A+Aj.

[0941] 96 pilot symbols in the second polarization direction are sequentially:

[0942] A-Aj, A-Aj, A-Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A-Aj, A-Aj, A-Aj, -A+Aj, A-Aj, -A+Aj, A+Aj, A-Aj, A+Aj, A+Aj, A+Aj, -A+Aj, A-Aj, -A+Aj, A-Aj, -A+Aj, A+Aj, A+Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, -A-Aj,

[0943] A-Aj, -A-Aj, A+Aj, -A-Aj, A-Aj, -A+Aj,
A+Aj, -A-Aj, -A+Aj, -A-Aj, -A-Aj, -A+Aj.

$-A+A_j$, $A-A_j$, $A+A_j$, $A+A_j$, $-A-A_j$, $-A+A_j$, $A+A_j$,
 $-A-A_j$, $-A-A_j$, $-A-A_j$, $-A-A_j$, $A-A_j$, $-A-A_j$,
 $-A+A_j$, $A-A_j$, $A+A_j$, $A-A_j$, $A-A_j$, $-A+A_j$, $-A+A_j$,
 $A+A_j$, $-A-A_j$, $A+A_j$, $A-A_j$, $-A+A_j$, $A-A_j$, $-A-A_j$,
 $A+A_j$, $A+A_j$, $-A-A_j$, $-A-A_j$, and $A+A_j$.

[0944] In some possible implementations, $N=6144$, $M=64$, $Q=96$, and the first polarization direction and the second polarization direction are orthogonal to each other.

[0945] 96 pilot symbols in the first polarization direction are sequentially:

[0946] -A+Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj,
-A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj,
-A+Aj, A+Aj, -A+Aj, -A-Aj, -A-Aj, A-Aj, -A+Aj,
-A+Aj, -A-Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A+Aj, A+Aj,
-A+Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, -A+Aj,
A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, -A+Aj, -A-Aj,
A+Aj, -A+Aj, A-Aj, -A-Aj, -A-Aj, A-Aj, A+Aj,
A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, A-Aj, A+Aj, A+Aj,
-A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, -A-Aj,
-A-Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A+Aj, A+Aj,
A+Aj, -A-Aj, A+Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj,
A+Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, -A-Aj, -A+Aj,
A-Aj, A+Aj, -A+Aj, -A-Aj, and -A+Aj.

[0947] 96 pilot symbols in the second polarization direction are sequentially:

[0948] A-Aj, -A+Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj,
 -A-Aj, -A+Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj,
 A-Aj, -A+Aj, -A-Aj, A-Aj, -A+Aj, -A-Aj, -A+Aj,
 A+Aj, -A-Aj, A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,
 A+Aj, -A-Aj, A+Aj, -A-Aj, -A-Aj, -A+Aj, A-Aj,
 -A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, A-Aj, A+Aj,
 -A-Aj, -A-Aj, A-Aj, A-Aj, -A+Aj, -A-Aj, A-Aj,
 A+Aj, -A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, -A-Aj,
 -A+Aj, A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, A-Aj,
 -A+Aj, A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, A-Aj,
 -A-Aj, -A+Aj, -A+Aj, -A+Aj, -A-Aj, -A-Aj, A+Aj,
 -A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, A+Aj, and A+Aj.

[0949] In some possible implementations, a quantity of data frames is W, W is an integer greater than 1, and the W data frames are carried in a plurality of optical signals for transmission.

[0950] In some possible implementations, the W data frames are respectively carried in W optical signals. Any two of the W optical signals have different wavelengths. Alternatively, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[0951] In some possible implementations, the W data frames include a first data frame and a second data frame, and the first polarization direction and the second polarization direction are orthogonal to each other. In the first polarization direction, Q pilot symbols in the first data frame are generated by using a first target polynomial and a first seed; and in the second polarization direction, the Q pilot symbols in the first data frame are generated by using the first target polynomial and a second seed. In the first polarization direction, Q pilot symbols in the second data frame are generated by using a second target polynomial and a third seed; and in the second polarization direction, the Q pilot symbols in the second data frame are generated by using the second target polynomial and a fourth seed.

[0952] In some possible implementations, a first pilot symbol sequence in the first data frame is the same as a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0953] In some possible implementations, the first target polynomial is the same as the second target polynomial, the first seed is the same as the third seed, the second seed is the same as the fourth seed, the first seed is different from the second seed, and the third seed is different from the fourth seed.

[0954] In some possible implementations, a first pilot symbol sequence in the first data frame is different from a second pilot symbol sequence in the second data frame, the first pilot symbol sequence includes the Q pilot symbols in the first data frame, and the second pilot symbol sequence includes the Q pilot symbols in the second data frame.

[0955] In some possible implementations, the first target polynomial is the same as the second target polynomial, and any two of the first seed, the second seed, the third seed, and the fourth seed are different from each other.

[0956] In some possible implementations, the first target polynomial is different from the second target polynomial.

[0957] In some possible implementations, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

[0958] In some possible implementations, in a polarization direction, the modulation format of the symbol in the data frame is QPSK, and $A=-1$ or 1 .

[0959] In embodiments of this disclosure, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N=M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that the simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, the simple hardware structure may be used for the implementation. In addition, the pilot symbols generated in the foregoing manner have the good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end restore the signal quality.

BRIEF DESCRIPTION OF DRAWINGS

[0960] FIG. 1 is a diagram of a communication system to which an embodiment of this disclosure is applied;

[0961] FIG. 2 is a diagram of an implementation of a transmitting-end DSP processor according to an embodiment of this disclosure;

[0962] FIG. 3 is a diagram of another implementation of a transmitting-end DSP processor according to an embodiment of this disclosure;

[0963] FIG. 4 is a schematic flowchart of a data transmission method according to an embodiment of this disclosure;

[0964] FIG. 5 is a diagram of a structure of a data frame according to an embodiment of this disclosure;

[0965] FIG. 6 is a diagram of symbols in a constellation diagram according to an embodiment of this disclosure;

[0966] FIG. 7 is a diagram of a first-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0967] FIG. 8 is a diagram of a second-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0968] FIG. 9 is a diagram of a structure of another data frame according to an embodiment of this disclosure;

[0969] FIG. 10 is a diagram of a third-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0970] FIG. 11 is a diagram of a first implementation procedure of generating pilot symbols based on dual-polarization (DP)-16QAM modulation according to an embodiment of this disclosure;

[0971] FIG. 12A to FIG. 12C are diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0972] FIG. 13A and FIG. 13B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0973] FIG. 14 is a diagram of a fourth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0974] FIG. 15 is a diagram of a second implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0975] FIG. 16A to FIG. 16C are other diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0976] FIG. 17A and FIG. 17B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0977] FIG. 18 is a diagram of a transmission scenario of a plurality of data frame streams according to an embodiment of this disclosure;

[0978] FIG. 19 is a diagram of a fifth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0979] FIG. 20 is a diagram of a third implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0980] FIG. 21A to FIG. 21C are other diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure;

[0981] FIG. 22A and FIG. 22B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0982] FIG. 23 is a diagram of a sixth-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0983] FIG. 24 is a diagram of a fourth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0984] FIG. 25A and FIG. 25B are other diagrams of an autocorrelation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure;

[0985] FIG. 26A and FIG. 26B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0986] FIG. 27 is a diagram of a seventh-generation structure of a pilot symbol according to an embodiment of this disclosure;

[0987] FIG. 28 is a diagram of a fifth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure;

[0988] FIG. 29A and FIG. 29B are other diagrams of a correlation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure;

[0989] FIG. 30A and FIG. 30B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure;

[0990] FIG. 31 is a diagram of a structure of a data transmission apparatus used at a transmitting end according to an embodiment of this disclosure;

[0991] FIG. 32 is a diagram of a structure of a data transmission apparatus used at a receiving end according to an embodiment of this disclosure; and

[0992] FIG. 33 is a diagram of another structure of a data transmission apparatus according to an embodiment of this disclosure.

DESCRIPTION OF EMBODIMENTS

[0993] Embodiments of this disclosure provide a data transmission method and a data transmission apparatus. A simple target polynomial is designed to generate pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the generated pilot symbols have good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps a receiving end restore signal quality.

[0994] It should be noted that in the specification, claims, and the foregoing accompanying drawings of this disclosure, the terms “first”, “second”, and the like are intended to distinguish between similar objects but do not limit a specific order or sequence. It should be understood that the foregoing terms may be interchanged in proper cases, so that embodiments described in this disclosure can be implemented in an order other than the content described in this disclosure. In addition, the terms “include”, “have”, and any other variant thereof are intended to cover a non-exclusive inclusion. For example, a process, a method, a system, a product, or a device that includes a series of steps or units is not necessarily limited to those expressly listed steps or units, but may include other steps or units not expressly listed or inherent to the process, the method, the product, or the device.

[0995] FIG. 1 is a diagram of a communication system to which an embodiment of this disclosure is applied. As shown in FIG. 1, at a transmitting end, a source provides a to-be-sent data stream. An encoder receives and encodes the data stream. Encoded codeword information combining a parity bit and an information bit is sent to a transmitting-end digital signal processing (DSP) processor for framing, and is transmitted through a channel to a receiving end. After receiving a signal with distortion resulting from noise or other impairments on the channel, the receiving end sends the signal to a receiving-end DSP processor for dispersion compensation, synchronization, phase recovery, and other operations. Then, a decoder performs decoding to restore original data, and sends the data to a sink.

[0996] FIG. 2 is a diagram of an implementation of a transmitting-end DSP processor according to an embodiment of this disclosure. As shown in FIG. 2, in a possible implementation, the transmitting-end DSP processor performs symbol mapping on a received data sequence. Usually, the received data sequence is an information sequence and a parity sequence obtained based on a forward error correction (FEC) code. A symbol mapping manner includes but is not limited to QPSK and QAM. The transmitting-end DSP processor further performs polarization symbol distribution on the data sequence to obtain a DP symbol, for example, a DP-QPSK, DP-8QAM, DP-16QAM, DP-32QAM, or DP-64QAM symbol. For ease of description, two polarization directions are collectively denoted as an X polarization direction and a Y polarization direction below, where the X polarization direction and the Y polarization direction are orthogonal to each other. It should be understood that the X polarization direction and the Y polarization direction are not two specified polarization directions, but are any two polarization directions that are orthogonal to each other.

[0997] Further, the transmitting-end DSP processor performs the following framing processing on a specific quantity of DP symbols. Pilot symbols are inserted in each of the X polarization direction and the Y polarization direction, to obtain a to-be-sent DP symbol sequence, which is referred to as a frame. In embodiments, the frame is referred to as a data frame, and may also be referred to as a DSP frame.

[0998] It should be noted that, after the symbol mapping, the symbols may be further interleaved, and the foregoing framing processing is performed on interleaved symbols. It should be understood that one DP symbol may be represented by two symbols, where one symbol is located in the X polarization direction, and the other symbol is located in the Y polarization direction. Each symbol may be represented by a complex number. For example, a symbol obtained through 16QAM modulation may be represented by any one of the following 16 complex numbers: $\pm 1 \pm 1j$, $\pm 1 \pm 3j$, $\pm 3 \pm 1j$, and $\pm 3 \pm 3j$, where $\pm$ herein represents a positive value or a negative value. For example, $+3$ represents 3 or -3 . j represents an imaginary unit. In some scenarios, the imaginary unit may alternatively be represented by another symbol like i . This is not limited herein. In some cases, real and imaginary parts are normalized, but the essence does not change. Further, a sequence having L DP symbols may be completely represented by two complex sequences whose lengths are L , where one complex sequence represents a symbol in the X polarization direction, and the other complex sequence represents a symbol in the Y polarization direction. Each complex sequence whose length is L is represented by a real part sequence (also referred to as an I-channel sequence) whose length is L and an imaginary part sequence (also referred to as a Q-channel sequence) whose length is L , where L is an integer greater than 1. Therefore, there are four different types of sequences, including an X-polarization I-channel sequence, an X-polarization Q-channel sequence, a Y-polarization I-channel sequence, and a Y-polarization Q-channel sequence.

[0999] FIG. 3 is a diagram of another implementation of a transmitting-end DSP processor according to an embodiment of this disclosure. As shown in FIG. 3, different from the framing operation performed on the symbols shown in FIG. 2, in another possible implementation, before symbol mapping, the transmitting-end DSP processor inserts, in a

received data sequence according to a used symbol mapping rule, bits corresponding to pilot symbols, and then may obtain, through the symbol mapping and polarization symbol distribution, a same frame as that in the operation shown in FIG. 2. In this case, before the symbol mapping, a bit sequence obtained by inserting the bits corresponding to the symbols may be further interleaved, and then the same frame as that in the operation shown in FIG. 2 is obtained through the symbol mapping and the polarization symbol distribution.

[1000] It should be noted that a specific framing manner used by the transmitting-end DSP processor is not limited in this disclosure. In addition to the framing manners described in FIG. 2 and FIG. 3, other similar framing manners are also applicable to this solution, and are not described one by one herein.

[1001] FIG. 4 is a schematic flowchart of a data transmission method according to an embodiment of this disclosure. It should be understood that the data transmission method is applied to a transmitting end, for example, may be implemented by the transmitting-end DSP processor shown in FIG. 1.

[1002] **401:** Generate data frames.

[1003] It should be noted that a specific implementation of generating the data frames is not limited in this disclosure. For example, the framing manner described in FIG. 2 or FIG. 3 may be used. Certainly, other similar framing manners are also applicable to this solution, and are not described one by one herein. It should be understood that the data frame includes symbols in two polarization directions, and structures of the data frame in the two polarization directions are similar. For example, the data frame includes N symbols in an X polarization direction and N symbols in a Y polarization direction. The following describes the structure of the data frame by using one of the polarization directions as an example.

[1004] FIG. 5 is a diagram of a structure of a data frame according to an embodiment of this disclosure. As shown in FIG. 5, in a polarization direction, the data frame includes N symbols, every M consecutive symbols in the N symbols include one pilot symbol at a fixed location and M-1 payload symbols, $N=M \times Q$, Q is an even number, and M is an integer greater than or equal to 1. In other words, the every M consecutive symbols in the N symbols may be considered as one group, and the N symbols include Q groups of symbols in total. For example, $N=6144$, $M=64$, $Q=96$, a symbol 1 to a symbol 64 are 64 consecutive symbols in a 1st group, a symbol 65 to a symbol 128 are 64 consecutive symbols in a 2nd group, . . . , and a symbol 6081 to a symbol 6144 are 64 consecutive symbols in a 96th group. It should be understood that the payload symbol may also be referred to as a symbol before framing, and includes an information symbol and a parity symbol that are obtained through FEC encoding. At a receiving end, a pilot symbol may be used to assist in carrier phase recovery, and may be further used to distinguish between the two polarization directions.

[1005] It should be understood that a specific location of a pilot symbol in each group of M symbols is not limited in this disclosure. In an example, each pilot symbol is at a start location of M consecutive symbols in which the pilot symbol is located. For example, a 1st symbol in the data frame shown in FIG. 5 is a 1st pilot symbol.

[1006] It should be noted that Q pilot symbols in the data frame are generated by using a target polynomial and a seed. Each pilot symbol is one of $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$, and A is a real number. A degree of the target polynomial is less than or equal to 10, and a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8, so that hardware implementation complexity is low. In addition, the Q pilot symbols are direct current balanced, that is, a sum of the Q pilot symbols is 0, to be specific, a sum of real parts and a sum of imaginary parts of complex numbers corresponding to the Q pilot symbols are both 0, so that the direct current balance can be achieved. This helps the receiving end restore signal quality.

[1007] **402:** Send the data frames.

[1008] The sent data frames are transmitted through a channel to the receiving end. A specific operation performed after the receiving end receives the data frames is not described in detail in this disclosure. For details, refer to the diagram of the structure of the system shown in FIG. 1.

[1009] The following describes in detail a characteristic of the pilot symbol and a specific manner of generating the pilot symbol.

[1010] In embodiments of this disclosure, a value of A is determined based on a modulation format used during symbol generation. In some actual application scenarios, $-A-Aj$, $-A+Aj$, $A-Aj$, and $A+Aj$ are symbols in a constellation diagram of the used modulation format. For example, if QPSK is used, there are only four symbols. In this case, $A=+1$, and each pilot symbol may be represented by one of $-1-1j$, $-1+1j$, $1-1j$, and $1+1j$. In one frame, all frame symbols represented by the four complex numbers exist. If 16QAM is used, there are 16 symbols in the constellation diagram. In this case, $A=+1$ or $+3$.

[1011] FIG. 6 is a diagram of symbols in a constellation diagram according to an embodiment of this disclosure. As shown in FIG. 6, when $A=3$ or -3 , each pilot symbol may be represented by one of $-3-3j$, $-3+3j$, $3-3j$, and $3+3j$, for example, a hollow symbol shown in FIG. 6. Similarly, if 64QAM is used, $A=+1$, $+3$, $+5$, or $+7$. It should be noted that a higher-order modulation format may alternatively be used. Details are not described in this disclosure. In an actual transmission process, four outer symbols in the constellation diagram are used as pilot symbols, so that a symbol error probability is low.

[1012] It should be noted that the symbols in the constellation diagram may alternatively be compressed, and correspondingly, a value of A is also compressed. The 16QAM is used as an example, and power normalization is performed on the 16 symbols in the 16QAM constellation diagram. In this case, values of the 16 symbols in the 16QAM constellation diagram are changed to

$$\left\{ \pm \frac{1}{\sqrt{10}} \pm \frac{1}{\sqrt{10}}j, \pm \frac{1}{\sqrt{10}} \pm \frac{3}{\sqrt{10}}j, \pm \frac{3}{\sqrt{10}} \pm \frac{1}{\sqrt{10}}j, \pm \frac{3}{\sqrt{10}} \pm \frac{3}{\sqrt{10}}j \right\},$$

and the value of A is

$$\frac{1}{\sqrt{10}} \text{ or } \frac{3}{\sqrt{10}}.$$

[1013] Alternatively, normalization in another manner may be used. This is not limited in this disclosure.

[1014] It should be understood that, when four outermost symbols in the constellation diagram are used as the pilot symbols $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, the pilot symbols have high sensitivity, but have a large peak to average power ratio. When four innermost symbols in the constellation diagram are used as the pilot symbols $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, the pilot symbols have small noise, but have low sensitivity.

[1015] It should be noted that, in some actual application scenarios, alternatively, the pilot symbols $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may not be the symbols in the constellation diagram of the used modulation format, and may be four symbols in an intermediate area between the four outermost symbols and the four innermost symbols in the constellation diagram. In this case, the pilot symbols have fair noise and sensitivity, but have a low peak to average power ratio. The 16QAM is used as an example, the values of the 16 symbols in the 16QAM constellation diagram are $\{\pm 1 \pm 1j, \pm 1 \pm 3j, \pm 3 \pm 1j, \pm 3 \pm 3j\}$, and the value of the real number A satisfies $1 \leq A \leq 3$. For example, the real number $A = \sqrt{5}$.

[1016] In addition, the two polarization directions are orthogonal to each other. To be specific, when one of the polarization directions is X polarization, the other polarization direction is Y polarization; or when one of the polarization directions is Y polarization, the other polarization direction is X polarization.

[1017] In the data frame, a sequence including Q pilot symbols in the X polarization direction is different from a sequence including Q pilot symbols in the Y polarization direction. For example, if a sequence including pilot symbols in the X polarization direction is $-A-A_j$, $-A-A_j$, $A+A_j$, and $A-A_j$, in a same order, a sequence including pilot symbols in the Y polarization direction cannot be the same as the sequence including the pilot symbols in the X polarization direction, and may be $-A-A_j$, $-A-A_j$, $A+A_j$, and $A+A_j$. There is one different symbol. This avoids a problem that the receiving end cannot distinguish between the two polarization directions in actual transmission.

[1018] In embodiments of this disclosure, when a 10-degree polynomial is used as the target polynomial, the 10-degree polynomial may be represented as:

[1019] $x^{10} + a_9 x^9 + a_8 x^8 + a_7 x^7 + a_6 x^6 + a_5 x^5 + a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + 1$, where values of $a_9, \dots$, and a_1 may be 0 or 1, and a quantity of non zeros in $a_9, \dots$, and a_1 is not greater than 6. FIG. 7 is a diagram of a first-generation structure of a pilot symbol according to an embodiment of this disclosure. As shown in FIG. 7, each block may be considered as one storage element, and a quantity of storage elements is the same as a quantity of bits in a preloaded seed, that is, each storage element is configured to input a corresponding bit that is in the seed. For example, if a seed length is 10 bits, and the seed may be represented as $m_9, m_8, m_7, m_6, m_5, m_4, m_3, m_2, m_1$, and m_0 in a binary form, 10 corresponding storage elements are used. Certainly, the seed may alternatively be represented in a hexadecimal or decimal form, and needs to be converted into the binary form during an operation with the target polynomial. For example, 0110111000 is represented as 0x1B8 in the hexadecimal form, and is represented as 440 in the decimal form.

[1020] It should be noted that the polynomial $x^{10} + a_9 x^9 + a_8 x^8 + a_7 x^7 + a_6 x^6 + a_5 x^5 + a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + 1$ is sometimes written as $x^{10} + a_9 x^9 + a_8 x^8 + a_7 x^7 + a_6 x^6 + a_5 x^5 + a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + 1$.

[1021] When a 9-degree polynomial is used as the target polynomial, the 9-degree polynomial may be represented as:

[1022] $x^9 + a_8 x^8 + a_7 x^7 + a_6 x^6 + a_5 x^5 + a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + 1$, where values of $a_8, \dots$, and a_1 may be 0 or 1, and a quantity of non zeros in $a_8, \dots$, and a_1 is not greater than 6.

[1023] FIG. 8 is a diagram of a second-generation structure of a pilot symbol according to an embodiment of this disclosure. As shown in FIG. 8, a seed length is 9 bits, and the seed may be represented as $m_8, m_7, m_6, m_5, m_4, m_3, m_2, m_1$, and m_0 in a binary form. Certainly, the seed may alternatively be represented in a hexadecimal or decimal form, and needs to be converted into the binary form during an operation with the target polynomial.

[1024] It should be noted that an algebraic expression formed through addition (or subtraction) of several monomials is referred to as a polynomial. Each monomial in the polynomial is referred to as a term of the polynomial, and a highest degree (degree, which is also referred to as an order) in degrees of these monomials is a degree of the polynomial. A quantity of terms of the polynomial is a quantity of the foregoing monomials whose coefficients are not 0. For example, a quantity of terms of the foregoing 10-degree polynomial is equal to 2 plus the quantity of non zeros in $a_9, \dots$, and a_1 .

[1025] In embodiments of this disclosure, a same target generator polynomial may be used for pilot symbols in two orthogonal polarization directions. However, because seeds used in the two polarization directions are different, correspondingly, pilot symbols output in the two polarization directions are not completely the same.

[1026] In FIG. 7 or FIG. 8, for a scenario in which Q pilot symbols need to be generated, a consecutive bit sequence $b_0, b_1, b_2, \dots$, and b_{2Q-1} whose bit length is $2Q$ is obtained based on the target polynomial and the seed. The foregoing bit sequence is also referred to as a pseudo random binary sequence (PRBS). A bit sequence generated by using the 9-degree polynomial is also referred to as a PRBS9, and a bit sequence generated by using the 10-degree polynomial is also referred to as a PRBS10. Every two consecutive bits in the bit sequence $b_0, b_1, b_2, \dots$, and b_{2Q-1} are mapped to one symbol, where b_{2t} and b_{2t+1} are mapped to one symbol $(2b_{2t-1})A + (2b_{2t+1} - 1)A_j$, and $0 \leq t < Q$.

[1027] In this embodiment of this disclosure, the target polynomial and the seed may be determined by designing the values of the coefficients $a_9, \dots$, and a_1 in the 10-degree polynomial or the values of the coefficients $a_8, \dots$, and a_1 in the 9-degree polynomial, so that a generated pilot symbol sequence in the X polarization direction or the Y polarization direction has a good autocorrelation characteristic, and pilot symbol sequences in the two polarization directions have a good cross-correlation characteristic. In addition, an appropriate target polynomial and seed are selected, so that the Q pilot symbols achieve the direct current balance. This helps the receiving end restore the signal quality.

[1028] It should be noted that the data frame considered in embodiments of the present disclosure includes only the pilot symbol and the payload symbol (also referred to as the symbol before framing). This DSP frame structure is different from another existing DSP superframe structure. A DSP

superframe usually includes a plurality of types of symbols, for example, a frame synchronization symbol used for frame synchronization and distinguishing between two polarization directions, a training symbol used for link training, and a pilot symbol used for carrier phase recovery. Therefore, in receiving-end processing corresponding to the DSP frame considered in the present disclosure, in addition to assisting in the carrier phase recovery, the pilot symbol further needs to be used to distinguish between two polarization directions and used for the frame synchronization. The receiving end obtains four data streams: a real part sequence data stream in the X polarization direction, an imaginary part sequence data stream in the X polarization direction, a real part sequence data stream in the Y polarization direction, and an imaginary part sequence data stream in the Y polarization direction, and needs to use the pilot symbol to determine whether each data stream corresponds to a real part sequence or an imaginary part sequence in the X polarization direction or the Y polarization direction. In this embodiment of the present disclosure, the target polynomial and the seed are determined by designing the values of the coefficients $a_9, \dots$, and a_1 in the 10-degree polynomial or the values of the coefficients $a_8, \dots$, and a_1 in the 9-degree polynomial, so that real part sequences (also referred to as I channels) and imaginary part sequences (also referred to as Q channels) of the generated pilot symbol sequences in the two polarization directions have good autocorrelation and cross-correlation characteristics. A total of four sequences with a length of Q bits: the real part sequence of the designed pilot symbol sequence in the X polarization direction, the imaginary part sequence of the designed pilot symbol sequence in the X polarization direction, the real part sequence of the designed pilot symbol sequence in the Y polarization direction, and the imaginary part sequence of the designed pilot symbol sequence in the Y polarization direction have a good autocorrelation characteristic and a good cross-correlation characteristic.

[1029] It should be noted that, it can be learned from FIG. 7 and FIG. 8 that the degree of the target polynomial and the quantity of non-zero terms in the coefficients affect complexity of the generation structure of the pilot symbol. In embodiments of this disclosure, when the target polynomial is designed, the degree of the polynomial needs to be constrained not to be greater than 10, and the quantity of non-zero terms of the polynomial needs to be constrained not to be greater than 8. When the polynomial is selected, the degree and the quantity of non-zero terms of the target polynomial need to be selected to be as small as possible, so that the complexity of the generation structure of the pilot symbol is low, and power consumption is low. It should be further noted that, when a target polynomial is given, there may be no seed that can cause a pilot symbol sequence, determined based on the target polynomial and the seed, to have good autocorrelation and cross-correlation. Therefore, in the target polynomial selection, not only a minimum degree and a minimum quantity of non-zero terms are selected, but also whether a corresponding seed can be selected to cause a generated pilot symbol sequence and a real part sequence and an imaginary part sequence of the pilot symbol sequence to have good autocorrelation and cross-correlation characteristics needs to be considered, to help the receiving end restore the signal quality.

[1030] In a possible implementation, the target polynomial is one in the following Table 1.

TABLE 1

Sequence number	Target polynomial
1	$x^9 + x^4 + x^3 + x + 1$
2	$x^9 + x^5 + x^4 + x + 1$
3	$x^9 + x^8 + x^5 + x^4 + 1$
4	$x^9 + x^8 + x^6 + x^5 + 1$
5	$x^{10} + x^4 + x^3 + x + 1$
6	$x^{10} + x^5 + x^2 + x + 1$
7	$x^{10} + x^8 + x^5 + x + 1$
8	$x^{10} + x^9 + x^4 + x + 1$
9	$x^{10} + x^9 + x^5 + x^2 + 1$
10	$x^{10} + x^9 + x^6 + x + 1$
11	$x^{10} + x^9 + x^7 + x^6 + 1$
12	$x^{10} + x^9 + x^8 + x^5 + 1$
13	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$
14	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$
15	$x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$
16	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$
17	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$
18	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$

[1031] In embodiments of this disclosure, that a polynomial whose degree is less than or equal to 10, especially the 9-degree or 10-degree polynomial is considered mainly because a higher-degree polynomial cannot significantly improve autocorrelation and cross-correlation characteristics of a generated pilot symbol sequence. Although implementation complexity of a lower-degree polynomial, for example, a 6-degree or 7-degree polynomial, is low, autocorrelation and cross-correlation characteristics of a generated pilot symbol sequence are usually not good enough. In addition, a polynomial whose quantity of terms is not greater than 8, especially a 9-degree or 10-degree polynomial whose quantity of terms is 5 or 7 is considered, and the polynomial has low hardware implementation complexity.

[1032] In a possible implementation, an example in which $N=6144$, $M=64$, and $Q=96$ is used, and a correspondence between the target polynomial, a seed in the X polarization direction, and a seed in the Y polarization direction is one in the following Table 2. In this way, generated 96 pilot symbols have good autocorrelation and cross-correlation characteristics in the X polarization direction or the Y polarization direction.

TABLE 2

Sequence number	Target polynomial	Seed in the X polarization direction	Seed in the Y polarization direction
1	$x^9 + x^4 + x^3 + x + 1$	0x002	0x115
2	$x^9 + x^4 + x^3 + x + 1$	0x002	0x02B
3	$x^9 + x^4 + x^3 + x + 1$	0x049	0x115
4	$x^9 + x^4 + x^3 + x + 1$	0x049	0x02B
5	$x^9 + x^4 + x^3 + x + 1$	0x115	0x08D
6	$x^9 + x^4 + x^3 + x + 1$	0x08D	0x02B
7	$x^9 + x^5 + x^4 + x + 1$	0x098	0x0FE
8	$x^9 + x^5 + x^4 + x + 1$	0x098	0x0BF
9	$x^9 + x^5 + x^4 + x + 1$	0x098	0x17F
10	$x^9 + x^5 + x^4 + x + 1$	0x14C	0x0FE
11	$x^9 + x^5 + x^4 + x + 1$	0x14C	0x0BF
12	$x^9 + x^5 + x^4 + x + 1$	0x14C	0x17F
13	$x^9 + x^5 + x^4 + x + 1$	0x0A6	0x0FE
14	$x^9 + x^5 + x^4 + x + 1$	0x0A6	0x0BF
15	$x^9 + x^5 + x^4 + x + 1$	0x0A6	0x17F
16	$x^9 + x^8 + x^5 + x^4 + 1$	0x1D4	0x11E
17	$x^9 + x^8 + x^5 + x^4 + 1$	0x1D4	0x03D

TABLE 2-continued

Se- quence num- ber	Target polynomial	Seed in the X polar- ization direction	Seed in the Y polar- ization direction
18	$x^9 + x^8 + x^5 + x^4 + 1$	0x1D4	0x08F
19	$x^9 + x^8 + x^5 + x^4 + 1$	0x0EA	0x11E
20	$x^9 + x^8 + x^5 + x^4 + 1$	0x0EA	0x03D
21	$x^9 + x^8 + x^5 + x^4 + 1$	0x0EA	0x08F
22	$x^9 + x^8 + x^5 + x^4 + 1$	0x11E	0x175
23	$x^9 + x^8 + x^5 + x^4 + 1$	0x175	0x03D
24	$x^9 + x^8 + x^5 + x^4 + 1$	0x175	0x08F
25	$x^9 + x^8 + x^6 + x^5 + 1$	0x16A	0x1E1
26	$x^9 + x^8 + x^6 + x^5 + 1$	0x16A	0x1C3
27	$x^9 + x^8 + x^6 + x^5 + 1$	0x1E1	0x069
28	$x^9 + x^8 + x^6 + x^5 + 1$	0x1E1	0x113
29	$x^9 + x^8 + x^6 + x^5 + 1$	0x069	0x1C3
30	$x^9 + x^8 + x^6 + x^5 + 1$	0x1C3	0x113
31	$x^{10} + x^4 + x^3 + x + 1$	0x0E6	0x36E
32	$x^{10} + x^5 + x^2 + x + 1$	0x3DC	0x36A
33	$x^{10} + x^5 + x^2 + x + 1$	0x36A	0x35E
34	$x^{10} + x^5 + x^2 + x + 1$	0x36A	0x1AF
35	$x^{10} + x^8 + x^5 + x + 1$	0x1FD	0x3 A7
36	$x^{10} + x^9 + x^4 + x + 1$	0x12A	0x039
37	$x^{10} + x^9 + x^4 + x + 1$	0x12A	0x107
38	$x^{10} + x^9 + x^4 + x + 1$	0x039	0x295
39	$x^{10} + x^9 + x^4 + x + 1$	0x295	0x107
40	$x^{10} + x^9 + x^5 + x^2 + 1$	0x26A	0x03A
41	$x^{10} + x^9 + x^6 + x + 1$	0x1A2	0x379
42	$x^{10} + x^9 + x^6 + x + 1$	0x1A2	0x3EF
43	$x^{10} + x^9 + x^6 + x + 1$	0x2D1	0x379
44	$x^{10} + x^9 + x^6 + x + 1$	0x2D1	0x3EF
45	$x^{10} + x^9 + x^7 + x^6 + 1$	0x3CC	0x1E2
46	$x^{10} + x^9 + x^8 + x^5 + 1$	0x170	0x14D
47	$x^{10} + x^9 + x^8 + x^5 + 1$	0x0B8	0x14D
48	$x^{10} + x^9 + x^8 + x^5 + 1$	0x299	0x14D
49	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x354	0x2AD
50	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x17C	0x2AD
51	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x1AA	0x2AD
52	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x06A	0x2AD
53	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x3E6	0x2AD
54	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2A9	0x2AD
55	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2F9	0x2AD
56	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x0D5	0x2AD
57	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2AD	0x1F3
58	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2AD	0x14B
59	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2AD	0x297
60	$x^{10} + x^8 + x^6 + x^5 + x^3 + x + 1$	0x2AD	0x12F
61	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x320	0x3 AC
62	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x320	0x1D6
63	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x320	0x075
64	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x320	0x0ED
65	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x320	0x0EB
66	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x0D4	0x3AC
67	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x0D4	0x1D6
68	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x0D4	0x075
69	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x0D4	0x0EB
70	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x3 AC	0x01A
71	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x01A	0x1D6
72	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x01A	0x075
73	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x01A	0x0EB
74	$x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$	0x1B0	0x3F4
75	$x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$	0x1B0	0x1FA
76	$x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$	0x3F4	0x13D
77	$x^{10} + x^8 + x^7 + x^6 + x^2 + x + 1$	0x1FA	0x13D
78	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x35C	0x2EE
79	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x0DC	0x2EE
80	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x33A	0x2EE
81	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x26E	0x2EE
82	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x2B9

TABLE 2-continued

Se- quence num- ber	Target polynomial	Seed in the X polar- ization direction	Seed in the Y polar- ization direction
83	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x1B9
84	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x275
85	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x39D
86	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x173
87	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x0EB
88	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x39B
89	$x^{10} + x^9 + x^7 + x^5 + x^4 + x^2 + 1$	0x2EE	0x337
90	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x0C6	0x157
91	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x0C6	0x2D7
92	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x263	0x157
93	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x263	0x2D7
94	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x350	0x130
95	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x350	0x298
96	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x350	0x34C
97	$x^{10} + x^9 + x^8 + x^4 + x^3 + x^2 + 1$	0x350	0x261
98	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x350	0x01F
99	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x130	0x014
100	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x130	0x282
101	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x298	0x014
102	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x298	0x282
103	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x014	0x34C
104	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x014	0x261
105	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x34C	0x282
106	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x282	0x261

[1033] It should be noted that, when the target polynomial is designed, the degree of the polynomial needs to be constrained not to be greater than 10, and the quantity of non-zero terms of the polynomial needs to be constrained not to be greater than 8. When the polynomial is selected, the degree and the quantity of non-zero terms of the target polynomial need to be selected to be as small as possible, so that the complexity of the generation structure of the pilot symbol is low, and power consumption is low. It is considered that when a target polynomial is given, there may be no seed that can cause a pilot symbol sequence, determined based on the target polynomial and the seed, to have good autocorrelation and cross-correlation. Therefore, in the target polynomial selection, not only a minimum degree and a minimum quantity of non-zero terms are selected, but also whether a corresponding seed can be selected to cause a generated pilot symbol sequence and a real part sequence and an imaginary part sequence of the pilot symbol sequence

to have good autocorrelation and cross-correlation characteristics needs to be considered, to help the receiving end restore the signal quality.

[1034] In another possible implementation, the target polynomial is one in the following Table 3.

TABLE 3

Sequence number	Target polynomial
1	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$
2	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$

[1035] It should be understood that, it is considered that there is a frequency difference between a laser at the transmitting end and a laser at the receiving end, namely, a frequency offset of the laser, and an effect of frame synchronization performed by the receiving end is affected. The polynomials in Table 3 have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is the frequency offset of the laser.

[1036] In a possible implementation, an example in which $N=6144$, $M=64$, and $Q=96$ is used, and a correspondence between the target polynomial, a seed in the X polarization direction, and a seed in the Y polarization direction is one in the following Table 4.

TABLE 4

Sequence number	Target polynomial	Seed in the X polarization direction	Seed in the Y polarization direction
1	$x^9 + x^8 + x^6 + x^5 + 1$	0x1D4	0x1E1
2	$x^9 + x^4 + x^3 + x + 1$	0x094	0x02B
3	$x^{10} + x^9 + x^7 + x^6 + 1$	0x058	0x3CC
4	$x^{10} + x^9 + x^7 + x^6 + 1$	0x3CC	0x316
5	$x^{10} + x^9 + x^7 + x^6 + 1$	0x3CC	0x18E
6	$x^{10} + x^9 + x^7 + x^6 + 1$	0x3CC	0x163
7	$x^{10} + x^9 + x^8 + x^5 + 1$	0x152	0x14D
8	$x^{10} + x^9 + x^8 + x^5 + 1$	0x2A4	0x14D
9	$x^{10} + x^9 + x^6 + x + 1$	0x0B4	0x2D1
10	$x^{10} + x^9 + x^6 + x + 1$	0x1A2	0x05A
11	$x^{10} + x^9 + x^6 + x + 1$	0x05A	0x2D1
12	$x^{10} + x^9 + x^4 + x + 1$	0x128	0x12A
13	$x^{10} + x^9 + x^4 + x + 1$	0x250	0x12A
14	$x^{10} + x^9 + x^4 + x + 1$	0x250	0x295
15	$x^{10} + x^4 + x^3 + x + 1$	0x39C	0x36E
16	$x^{10} + x^4 + x^3 + x + 1$	0x18E	0x36E
17	$x^{10} + x^4 + x^3 + x + 1$	0x36E	0x239
18	$x^{10} + x^4 + x^3 + x + 1$	0x36E	0x0E7
19	$x^{10} + x^5 + x^2 + x + 1$	0x232	0x36A
20	$x^{10} + x^5 + x^2 + x + 1$	0x36A	0x119
21	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x282	0x01F
22	$x^{10} + x^9 + x^8 + x^7 + x^3 + x^2 + 1$	0x014	0x01F
23	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x1BA	0x2CE
24	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2A6	0x2CE
25	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x196	0x2CE
26	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2CE	0x259
27	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x380	0x2CE
28	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x380	0x081
29	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2CE	0x375
30	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2CE	0x353
31	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2CE	0x2FF
32	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x2CE	0x3FF
33	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x301	0x2D9
34	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x301	0x3D9
35	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x1BA	0x3D9

TABLE 4-continued

Sequence number	Target polynomial	Seed in the X polarization direction	Seed in the Y polarization direction
36	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x1BA	0x0AD
37	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x19C	0x301
38	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x0D4	0x0ED
39	$x^{10} + x^8 + x^7 + x^3 + x^2 + x + 1$	0x01A	0x0ED
40	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x186
41	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x261
42	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x1A3
43	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x0C7
44	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x346
45	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x1C5
46	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x063
47	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x1C5	0x327
48	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x240	0x3C4
49	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x3C4	0x103
50	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x2E2	0x226
51	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x2E2	0x165
52	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x165	0x0C7
53	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x266	0x0C7
54	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x2E2	0x3E2
55	$x^{10} + x^9 + x^6 + x^3 + x^2 + x + 1$	0x0C4	0x226
56	$x^{10} + x^9 + x^8 + x^7 + x^4 + x + 1$	0x1D8	0x2D9

[1037] It should be understood that, it is considered that there is the frequency difference between the laser at the transmitting end and the laser at the receiving end, namely, the frequency offset of the laser, and the effect of the frame synchronization performed by the receiving end is affected. The polynomials and the seeds in Table 4 have good differential autocorrelation and differential cross-correlation characteristics, and good correlation effects can still be achieved when there is the frequency offset of the laser.

[1038] Further, in a polarization direction, a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are $-A-A_j$ is the same as the quantity of pilot symbols that are $A+A_j$, and the quantity of pilot symbols that are $-A+A_j$ is the same as the quantity of pilot symbols that are $A-A_j$. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality. For example, $Q=96$, the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 23, 25, 25, and 23 in sequence, and the difference between the every two of quantities is less than or equal to 2.

[1039] In a possible implementation, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$; the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$; the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor + 1$; or the quantity of pilot symbols that are $-A-A_j$ is $Q/2 \lfloor Q/4 \rfloor - 1$, the

quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$. $\lfloor a \rfloor$ represents rounding down a positive real number a to the nearest integer. For example, $Q=96$, and the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 24, 24, 24, and 24 in sequence, the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 23, 25, 25, and 23 in sequence, or the quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ may be 25, 23, 23, and 25 in sequence.

[1040] Further, in two polarization directions, a total quantity of pilot symbols that are $-A-A_j$ is $Q/2$, a total quantity of pilot symbols that are $-A+A_j$ is $Q/2$, a total quantity of pilot symbols that are $A-A_j$ is $Q/2$, and a total quantity of pilot symbols that are $A+A_j$ is $Q/2$. This effectively ensures balance between symbol quantities. In addition, it can be further ensured that the sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[1041] In a possible implementation, in the X polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$. In the Y polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor$.

[1042] In another possible implementation, in the X polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $-A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, and the quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor + 1$. In the Y polarization direction, the quantity of pilot symbols that are $-A-A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$, the quantity of pilot symbols that are $-A+A_j$ is $\lfloor Q/4 \rfloor + 1$, the quantity of pilot symbols that are $A-A_j$ is $\lfloor Q/4 \rfloor + 1$, and the quantity of pilot symbols that are $A+A_j$ is $Q/2 - \lfloor Q/4 \rfloor - 1$.

[1043] Further, in a polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4. A pilot symbol sequence obtained under this condition helps improve the signal quality restored by the receiving end.

[1044] The following describes several specific embodiments by using an example in which $N=6144$, $M=64$, and $Q=96$.

Embodiment 1: Implementation in which the Degree of the Polynomial is 9

[1045] FIG. 9 is a diagram of a structure of another data frame according to an embodiment of this disclosure. Considering DP-16QAM modulation, $96 \times 63 = 6048$ DP payload symbols are obtained through Bose-Chaudhuri-Hocquenghem (BCH) (126, 110) encoding and the DP-16QAM DP modulation. One pilot symbol is inserted before every 63 symbols in the 6048 payload symbols, and a total of 96 pilot symbols are inserted, to obtain the structure of the data frame (also referred to as a DSP frame) shown in FIG. 9, that is, $N=6144$, $M=64$, and $Q=96$. In the X polarization direction or the Y polarization direction, the 96 pilot symbols are generated based on the target polynomial and the seed. For

a scenario in which the 96 pilot symbols need to be generated, a consecutive bit sequence $b_0, b_1, b_2, \dots$, and b_{191} whose bit length is $96 \times 2 = 192$ is obtained based on the target polynomial and the seed. Every two consecutive bits in the bit sequence $b_0, b_1, b_2, \dots$, and b_{191} are mapped to one symbol, where b_{2r} and b_{2r+1} are mapped to one symbol $(2b_{2r-1})A + (2b_{2r+1}-1)A_j$. It should be noted that, alternatively, the symbol $(2b_{2r-1})A + (2b_{2r+1}-1)A_j$ may not be a symbol in a constellation diagram of a used modulation format, and may be one of four symbols in an intermediate area between four outermost symbols and four innermost symbols in the constellation diagram of the used modulation format.

[1046] FIG. 10 is a diagram of a third-generation structure of a pilot symbol according to an embodiment of this disclosure. If the target polynomial is $x^9 + x^4 + x^3 + x + 1$, the seed in the X polarization direction is 0x049, and the seed in the Y polarization direction is 0x115, that is, an item whose sequence number is 3 in Table 2 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 10.

[1047] In the X polarization direction, an input seed is 0x049, and is converted into a binary sequence 001001001, that is, values of m_8 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is $A-A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is $-A-A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is $A+A_j$. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is $-A+A_j$. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1048] In the Y polarization direction, the input polarization seed is 0x115, and is converted into a binary sequence 100010101, that is, values of m_8 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the Y polarization direction is $A-A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the Y polarization direction is $-A-A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the Y polarization direction is $A+A_j$. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the Y polarization direction is $-A+A_j$. By analogy, the 96 pilot symbols in the Y polarization direction may be obtained.

[1049] The 96 pilot symbols in each of the two polarization directions are shown in the following Table 5.

TABLE 5

Polarization direction	Pilot symbol
X polarization direction	$A - A_j, -A + A_j, -A - A_j, A - A_j, -A - A_j,$ $-A - A_j, -A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A + A_j, A - A_j, -A + A_j, A + A_j, -A + A_j,$ $A + A_j, -A - A_j, A - A_j, A - A_j, -A + A_j,$ $-A + A_j, -A + A_j, -A - A_j, A - A_j, -A - A_j,$ $A - A_j, -A - A_j, A - A_j, A - A_j, A - A_j,$ $A + A_j, -A - A_j, -A - A_j, -A - A_j, A - A_j,$ $A + A_j, A + A_j, -A - A_j, A + A_j, A + A_j,$ $-A + A_j, A - A_j, A + A_j, A - A_j, -A - A_j,$ $-A + A_j, -A - A_j, -A + A_j, A + A_j, -A + A_j,$ $-A - A_j, -A - A_j, A + A_j, A + A_j, A + A_j,$

TABLE 5-continued

Polarization direction	Pilot symbol
Y polarization direction	$-A - A_j, A - A_j, -A + A_j, -A + A_j, -A - A_j,$
	$-A - A_j, A - A_j, A + A_j, A + A_j, A + A_j,$
	$A + A_j, A - A_j, -A - A_j, -A - A_j, A - A_j, -A + A_j,$
	$-A + A_j, A - A_j, -A - A_j, A + A_j, -A + A_j,$
	$-A - A_j, -A + A_j, -A + A_j, A + A_j, -A + A_j,$
	$A + A_j, A - A_j, A - A_j, A + A_j, A + A_j, A - A_j,$
	$A + A_j, A - A_j, A + A_j, -A + A_j, -A + A_j,$
	$-A - A_j, -A + A_j, -A - A_j, \text{ and } A + A_j$
	$A - A_j, A - A_j, A - A_j, -A - A_j, A - A_j, -A + A_j,$
	$A - A_j, A - A_j, -A + A_j, A - A_j, A + A_j,$
	$-A + A_j, A - A_j, -A - A_j, -A - A_j, -A + A_j,$
	$-A + A_j, -A - A_j, A + A_j, A - A_j, A - A_j,$
	$A - A_j, -A + A_j, A - A_j, -A + A_j, -A + A_j,$
	$A + A_j, A - A_j, A + A_j, A + A_j, A + A_j,$
	$-A + A_j, A - A_j, -A + A_j, A - A_j, -A + A_j,$
	$A + A_j, A + A_j, A + A_j, -A + A_j, -A - A_j,$
	$-A + A_j, A + A_j, A + A_j, -A - A_j, -A + A_j,$
	$-A + A_j, -A - A_j, -A + A_j, A - A_j, A + A_j,$
	$-A - A_j, A - A_j, -A - A_j, A + A_j, -A - A_j,$
	$A - A_j, -A + A_j, A + A_j, -A - A_j, -A + A_j,$
	$A - A_j, -A - A_j, -A + A_j, -A + A_j, A - A_j,$
	$A + A_j, A + A_j, -A - A_j, -A + A_j, A + A_j,$
	$-A - A_j, -A - A_j, -A - A_j, -A - A_j, A - A_j,$
	$A + A_j, A + A_j, -A - A_j, -A + A_j, A + A_j,$
	$-A - A_j, -A - A_j, -A - A_j, -A - A_j, A - A_j,$
	$-A - A_j, -A + A_j, A - A_j, A + A_j, A - A_j, A -$
	$A_j, -A + A_j, -A + A_j, A + A_j, -A - A_j, A +$
	$A_j, A - A_j, -A + A_j, A - A_j, -A - A_j, A + A_j,$
	$A + A_j, -A - A_j, -A - A_j, \text{ and } A + A_j$

[1050] In the X polarization direction, quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence. In the Y polarization direction, quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 23, 25, 25, and 23 in sequence. In a polarization direction, a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are $-A-A_j$ is the same as the quantity of pilot symbols that are $A+A_j$, and the quantity of pilot symbols that are $-A+A_j$ is the same as the quantity of pilot symbols that are $A-A_j$. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance. Further, in the X and Y polarization directions, a total quantity of pilot symbols that are $-A-A_j$, a total quantity of pilot symbols that are $-A+A_j$, a total quantity of pilot symbols that are $A-A_j$, and a total quantity of pilot symbols that are $A+A_j$ are all 48. This effectively ensures balance between symbol quantities, to help the receiving end restore the signal quality.

[1051] FIG. 11 is a diagram of a first implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure. Considering the DP-16QAM modulation, a manner of generating the pilot symbols in the X polarization direction and the Y polarization direction is shown in FIG. 11, where a degree of a generator polynomial is 9, and a bit sequence generated by using the 9-degree polynomial is referred to as a PRBS9.

[1052] FIG. 12A to FIG. 12C are diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure. FIG. 12A shows a periodic autocorrelation result of a pilot symbol sequence in the X polarization direction. FIG. 12B shows a periodic auto-

correlation result of a pilot symbol sequence in the Y polarization direction. FIG. 12C shows a periodic cross-correlation result of pilot symbol sequences in the X and Y polarization directions. Normalized amplitudes of side lobe values of periodic autocorrelation functions of the symbol sequences in the two polarization directions are not greater than 0.222, and a normalized amplitude of a side lobe value of a periodic cross-correlation function of the symbol sequences in the two polarization directions is not greater than 0.193.

[1053] FIG. 13A and FIG. 13B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure. A real part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_I) , an imaginary part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_Q) , a real part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_I) , and an imaginary part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_Q) . FIG. 13A and FIG. 13B show autocorrelation characteristics and cross-correlation characteristics of the four sequences: X_I , X_Q , Y_I , and Y_Q whose lengths are Q bits. FIG. 13A shows periodic autocorrelation results of the real part sequence in the X polarization direction (X_I), the imaginary part sequence in the X polarization direction (X_Q), the real part sequence in the Y polarization direction (Y_I), and the imaginary part sequence in the Y polarization direction (Y_Q). FIG. 13B shows periodic cross-correlation results of the real part sequence in the X polarization direction and the real part sequence in the Y polarization direction ($X_I \& Y_I$), the real part sequence in the X polarization direction and the imaginary part sequence in the X polarization direction ($X_I \& X_Q$), the real part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction ($X_I \& Y_Q$), the imaginary part sequence in the X polarization direction and the real part sequence in the Y polarization direction ($X_Q \& Y_I$), the real part sequence in the Y polarization direction and the imaginary part sequence in the Y polarization direction ($Y_I \& Y_Q$), and the imaginary part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction ($X_Q \& Y_Q$). Neither a side lobe value of a periodic autocorrelation function nor an absolute value of a normalized amplitude of a periodic cross-correlation function is greater than 0.2917.

[1054] Based on received signals in the two polarization directions, the receiving end performs DSP by using a pilot sequence (PS), to restore the signals. For example, correlation values between the received signals and a real part sequence/an imaginary part sequence of the PS in the X polarization direction and a real part sequence/an imaginary part sequence of the PS in the Y polarization direction are separately calculated, so that the X/Y polarization direction and a real part/an imaginary part can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

Embodiment 2: Implementation in which the Degree of the Polynomial is 10

[1055] FIG. 14 is a diagram of a fourth-generation structure of a pilot symbol according to an embodiment of this disclosure. If the target polynomial is $x^{10}+x^9+x^4+x+1$, the seed in the X polarization direction is 0x12A, and the seed in the Y polarization direction is 0x039, that is, an item whose sequence number is 36 in Table 2 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 14.

[1056] In the X polarization direction, the input polarization seed is 0x12A, and is converted into a binary sequence 0100101010, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is A-Aj. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is -A-Aj. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is A+Aj. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is -A+Aj. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1057] In the Y polarization direction, the input polarization seed is 0x039, and is converted into a binary sequence 0000111001, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the Y polarization direction is A-Aj. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the Y polarization direction is -A-Aj. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the Y polarization direction is A+Aj. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the Y polarization direction is -A+Aj. By analogy, the 96 pilot symbols in the Y polarization direction may be obtained.

[1058] The 96 pilot symbols in each of the two polarization directions are shown in the following Table 6.

TABLE 6

Polarization direction	Pilot symbol
X polarization direction	$-A + A_j, -A + A_j, -A + A_j, -A - A_j, A - A_j,$ $A + A_j, -A + A_j, A - A_j, A + A_j, -A - A_j,$ $A - A_j, -A + A_j, -A + A_j, -A + A_j, A + A_j,$ $-A + A_j, -A - A_j, -A - A_j, A - A_j, -A + A_j,$ $-A + A_j, -A - A_j, -A - A_j, A - A_j, A - A_j,$ $A - A_j, A + A_j, -A + A_j, A - A_j, -A - A_j,$ $A + A_j, -A + A_j, -A - A_j, -A + A_j, A + A_j,$ $A - A_j, A + A_j, -A + A_j, A + A_j, -A + A_j,$ $-A - A_j, A + A_j, -A + A_j, A - A_j, -A - A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, A - A_j, -A -$ $A_j, A - A_j, A + A_j, A - A_j, A - A_j, A + A_j,$ $A - A_j, A + A_j, A + A_j, A - A_j, A - A_j, -A -$ $A_j, A + A_j, -A - A_j, A - A_j, A - A_j, A - A_j,$ $A + A_j, A + A_j, -A - A_j, A + A_j, A + A_j,$ $A + A_j, -A - A_j, A - A_j, -A + A_j, A + A_j,$ $-A - A_j, -A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A + A_j, A - A_j, -A - A_j, -A + A_j, A - A_j,$ $A + A_j, -A + A_j, -A - A_j, \text{and } -A + A_j$
Y polarization direction	$A - A_j, -A + A_j, A + A_j, -A - A_j, -A - A_j,$ $-A + A_j, -A - A_j, -A + A_j, -A - A_j, -A - A_j,$ $A + A_j, -A - A_j, A + A_j, A - A_j, -A + A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A - A_j, -A - A_j, A + A_j, A - A_j, -A + A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, -A + A_j,$ $A + A_j, -A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, -A + A_j,$ $A + A_j, -A - A_j, A + A_j, -A + A_j, -A - A_j,$

TABLE 6-continued

Polarization direction	Pilot symbol
	$A + A_j, A + A_j, A - A_j, -A - A_j, A + A_j,$ $-A - A_j, -A - A_j, -A + A_j, A - A_j, -A + A_j,$ $A - A_j, A + A_j, -A + A_j, A + A_j, A - A_j, A + A_j,$ $-A - A_j, -A - A_j, A - A_j, A - A_j, -A + A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A - A_j, A - A_j, -A - A_j, -A - A_j, -A + A_j,$ $A - A_j, A + A_j, A - A_j, A + A_j, A - A_j,$ $-A + A_j, -A + A_j, A - A_j, -A + A_j, -A + A_j,$ $A - A_j, A - A_j, A - A_j, -A + A_j, A + A_j, A - A_j,$ $A + A_j, A + A_j, A - A_j, -A - A_j, -A + A_j,$ $A + A_j, A + A_j, A + A_j, A - A_j, -A + A_j,$ $-A + A_j, -A - A_j, -A + A_j, -A + A_j, -A + A_j,$ $-A - A_j, -A + A_j, -A - A_j, A - A_j, A + A_j,$ $A + A_j, A - A_j, A + A_j, \text{ and } A + A_j$

[1059] In the X polarization direction, quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence. In the Y polarization direction, quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 23, 25, 25, and 23 in sequence. In a polarization direction, a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are $-A-A_j$ is the same as the quantity of pilot symbols that are $A+A_j$, and the quantity of pilot symbols that are $-A+A_j$ is the same as the quantity of pilot symbols that are $A-A_j$. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance. Further, in the X and Y polarization directions, a total quantity of pilot symbols that are $-A-A_j$, a total quantity of pilot symbols that are $-A+A_j$, a total quantity of pilot symbols that are $A-A_j$, and a total quantity of pilot symbols that are $A+A_j$ are all 48. This effectively ensures balance between symbol quantities, to help the receiving end restore the signal quality.

[1060] FIG. 15 is a diagram of a second implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure. Considering the DP-16QAM modulation, a manner of generating the pilot symbols in the X polarization direction and the Y polarization direction is shown in FIG. 15, where a degree of a generator polynomial is 10, and a bit sequence generated by using the 10-degree polynomial is referred to as a PRBS10.

[1061] FIG. 16A to FIG. 16C are diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure. A in FIG. 16 shows a periodic autocorrelation result of a pilot symbol sequence in the X polarization direction. B in FIG. 16 shows a periodic autocorrelation result of a pilot symbol sequence in the Y polarization direction. C in FIG. 16 shows a periodic cross-correlation result of pilot symbol sequences in the X and Y polarization directions. Normalized amplitudes of side lobe values of periodic autocorrelation functions of the symbol sequences in the two polarization directions are not greater than 0.193, and a normalized amplitude of a side lobe value of a periodic cross-correlation function of the symbol sequences in the two polarization directions is not greater than 0.251.

[1062] FIG. 17A and FIG. 17B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure. A real part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_I), an imaginary part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_Q), a real part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_I), and an imaginary part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_Q). FIG. 17A and FIG. 17B show autocorrelation characteristics and cross-correlation characteristics of the four sequences: X_I , X_Q , Y_I , and Y_Q whose lengths are Q bits. FIG. 17A shows periodic autocorrelation results of the real part sequence in the X polarization direction (X_I), the imaginary part sequence in the X polarization direction (X_Q), the real part sequence in the Y polarization direction (Y_I), and the imaginary part sequence in the Y polarization direction (Y_Q). FIG. 17B shows periodic cross-correlation results of the real part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_I & Y_I), the real part sequence in the X polarization direction and the imaginary part sequence in the X polarization direction (X_I & X_Q), the real part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_I & Y_Q), the imaginary part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_Q & Y_I), the real part sequence in the Y polarization direction and the imaginary part sequence in the Y polarization direction (Y_I & Y_Q), and the imaginary part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_Q & Y_Q). Neither a side lobe value of a periodic autocorrelation function nor an absolute value of a normalized amplitude of a periodic cross-correlation function is greater than 0.2917.

[1063] Based on received signals in the two polarization directions, the receiving end performs DSP by using a PS, to restore the signals. For example, correlation values between the received signals and a real part sequence/an imaginary part sequence of the PS in the X polarization direction and a real part sequence/an imaginary part sequence of the PS in the Y polarization direction are separately calculated, so that the X/Y polarization direction and a real part/an imaginary part can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

[1064] Based on the foregoing descriptions, in embodiments of this disclosure, in the data frame including the N symbols, the every M consecutive symbols include the pilot symbol at the fixed location and the M-1 payload symbols, and $N=M \times Q$. The Q pilot symbols in the data frame are generated by using the target polynomial and the seed, the Q pilot symbols are direct current balanced, the degree of the target polynomial is less than or equal to 10, and the quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8. It can be learned that a simple target polynomial is designed in this disclosure to generate the pilot symbols. Correspondingly, a simple hardware structure may be used for implementation. In addition, the pilot symbols generated in the foregoing manner have the

good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end restore the signal quality.

[1065] It should be noted that, currently, communication architectures of some optical transport networks support transmission rates of 400 Gbps and 800 Gbps. At the transmission rate of 400 Gbps, a correspondingly needed baud rate is usually about 60 gigabaud (Gbaud) when the DP-16QAM (DP-16QAM) modulation is used. At the transmission rate of 800 Gbps, a correspondingly needed baud rate is usually about 120 Gbaud when the DP-16QAM modulation is used. With the growth of services, metro telecommunication transmission and data center transmission scenarios have increasingly high requirements on transmission rates. For example, at transmission rates such as 1.2 terabits per second (Tbps) and 1.6 Tbps, when the DP-16QAM modulation and a single-wavelength transmission manner are used, corresponding baud rates are about 180 Gbaud and 240 Gbaud. At a same transmission rate, a higher baud rate is needed when a lower-order modulation, such as DP-QPSK, is used. When higher-order modulation, such as DP-32QAM or DP-64QAM, is used, although a needed baud rate is lower, a transmission distance is limited. A higher transmission rate needed in the optical transport network corresponds to a higher needed baud rate, and corresponds to higher power consumption of a device. Currently, there is no low-power device with a baud rate higher than 140 Gbaud. In the metro telecommunication transmission and data center transmission scenarios, low power consumption is usually required for implementation. A plurality of data frames are carried in a plurality of optical signals for transmission, so that data streams are transmitted in parallel, and an overall transmission rate can be improved while a current baud rate and a current modulation order are maintained. For example, wavelengths of the plurality of optical signals are different, and the plurality of optical signals with the different wavelengths are transmitted on a same optical fiber, and is also referred to as wavelength-division multiplexing (WDM) transmission. For another example, the plurality of optical signals are respectively transmitted through a plurality of different optical fibers, and is also referred to as a parallel single mode (PSM). The following embodiment does not need a device with a higher baud rate, and is applicable to the metro telecommunication transmission and data center transmission scenarios that require the low power consumption.

Embodiment 3

[1066] Two data frames: a first data frame and a second data frame are considered. For structures of the first data frame and the second data frame, refer to the diagram of the structure of the data frame in FIG. 9. Considering DP-16QAM modulation, $96 \times 63 = 6048$ DP payload symbols are obtained through BCH (126, 110) encoding and the DP-16QAM modulation. One pilot symbol is inserted before every 63 symbols in the 6048 payload symbols, and a total of 96 pilot symbols are inserted, to obtain the structure of the data frame (also referred to as a DSP frame) shown in FIG. 9, that is, $N=6144$, $M=64$, and $Q=96$. In the X polarization direction or the Y polarization direction, the 96 pilot symbols are generated based on the target polynomial and the seed. For the first data frame, in the X polarization direction, 96 pilot symbols are generated based on a first target polynomial and a first seed; and in the Y polarization

direction, the 96 pilot symbols are generated based on the first target polynomial and a second seed. For the second data frame, in the X polarization direction, 96 pilot symbols are generated based on a second target polynomial and a third seed; and in the Y polarization direction, the 96 pilot symbols are generated based on the second target polynomial and a fourth seed.

[1067] FIG. 18 is a diagram of a transmission scenario of a plurality of data frame streams according to an embodiment of this disclosure. In this embodiment, two DP symbol streams are obtained through the BCH (126, 110) encoding and the DP-16QAM modulation. In each DP symbol stream, one pilot symbol is inserted before every 63 symbols in every 6048 payload symbols according to the foregoing framing solution, and a total of 96 pilot symbols are inserted, to obtain one data frame. In other words, the first data frame is located in one data frame stream obtained by performing framing on one of the two DP symbol streams, and the second data frame is located in another data frame stream obtained by performing framing on the other DP symbol stream in the two DP symbol streams. The two data frame streams are carried in the plurality of optical signals for transmission, as shown in an example in A in FIG. 18. In this embodiment, the two data frame streams are transmitted on different wavelengths. For a 1.6 TE scenario, a rate of each data frame stream is about 800 G bits per second.

[1068] In this embodiment, the first target polynomial and the second target polynomial are a same target polynomial. The first seed and the third seed are the same, and are referred to as seeds in the X polarization direction. The second seed and the fourth seed are the same, and are referred to as seeds in the Y polarization direction. In other words, same 96 pilot symbols are used in the first data frame and the second data frame. Considering that a 9-degree target polynomial is $x^9 + x^8 + x^5 + x^{4+1}$, the seed in the X polarization direction is 0x175, and the seed in the Y polarization direction is 0x03D, that is, an item whose sequence number is 23 in Table 2 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 19.

[1069] FIG. 19 is a diagram of a fifth-generation structure of a pilot symbol according to an embodiment of this disclosure. In the X polarization direction, the input seed is 0x175, and is converted into a binary sequence 101110101, that is, values of m_8 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is A-Aj. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is -A-Aj. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is A+Aj. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is -A+Aj. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1070] In the Y polarization direction, the input polarization seed is 0x03D, and is converted into a binary sequence 000111101, that is, values of m_8 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the Y polarization direction is A-Aj. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the Y polarization direction is -A-Aj. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the Y polarization direction is A+Aj. If two

bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the Y polarization direction is -A+Aj. By analogy, the 96 pilot symbols in the Y polarization direction may be obtained.

[1071] The 96 pilot symbols in each of the two polarization directions are shown in the following Table 7.

TABLE 7

Polarization direction	Pilot symbol
X polarization direction	A - Aj, A - Aj, A + Aj, A - Aj, A - Aj, -A - Aj, A + Aj, -A + Aj, -A + Aj, -A + Aj, -A - Aj, Aj, -A + Aj, A + Aj, -A + Aj, -A - Aj, A - Aj, -A + Aj, A - Aj, A - Aj, -A - Aj, A - Aj, A - Aj, A - Aj, A + Aj, -A + Aj, -A + Aj, -A - Aj, A - Aj, A + Aj, A + Aj, A + Aj, -A - Aj, -A + Aj, A - Aj, A + Aj, -A + Aj, -A - Aj, -A - Aj, A - Aj, A - Aj, A + Aj, A - Aj, A + Aj, A + Aj, -A + Aj, -A + Aj, -A - Aj, -A - Aj, -A - Aj, A + Aj, -A - Aj, A + Aj, -A - Aj, -A - Aj, -A - Aj, A - Aj, A + Aj, -A - Aj, A - Aj, A + Aj, -A - Aj, -A - Aj, A - Aj, -A + Aj, A - Aj, -A + Aj, A + Aj, A - Aj, A - Aj, -A - Aj, -A + Aj, -A - Aj, -A + Aj, -A - Aj, A + Aj, A + Aj, -A - Aj, A + Aj, A - Aj, A - Aj, -A - Aj, -A + Aj, A + Aj, -A + Aj, -A + Aj, -A + Aj, A + Aj, A - Aj, A + Aj, -A + Aj, A - Aj, -A - Aj, -A - Aj, A + Aj, -A + Aj, -A - Aj, A - Aj, A - Aj, -A + Aj, A - Aj, and -A - Aj
Y polarization direction	A - Aj, A + Aj, A + Aj, -A - Aj, -A + Aj, -A + Aj, -A - Aj, -A + Aj, -A - Aj, A - Aj, -A - Aj, A - Aj, A + Aj, -A - Aj, A + Aj, -A + Aj, -A + Aj, A - Aj, A + Aj, -A + Aj, A + Aj, A + Aj, A - Aj, A + Aj, -A - Aj, -A + Aj, Aj, A - Aj, -A - Aj, A - Aj, -A - Aj, A - Aj, -A - Aj, -A - Aj, -A + Aj, A + Aj, A - Aj, -A - Aj, -A + Aj, -A + Aj, -A - Aj, A - Aj, A + Aj, A - Aj, -A - Aj, -A - Aj, -A + Aj, -A + Aj, A - Aj, A - Aj, -A + Aj, A - Aj, A + Aj, A + Aj, -A - Aj, A - Aj, A - Aj, -A - Aj, -A + Aj, A - Aj, -A - Aj, -A + Aj, A + Aj, -A - Aj, -A - Aj, A + Aj, -A + Aj, A + Aj, -A + Aj, A - Aj, -A + Aj, -A - Aj, A - Aj, -A + Aj, -A + Aj, -A + Aj, -A - Aj, A - Aj, -A - Aj, -A + Aj, -A - Aj, -A - Aj, A + Aj, A + Aj, A + Aj, A + Aj, A - Aj, A - Aj, A + Aj, A + Aj, A - Aj, -A + Aj, A + Aj, -A - Aj, A - Aj, A + Aj, A + Aj, and A + Aj

[1072] In the X polarization direction, quantities of pilot symbols that are respectively -A-Aj, -A+Aj, A-Aj, and A+Aj are all 24. In the Y polarization direction, quantities of pilot symbols that are respectively -A-Aj, -A+Aj, A-Aj, and A+Aj are 25, 23, 23, and 25 in sequence. In a polarization direction, a difference between every two of quantities of pilot symbols that are respectively -A-Aj, -A+Aj, A-Aj, and A+Aj in the data frame is less than or equal to 2. In addition, the quantity of pilot symbols that are -A-Aj is the same as the quantity of pilot symbols that are A+Aj, and the quantity of pilot symbols that are -A+Aj is the same as the quantity of pilot symbols that are A-Aj. This effectively ensures that symbol quantities in each polarization direction are approximately balanced, and can further ensure that a sequence including the pilot symbols achieves the direct current balance, to help the receiving end restore the signal quality.

[1073] FIG. 20 is a diagram of a third implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure. Considering the DP-16QAM modulation, a manner of gen-

erating the pilot symbols in the X polarization direction and the Y polarization direction is shown in FIG. 20, where a degree of a generator polynomial is 9, and a bit sequence generated by using the 9-degree polynomial is referred to as a PRBS9.

[1074] FIG. 21A to FIG. 21C are other diagrams of correlation characteristics corresponding to pilot symbols according to an embodiment of this disclosure. FIG. 21A shows a periodic autocorrelation result of a pilot symbol sequence in the X polarization direction. FIG. 21B shows a periodic autocorrelation result of a pilot symbol sequence in the Y polarization direction. FIG. 21C shows a periodic cross-correlation result of pilot symbol sequences in the X and Y polarization directions. Normalized amplitudes of side lobe values of periodic autocorrelation functions of the symbol sequences in the two polarization directions are not greater than 0.180, and a normalized amplitude of a side lobe value of a periodic cross-correlation function of the symbol sequences in the two polarization directions is not greater than 0.225.

[1075] FIG. 22A and FIG. 22B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure. A real part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_I), an imaginary part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_Q), a real part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_I), and an imaginary part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_Q). FIG. 22A and FIG. 22B show autocorrelation characteristics and cross-correlation characteristics of the four sequences: X_I, X_Q, Y_I, and Y_Q whose lengths are Q bits. FIG. 22A shows periodic autocorrelation results of the real part sequence in the X polarization direction (X_I), the imaginary part sequence in the X polarization direction (X_Q), the real part sequence in the Y polarization direction (Y_I), and the imaginary part sequence in the Y polarization direction (Y_Q). FIG. 22B shows periodic cross-correlation results of the real part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_I&Y_I), the real part sequence in the X polarization direction and the imaginary part sequence in the X polarization direction (X_I&X_Q), the real part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_I&Y_Q), the imaginary part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_Q&Y_I), the real part sequence in the Y polarization direction and the imaginary part sequence in the Y polarization direction (Y_I&Y_Q), and the imaginary part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_Q&Y_Q). Neither a side lobe value of a periodic autocorrelation function nor an absolute value of a normalized amplitude of a periodic cross-correlation function is greater than 0.292.

[1076] Based on received signals in the two polarization directions, the receiving end performs DSP by using a PS, to restore the signals. For example, correlation values between the received signals and a real part sequence and an imaginary part sequence of the PS in the X polarization direction and correlation values between the received signals and a real part sequence and an imaginary part sequence of the PS in the Y polarization direction are separately calculated, so

that the X polarization direction and the Y polarization direction and a real part and an imaginary part can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good autocorrelation and cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

[1077] It should be noted that, in this embodiment, it is considered that the two data frame streams are transmitted on the different wavelengths, and 96 pilot symbols in each data frame stream are generated by using a same target polynomial. In some other specific applications, pilot symbol sequences in the two data frame streams are generated by using different target polynomials. In some other specific applications, pilot symbol sequences in the two data frame streams are generated by using a same target polynomial and different seeds. In some other specific applications, the two data frame streams are transmitted on two optical fibers.

[1078] It should be noted that, in some other specific applications, a quantity of data frame streams is greater than 2. As shown in B in FIG. 18, W DP symbol streams are obtained through FEC encoding and modulation. In each DP symbol stream, one pilot symbol is inserted before every M-1 symbols in every (M-1)×Q payload symbols according to the foregoing framing solution, and a total of Q pilot symbols are inserted, to obtain one data frame. In this way, a total of W data frame streams are obtained. The W data frame streams are carried in a plurality of optical signals for transmission. For example, the W data frame streams are respectively carried in W optical signals. In an example, any two of the W optical signals have different wavelengths, and the W optical signals are transmitted on a same optical fiber. In another example, all of the W optical signals have a same wavelength, and the W optical signals are respectively transmitted through W optical fibers.

[1079] The following describes several specific embodiments based on Table 4.

Embodiment 4: The Target Polynomial is $x^{10} + x^9 + x^6 + x + 1$, the Seed in the X Polarization Direction is 0x0B4, and the Seed in the Y Polarization Direction is 0x2D1

[1080] As shown in FIG. 9, considering DP-16QAM modulation, $96 \times 63 = 6048$ DP payload symbols are obtained through BCH (126, 110) encoding and the DP-16QAM modulation. One pilot symbol is inserted before every 63 symbols in the 6048 payload symbols, and a total of 96 pilot symbols are inserted, to obtain the structure of the data frame (also referred to as a DSP frame) shown in FIG. 14, that is, $N=6144$, $M=64$, and $Q=96$. In the X polarization direction or the Y polarization direction, the 96 pilot symbols are generated based on the target polynomial and the seed. For a scenario in which the 96 pilot symbols need to be generated, a consecutive bit sequence $b_0, b_1, b_2, \dots$, and b_{191} whose bit length is $96 \times 2 = 192$ is obtained based on the target polynomial and the seed. Every two consecutive bits in the bit sequence $b_0, b_1, b_2, \dots$, and b_{191} are mapped to one symbol, where b_{2r} and b_{2r+1} are mapped to one symbol $(2b_{2r-1})A + (2b_{2r+1}-1)A_j$. It should be noted that, alternatively, the symbol $(2b_{2r-1})A + (2b_{2r+1}-1)A_j$ may not be a symbol in a constellation diagram of a used modulation format, and may be one of four symbols in an intermediate

area between four outermost symbols and four innermost symbols in the constellation diagram of the used modulation format.

[1081] FIG. 23 is a diagram of a sixth-generation structure of a pilot symbol according to an embodiment of this disclosure. If the target polynomial is $x^{10}+x^9+x^6+x+1$, the seed in the X polarization direction is 0x0B4, and the seed in the Y polarization direction is 0x2D1, that is, an item whose sequence number is 9 in Table 4 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 23.

[1082] In the X polarization direction, the input seed is 0x0B34, and is converted into a binary sequence 0010110100, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is $A-A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is $-A-A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is $A+A_j$. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is $-A+A_j$. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1083] In the Y polarization direction, the input seed is 0x2D1, and is converted into a binary sequence 1011010001, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the Y polarization direction is A-Aj. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the Y polarization direction is -A-Aj. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the Y polarization direction is A+Aj. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the Y polarization direction is -A+Aj. By analogy, the 96 pilot symbols in the Y polarization direction may be obtained.

[1084] The 96 pilot symbols in each of the two polarization directions are shown in the following Table 8.

TABLE 8

Polarization direction	Pilot symbol
X polarization direction	$-A - A_j, A - A_j, A + A_j, -A + A_j, -A - A_j,$ $-A - A_j, A - A_j, -A + A_j, A - A_j, -A + A_j,$ $-A + A_j, -A + A_j, -A + A_j, -A + A_j, -A - A_j,$ $A + A_j, -A + A_j, A - A_j, -A - A_j, A + A_j,$ $A - A_j, -A + A_j, -A + A_j, A + A_j, A - A_j,$ $-A + A_j, A - A_j, A - A_j, A + A_j, A + A_j, A - A_j,$ $-A - A_j, -A + A_j, A + A_j, A - A_j, -A + A_j,$ $-A - A_j, A - A_j, -A - A_j, -A - A_j, A - A_j,$ $-A + A_j, A + A_j, -A - A_j, -A + A_j, -A + A_j,$ $-A - A_j, A + A_j, A - A_j, A + A_j, A + A_j,$ $-A - A_j, A - A_j, -A - A_j, -A - A_j, -A + A_j,$ $-A + A_j, -A + A_j, -A - A_j, -A + A_j, -A + A_j,$ $-A - A_j, A + A_j, A + A_j, A + A_j, A + A_j,$ $-A - A_j, A - A_j, -A - A_j, A - A_j, A + A_j,$ $-A + A_j, A - A_j, A - A_j, A - A_j, A - A_j,$ $-A + A_j, A - A_j, A - A_j, A - A_j, A - A_j,$ $-A - A_j, A - A_j, A - A_j,$ and $-A - A_j$
Y polarization direction	$A - A_j, -A - A_j, A - A_j, A + A_j, -A + A_j,$ $A - A_j, -A - A_j, -A + A_j, A - A_j, A - A_j,$ $A + A_j, A + A_j, -A - A_j, -A - A_j, A - A_j,$ $A + A_j, -A - A_j, A + A_j, A + A_j, A + A_j,$ $-A - A_j, A + A_j, A + A_j, -A + A_j, -A + A_j,$

TABLE 8-continued

Polarization direction	Pilot symbol
	-A - Aj, -A - Aj, -A + Aj, -A + Aj, A - Aj,
	-A + Aj, -A - Aj, A + Aj, -A - Aj, -A + Aj,
	-A + Aj, A + Aj, A + Aj, -A + Aj, A + Aj,
	-A + Aj, -A + Aj, A + Aj, -A + Aj, -A - Aj,
	A + Aj, -A - Aj, A + Aj, -A + Aj, -A - Aj,
	-A - Aj, -A + Aj, A - Aj, A + Aj, -A - Aj,
	A - Aj, A + Aj, A - Aj, A + Aj, -A + Aj, A - Aj,
	-A - Aj, -A - Aj, A - Aj, A + Aj, -A - Aj,
	-A + Aj, A - Aj, A + Aj, -A + Aj, -A + Aj,
	-A + Aj, -A - Aj, -A - Aj, A - Aj, A - Aj,
	-A + Aj, -A - Aj, -A - Aj, A - Aj, A -
	Aj, A - Aj, A - Aj, -A + Aj, -A - Aj, A + Aj,
	-A + Aj, A - Aj, A + Aj, -A + Aj, -A - Aj,
	A - Aj, A - Aj, and A - Aj

[1085] FIG. 24 is a diagram of a fourth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure. Considering the DP-16QAM modulation, a manner of generating the pilot symbols in the X polarization direction and the Y polarization direction is shown in FIG. 16A to FIG. 16C, where a degree of a generator polynomial is 10, and a bit sequence generated by using the 10-degree polynomial is referred to as a PRBS10.

[1086] FIG. 25A and FIG. 25B are other diagrams of a correlation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure. FIG. 25A shows periodic autocorrelation results obtained by performing differencing on every two adjacent symbols in pilot symbol sequences in the X and Y polarization directions. FIG. 25B shows a periodic differential cross-correlation result of the pilot symbol sequences in the X and Y polarization directions. Normalized amplitudes of side lobe values of periodic differential autocorrelation functions of the symbol sequences in the two polarization directions are not greater than 0.179, and a normalized amplitude of a side lobe value of a periodic differential cross-correlation function of the symbol sequences in the two polarization directions is not greater than 0.222.

[1087] FIG. 26A and FIG. 26B are diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure. A real part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_I), an imaginary part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_Q), a real part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_I), and an imaginary part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_Q). FIG. 26A and FIG. 26B show autocorrelation characteristics and cross-correlation characteristics of the four sequences: X_I, X_Q, Y_I, and Y_Q whose lengths are Q bits. FIG. 26A shows periodic differential autocorrelation results of the real part sequence in the X polarization direction (X_I), the imaginary part sequence in the X polarization direction (X_Q), the real part sequence in the Y polarization direction (Y_I), and the imaginary part sequence in the Y polarization direction (Y_Q). FIG. 26B shows periodic differential cross-correlation results of the real part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_I & Y_I), the

real part sequence in the X polarization direction and the imaginary part sequence in the X polarization direction ($X_I \& X_Q$), the real part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction ($X_I \& Y_Q$), the imaginary part sequence in the X polarization direction and the real part sequence in the Y polarization direction ($X_Q \& Y_I$), the real part sequence in the Y polarization direction and the imaginary part sequence in the Y polarization direction ($Y_I \& Y_Q$), and the imaginary part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction ($X_Q \& Y_Q$). Neither a side lobe value of a periodic differential autocorrelation function nor a normalized amplitude of a periodic differential cross-correlation function is greater than 0.2709.

[1088] Based on received signals in the two polarization directions, the receiving end performs DSP by using a PS, to restore the signals. For example, differential correlation values between the received signals and sequence symbols of the PS in the X polarization and the Y polarization are separately calculated, so that the polarization directions can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good differential autocorrelation and differential cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

Embodiment 5: The Target Polynomial is $x^{10} + x^9 + x^4 + x + 1$, the Seed in the X Polarization Direction is 0x128, and the Seed in the Y Polarization Direction is 0x12A

[1089] FIG. 27 is a diagram of a seventh-generation structure of a pilot symbol according to an embodiment of this disclosure. If the target polynomial is $x^{10} + x^9 + x^4 + x + 1$, the seed in the X polarization direction is 0x128, and the seed in the Y polarization direction is 0x12A, that is, an item whose sequence number is 12 in Table 4 is used, a process of generating the 96 pilot symbols in each of the two polarization directions is shown in FIG. 27.

[1090] In the X polarization direction, the input seed is 0x128, and is converted into a binary sequence 0100101000, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the X polarization direction is $A - A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the X polarization direction is $-A - A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the X polarization direction is $A + A_j$. If two bits that are consecutively output in sequence are 0 and 1, the pilot symbol in the X polarization direction is $-A + A_j$. By analogy, the 96 pilot symbols in the X polarization direction may be obtained.

[1091] In the Y polarization direction, the input seed is 0x12A, and is converted into a binary sequence 0100101010, that is, values of m_9 to m_0 . If two bits that are consecutively output in sequence are 1 and 0, the pilot symbol in the Y polarization direction is $A - A_j$. If two bits that are consecutively output in sequence are 0 and 0, the pilot symbol in the Y polarization direction is $-A - A_j$. If two bits that are consecutively output in sequence are 1 and 1, the pilot symbol in the Y polarization direction is $A + A_j$. If two bits that are consecutively output in sequence are 0 and 1, the

pilot symbol in the Y polarization direction is $-A + A_j$. By analogy, the 96 pilot symbols in the Y polarization direction may be obtained.

[1092] The 96 pilot symbols in each of the two polarization directions are shown in the following Table 9.

TABLE 9

Polarization direction	Pilot symbol
X polarization direction	$-A - A_j, -A + A_j, -A + A_j, -A - A_j, A - A_j,$ $-A + A_j, -A + A_j, A - A_j, -A - A_j, A - A_j,$ $-A + A_j, -A - A_j, -A - A_j, -A - A_j, A + A_j,$ $-A + A_j, A + A_j, -A + A_j, A + A_j, -A - A_j,$ $A - A_j, A + A_j, -A - A_j, A - A_j, A + A_j,$ $-A + A_j, -A + A_j, -A - A_j, A + A_j, A + A_j,$ $-A + A_j, A + A_j, -A - A_j, -A + A_j, -A - A_j,$ $A + A_j, A + A_j, A + A_j, A + A_j, -A - A_j, A -$ $A_j, A - A_j, -A - A_j, A - A_j, A - A_j, A - A_j,$ $-A - A_j, A - A_j, -A + A_j, -A + A_j, A + A_j,$ $A + A_j, -A + A_j, A + A_j, A - A_j, -A + A_j,$ $-A - A_j, -A - A_j, -A + A_j, -A - A_j, A - A_j,$ $-A + A_j, A + A_j, A - A_j, -A - A_j, -A + A_j,$ $A + A_j, A + A_j, -A + A_j, -A + A_j, A + A_j,$ $A + A_j, -A - A_j, A + A_j, A - A_j, A + A_j, -A - A_j,$ $A - A_j, A - A_j, A - A_j, A - A_j, A - A_j, A -$ $A_j, A - A_j, -A + A_j, -A + A_j, -A - A_j, A -$ $A_j, A + A_j, -A + A_j, and -A - A_j$
Y polarization direction	$-A + A_j, -A + A_j, -A + A_j, -A - A_j, A - A_j,$ $A + A_j, -A + A_j, A - A_j, A + A_j, -A - A_j,$ $A - A_j, -A + A_j, -A + A_j, -A + A_j, A + A_j,$ $-A + A_j, -A - A_j, -A - A_j, A - A_j, -A + A_j,$ $-A + A_j, -A - A_j, -A - A_j, A - A_j, A - A_j,$ $A - A_j, A + A_j, -A + A_j, A - A_j, A - A_j,$ $A + A_j, -A + A_j, -A - A_j, -A + A_j, A + A_j,$ $A - A_j, A + A_j, -A + A_j, A + A_j, -A + A_j,$ $-A - A_j, A + A_j, -A + A_j, A - A_j, -A - A_j,$ $-A - A_j, A - A_j, A + A_j, -A - A_j, A + A_j, -A -$ $A_j, A - A_j, A + A_j, A - A_j, A - A_j, A + A_j,$ $A - A_j, A + A_j, A + A_j, A - A_j, A - A_j, -A -$ $A_j, A + A_j, -A - A_j, -A - A_j, A - A_j, A - A_j,$ $A + A_j, A + A_j, -A - A_j, A - A_j, A + A_j,$ $A + A_j, -A - A_j, A + A_j, -A + A_j, A + A_j, -A -$ $A_j, -A - A_j, A + A_j, -A - A_j, -A + A_j,$ $-A + A_j, A - A_j, -A - A_j, -A + A_j, A - A_j,$ $A + A_j, -A + A_j, -A - A_j, and -A + A_j$

[1093] FIG. 28 is a diagram of a fifth implementation procedure of generating pilot symbols based on DP-16QAM modulation according to an embodiment of this disclosure. Considering the DP-16QAM modulation, a manner of generating the pilot symbols in the X polarization direction and the Y polarization direction is shown in FIG. 28, where a degree of a generator polynomial is 10, and a bit sequence generated by using the 10-degree polynomial is referred to as a PRBS10.

[1094] FIG. 29A and FIG. 29B are other diagrams of a correlation characteristic and a cross-correlation characteristic that correspond to pilot symbols according to an embodiment of this disclosure. FIG. 29A shows periodic autocorrelation results obtained by performing differencing on every two adjacent symbols in pilot symbol sequences in the X and Y polarization directions. FIG. 29B shows a periodic differential cross-correlation result of the pilot symbol sequences in the X and Y polarization directions. Normalized amplitudes of side lobe values of periodic differential autocorrelation functions of the symbol sequences in the two polarization directions are not greater than 0.178, and a normalized amplitude of a side lobe value

of a periodic differential cross-correlation function of the symbol sequences in the two polarization directions is not greater than 0.222.

[1095] FIG. 30A and FIG. 30B are other diagrams of reflecting an autocorrelation characteristic and a cross-correlation characteristic according to an embodiment of this disclosure. A real part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_I), an imaginary part sequence of the pilot symbol sequence in the X polarization direction is denoted as (X_Q), a real part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_I), and an imaginary part sequence of the pilot symbol sequence in the Y polarization direction is denoted as (Y_Q). FIG. 30A and FIG. 30B show autocorrelation characteristics and cross-correlation characteristics of the four sequences: X_I, X_Q, Y_I, and Y_Q whose lengths are Q bits. FIG. 30A shows periodic differential autocorrelation results of the real part sequence in the X polarization direction (X_I), the imaginary part sequence in the X polarization direction (X_Q), the real part sequence in the Y polarization direction (Y_I), and the imaginary part sequence in the Y polarization direction (Y_Q). FIG. 30B shows periodic differential cross-correlation results of the real part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_I&Y_I), the real part sequence in the X polarization direction and the imaginary part sequence in the X polarization direction (X_I&X_Q), the real part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_I&Y_Q), the imaginary part sequence in the X polarization direction and the real part sequence in the Y polarization direction (X_Q&Y_I), the real part sequence in the Y polarization direction and the imaginary part sequence in the Y polarization direction (Y_I&Y_Q), and the imaginary part sequence in the X polarization direction and the imaginary part sequence in the Y polarization direction (X_Q&Y_Q). Neither a side lobe value of a periodic differential autocorrelation function nor a normalized amplitude of a periodic differential cross-correlation function is greater than 0.2709.

[1096] Based on received signals in the two polarization directions, the receiving end performs DSP by using a PS, to restore the signals. For example, differential correlation values between the received signals and sequence symbols of the PS in the X polarization and the Y polarization are separately calculated, so that the polarization directions can be distinguished, frame synchronization and alignment can be performed, and carrier phase recovery is performed by using the PS. The designed pilot symbol sequence has good differential autocorrelation and differential cross-correlation characteristics, and are direct current balanced. This helps the receiving end improve quality of the restored signals.

[1097] The following describes data transmission apparatuses provided in embodiments of this disclosure.

[1098] FIG. 31 is a diagram of a structure of a data transmission apparatus used at a transmitting end according to an embodiment of this disclosure. As shown in FIG. 31, the data transmission apparatus includes a processing unit 101 and a sending unit 102. The processing unit 101 is configured to perform the operation in step 401 in the embodiment shown in FIG. 4. The sending unit 102 is configured to perform the operation in step 402 in the embodiment shown in FIG. 4. For specific operations, refer

to related descriptions of the embodiment shown in FIG. 4. Details are not described herein again.

[1099] FIG. 32 is a diagram of a structure of a data transmission apparatus used at a receiving end according to an embodiment of this disclosure. As shown in FIG. 32, the data transmission apparatus includes a processing unit 201 and a receiving unit 202. The receiving unit 202 is configured to receive data frames transmitted from a transmitting end through a channel. The processing unit 201 is configured to perform dispersion compensation, synchronization, phase recovery, and other operations.

[1100] It should be understood that the apparatus provided in this disclosure may alternatively be implemented in another manner. For example, division into the units in the foregoing apparatus is merely logical function division and may be other division in actual implementation. For example, a plurality of units or components may be combined or integrated into another system. In addition, functional units in embodiments of this disclosure may be integrated into one processing unit, may be independent physical units, or two or more functional units may be integrated into one processing unit. The integrated unit may be implemented in a form of hardware, or may be implemented in a form of a software functional unit.

[1101] FIG. 33 is a diagram of another structure of a data transmission apparatus according to an embodiment of this disclosure. As shown in FIG. 33, the data transmission apparatus includes a processor 301, a memory 302, and a transceiver 303. The processor 301, the memory 302, and the transceiver 303 are interconnected through lines. The memory 302 is configured to store program instructions and data. The transceiver 303 is configured to perform data sending and receiving operations, and the processor 301 is configured to perform other operations than data sending and receiving. In a possible implementation, the processor 301 may include the processing unit 101 shown in FIG. 31, and the transceiver 303 includes the sending unit 102 shown in FIG. 31. In another possible implementation, the processor 301 may include the processing unit 201 shown in FIG. 32, and the transceiver 303 includes the receiving unit 202 shown in FIG. 32.

[1102] It should be noted that the processor shown in FIG. 33 may be a general-purpose central processing unit (CPU), a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or another programmable logic device, a transistor logic device, a hardware component, or any combination thereof. The memory shown in FIG. 33 may store an operating system and another application program. When the technical solutions provided in embodiments of this disclosure are implemented by using software or firmware, program code used for implementing the technical solutions provided in embodiments of this disclosure is stored in the memory and is executed by the processor. In an embodiment, the processor may include the memory inside. In another embodiment, the processor and the memory are two independent structures.

[1103] It may be clearly understood by a person skilled in the art that, for the purpose of convenient and brief description, for detailed working processes of the foregoing described system, apparatus, and unit, refer to corresponding processes in the foregoing method embodiments. Details are not described herein again.

[1104] A person of ordinary skill in the art may understand that all or some of the steps in the foregoing embodiments may be implemented by hardware or a program instructing related hardware. The program may be stored in a computer-readable storage medium. The storage medium mentioned above may be a read-only memory, a random-access memory, or the like. Whether the functions are performed by hardware or software depends on particular applications and design constraint conditions of the technical solutions. A person skilled in the art may use different methods to implement the described functions for each particular application, but it should not be considered that the implementation goes beyond the scope of this disclosure.

[1105] When software is used to implement the functions, all or some of the method steps described in the foregoing embodiments may be implemented in a form of a computer program product. The computer program product includes one or more computer instructions. When the computer program instructions are loaded and executed on a computer, the procedures or functions according to embodiments of this disclosure are all or partially generated. The computer may be a general-purpose computer, a dedicated computer, a computer network, or another programmable apparatus. The computer instructions may be stored in a computer-readable storage medium or may be transmitted from a computer-readable storage medium to another computer-readable storage medium. For example, the computer instructions may be transmitted from a website, computer, server, or data center to another website, computer, server, or data center in a wired (for example, a coaxial cable, an optical fiber, or a digital subscriber line (DSL)) or wireless (for example, infrared, radio, or microwave) manner. The computer-readable storage medium may be any usable medium accessible by the computer, or a data storage device, for example, a server or a data center, integrating one or more usable media. The usable medium may be a magnetic medium (for example, a floppy disk, a hard disk, or a magnetic tape), an optical medium (for example, a digital versatile disc (DVD), a semiconductor medium (for example, a solid-state drive (SSD)), or the like.

1. A data transmission method, comprising:

generating data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and M-1 payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-Aj$, $-A+Aj$, $A-Aj$, or $A+Aj$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and

sending the data frames.

2. The method of claim 1, wherein the target polynomial is $x^9+x^8+x^5+x^4+1$.

3. The method of claim 1, wherein each of the Q pilot symbols is at a start location of one of the M consecutive symbols in which the pilot symbol is located.

4. The method of claim 1, wherein a first sequence comprising Q pilot symbols in a first polarization direction

is different from a second sequence comprising Q pilot symbols in a second polarization direction, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

5. The method of claim 1, wherein a correspondence among the target polynomial, a first seed in a first polarization direction, and a second seed in a second polarization direction is: the target polynomial is $x^9 + x^8 + x^5 + x^4 + 1$; the first seed is 0x175; and the second seed is 0x03D.

6. The method of claim 5, wherein Q=96.

7. The method of claim 6, wherein $N=6144$, and wherein $M=64$.

8. The method of claim 1, wherein in each polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

9. The method of claim 1, wherein in each polarization direction, a modulation format of the N symbols is 16 quadrature amplitude modulation (QAM), and wherein A=-1, 1, -3, 3, $-\sqrt{5}$, or $\sqrt{5}$.

10. The method of claim 1, wherein a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in each data frame is less than or equal to 2.

11. The method of claim 10, wherein in a first polarization direction in each data frame:

a first quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$;

a second quantity of pilot symbols that are $-A+A_j$ is $Q/2-|Q/4|$;

a third quantity of pilot symbols that are A-Aj is $Q/2 - \lfloor Q/4 \rfloor$; and

a fourth quantity of pilot symbols that are $A+A_j$ is $\lfloor Q/4 \rfloor$,
or

wherein in a second polarization direction in each data frame: direction, in the data frame,

the first quantity is $\lfloor Q/4 \rfloor + 1$;

the second quantity is $Q/2 - \lfloor Q/4 \rfloor - 1$;

the third quantity is $Q/2 - \lfloor Q/4 \rfloor - 1$; and

the fourth quantity is $\lfloor Q/4 \rfloor + 1$; and

wherein $[a]$ represents rounding down a positive real number a to the nearest integer.

12. The method of claim 10, wherein in a first polarization direction in each data frame, a first quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 24, 24, 24, and 24 in sequence, wherein in a second polarization direction in each data frame, a second quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

13. The method of claim 1, wherein a sum of the Q pilot symbols is 0.

14. The method of claim 6, wherein 96 pilot symbols in the first polarization direction are sequentially:

A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, -A+Aj,
-A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj
-A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A-Aj,
A-Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,
A+Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, -A-Aj,
-A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, A+Aj, -A+Aj,
-A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj,
-A-Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A-Aj, A+Aj,
-A-Aj, -A-Aj, A-Aj, A+Aj, A-Aj, -A+Aj, A+Aj,
A-Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, -A+Aj, -A-Aj,
A+Aj, A-Aj, -A-Aj, A+Aj, -A+Aj, A-Aj, -A+Aj,

A+A_j, -A+A_j, -A+A_j, -A+A_j, A+A_j, A-A_j, A+A_j,
-A+A_j, A-A_j, -A-A_j, -A-A_j, A+A_j, -A+A_j, -A-A_j,
A-A_j, A-A_j, -A+A_j, A-A_j, and -A-A_j, and
wherein 96 pilot symbols in the second polarization
direction are sequentially:

A-Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj,
 -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, -A-Aj,
 A+Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj,
 A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, A-Aj, -A-Aj,
 A-Aj, -A-Aj, -A-Aj, -A-Aj, -A+Aj, A+Aj, A-Aj,
 -A-Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj, A-Aj,
 -A-Aj, -A-Aj, -A+Aj, -A-Aj, A-Aj, A-Aj,
 -A+Aj, A-Aj, A+Aj, A-Aj, -A-Aj, A-Aj, A-Aj,
 -A-Aj, -A+Aj, A-Aj, -A-Aj, -A+Aj, A+Aj,
 -A-Aj, -A-Aj, -A-Aj, -A+Aj, -A-Aj, -A-Aj,
 A-Aj, A+Aj, A+Aj, A+Aj, A-Aj, A-Aj, A+Aj,
 A+Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A+Aj,
 A+Aj, and A+Aj.

15. A method, comprising:

receiving data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and M-1 payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of $-A-A_j$, $-A+A_j$, $A-A_j$, or $A+A_j$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and

processing the data frames.

16. The method of claim **15**, wherein each of the Q pilot symbols is at a start location of one of the M consecutive symbols in which the pilot symbol is located.

17. The method of claim 15, wherein a first sequence comprising Q pilot symbols in a first polarization direction is different from a second sequence comprising Q pilot symbols in a second polarization direction, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

18. The method of claim **15**, wherein a correspondence among the target polynomial, a first seed in a first polarization direction, and a second seed in a second polarization direction is: the target polynomial is $x^9 + x^8 + x^5 + x^4 + 1$; the first seed is 0x175; and the second seed is 0x03D.

19. The method of claim **18**, wherein Q=96.

20. The method of claim 19, wherein N=6144, and wherein M=64.

21. The method of claim 15, wherein in each polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

22. The method of claim 15, wherein a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in each data frame is less than or equal to 2.

23. The method of claim 22, wherein in a first polarization direction in each data frame, a first quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 24, 24, 24, and 24 in sequence, wherein in a second

polarization direction in each data frame, a second quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

24. The method of claim **15**, wherein a sum of the Q pilot symbols is 0.

25. The method of claim **19**, wherein 96 pilot symbols in the first polarization direction are sequentially:

A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, -A+Aj,
 -A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj,
 -A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A-Aj,
 A-Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,
 A+Aj, -A-Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, -A-Aj,
 -A-Aj, A-Aj, A+Aj, A-Aj, A+Aj, A+Aj, -A+Aj,
 -A+Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj,
 -A-Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A-Aj, A+Aj,
 -A-Aj, -A-Aj, A-Aj, -A+Aj, A-Aj, -A-Aj, A+Aj,
 A-Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, -A+Aj, -A-Aj,
 A+Aj, A+Aj, -A-Aj, A+Aj, -A+Aj, A-Aj, -A+Aj,
 A+Aj, A-Aj, -A-Aj, -A-Aj, -A-Aj, A+Aj, A+Aj,
 A-Aj, A-Aj, -A+Aj, A-Aj, and -A-Aj, and

wherein 96 pilot symbols in the second polarization direction are sequentially:

A-Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A+Aj, -A-Aj, A+Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, -A+Aj, A+Aj, A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, -A+Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A-Aj, A+Aj, A+Aj, -A-Aj, A-Aj, A+Aj, -A-Aj, A-Aj, A+Aj, A+Aj, -A-Aj, A-Aj, A+Aj, A+Aj, and A+Aj.

26. A data transmission apparatus, comprising:

a processor configured to generate data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and M-1 payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-A_j$, $-A+A_j$, $A-A_j$, or $A+A_j$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and

a transmitter coupled to the processor and configured to send the data frames.

27. The data transmission apparatus of claim 26, wherein the target polynomial is $x^9+x^8+x^5+x^4+1$.

28. The data transmission apparatus of claim **26**, wherein each of the Q pilot symbols is at a start location of one of the M consecutive symbols in which the pilot symbol is located.

29. The data transmission apparatus of claim **26**, wherein a first sequence comprising Q pilot symbols in a first polarization direction is different from a second sequence comprising Q pilot symbols in a second polarization direction, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

30. The data transmission apparatus of claim **26**, wherein a correspondence among the target polynomial, a first seed in a first polarization direction, and a second seed in a second polarization direction is: the target polynomial is $x^9 + x^8 + x^5 + x^4 + 1$; the first seed is 0x175; and the second seed is 0x03D.

31. The data transmission apparatus of claim 30, wherein
Q=96.

32. The data transmission apparatus of claim 31, wherein N=6144, and wherein M=64.

33. The data transmission apparatus of claim 26, wherein in each polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

34. The data transmission apparatus of claim 26, wherein in each polarization direction, a modulation format of the N symbols is 16 quadrature amplitude modulation (QAM), and wherein A=-1, 1, -3, 3, $- \sqrt{5}$, or $\sqrt{5}$.

35. The data transmission apparatus of claim 26, wherein a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in each data frame is less than or equal to 2.

36. The data transmission apparatus of claim 35, wherein in a first polarization direction in each data frame:

a first quantity of pilot symbols that are $-A-A_j$ is $\lfloor Q/4 \rfloor$;

a second quantity of pilot symbols that are $-A + A_j$ is $Q/2 - \lfloor Q/4 \rfloor$;

a third quantity of pilot symbols that are A-A_j is $Q/2 - \lfloor Q/4 \rfloor$; and

a fourth quantity of pilot symbols that are A+A_j is $\lfloor Q/4 \rfloor$,
or

wherein in a second polarization direction in each data frame:

the first quantity is $\lfloor Q/4 \rfloor + 1$;

the second quantity is $Q/2 - \lfloor Q/4 \rfloor - 1$;

the third quantity is $Q/2 - |Q/4| - 1$; and

the fourth quantity is $\lfloor Q/4 \rfloor + 1$; and

wherein $[a]$ represents rounding down a positive real number a to the nearest integer.

37. The data transmission apparatus of claim 35, wherein in a first polarization direction in each data frame, a first quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 24, 24, 24, and 24 in sequence, wherein in a second polarization direction in each data frame, a second quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

38. The data transmission apparatus of claim 26, wherein a sum of the Q pilot symbols is 0.

39. The data transmission apparatus of claim 31, wherein 96 pilot symbols in the first polarization direction are sequentially:

A-Aj, A-Aj, A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, -A+Aj,
-A+Aj, -A+Aj, -A-Aj, -A+Aj, A+Aj, -A+Aj,
-A-Aj, A-Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A-Aj,
A-Aj, A+Aj, -A+Aj, -A+Aj, -A-Aj, A+Aj, A+Aj,

A+A_j, -A-A_j, -A+A_j, A-A_j, A+A_j, -A+A_j, -A-A_j,
 -A-A_j, A-A_j, A+A_j, A-A_j, A+A_j, A+A_j, -A+A_j,
 -A+A_j, -A-A_j, -A-A_j, -A-A_j, A+A_j, -A-A_j, A+A_j,
 -A-A_j, A-A_j, -A-A_j, A+A_j, -A-A_j, A-A_j, A+A_j,
 -A-A_j, -A-A_j, A-A_j, -A+A_j, A-A_j, -A+A_j, A+A_j,
 A-A_j, A-A_j, -A+A_j, A+A_j, -A-A_j, -A+A_j, -A-A_j,
 A+A_j, A+A_j, -A-A_j, A+A_j, -A+A_j, A-A_j, -A+A_j,
 A+A_j, -A+A_j, -A+A_j, -A+A_j, -A+A_j, A+A_j, A+A_j,
 -A+A_j, A-A_j, -A-A_j, -A-A_j, A+A_j, -A+A_j, -A-A_j,
 A-A_j, A-A_j, -A+A_j, A-A_j, and -A-A_j, and

wherein 96 pilot symbols in the second polarization direction are sequentially:

A-Aj, A+Aj, A+Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, -A-Aj, A+Aj, -A+Aj, -A+Aj, A-Aj, A+Aj, A+Aj, A+Aj, A-Aj, A-Aj, A-Aj, A-Aj, A-Aj, A-Aj, A+Aj, -A-Aj, A-Aj, -A+Aj, -A+Aj, -A-Aj, -A+Aj, -A+Aj, A+Aj, A-Aj, -A-Aj, -A+Aj, -A+Aj, -A-Aj, A-Aj, A+Aj, A-Aj, -A-Aj, -A-Aj, -A+Aj, -A+Aj, A-Aj, A-Aj, -A-Aj, A+Aj, A+Aj, -A-Aj, A-Aj, A-Aj, -A-Aj, A-Aj, A+Aj, A+Aj, A+Aj, A+Aj, A-Aj, A-Aj, A+Aj, A+Aj, A-Aj, -A+Aj, A+Aj, -A-Aj, A-Aj, A+Aj, A+Aj, and A+Aj.

40. A data transmission apparatus, comprising:

a receiver configured to receive data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and M-1 payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-A_j$, $-A+A_j$, $A-A_j$, or $A+A_j$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and

a processor coupled to the receiver and configured to process the data frames.

41. The data transmission apparatus of claim 40, wherein each of the Q pilot symbols is at a start location of one of the M consecutive symbols in which the pilot symbol is located.

42. The data transmission apparatus of claim **40**, wherein a first sequence comprising Q pilot symbols in a first polarization direction is different from a second sequence comprising Q pilot symbols in a second polarization direction, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

43. The data transmission apparatus of claim **40**, wherein a correspondence among the target polynomial, a first seed in a first polarization direction, and a second seed in a second polarization direction is: the target polynomial is $x^9+x^8+x^5+x^4+1$; the first seed is 0x175; and the second seed is 0x03D.

44. The data transmission apparatus of claim 43, wherein O=96.

45. The data transmission apparatus of claim **44**, wherein N=6144, and wherein M=64.

46. The data transmission apparatus of claim 40, wherein in each polarization direction, a quantity of consecutive identical pilot symbols in the data frame is less than or equal to 4.

47. The data transmission apparatus of claim 40, wherein a difference between every two of quantities of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ in each data frame is less than or equal to 2.

48. The data transmission apparatus of claim 47, wherein in a first polarization direction in each data frame, a first quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 24, 24, 24, and 24 in sequence, wherein in a second polarization direction in each data frame, a second quantity of pilot symbols that are respectively $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$ are 25, 23, 23, and 25 in sequence, and wherein the first polarization direction and the second polarization direction are orthogonal to each other.

49. The data transmission apparatus of claim 40, wherein a sum of the Q pilot symbols is 0.

50. The data transmission apparatus of claim 44, wherein 96 pilot symbols in the first polarization direction are sequentially:

$A-A_j$, $A-A_j$, $A+A_j$, $A-A_j$, $A-A_j$, $-A-A_j$, $A+A_j$, $-A+A_j$,
 $-A+A_j$, $-A+A_j$, $-A-A_j$, $-A+A_j$, $A+A_j$, $-A+A_j$,
 $-A-A_j$, $A-A_j$, $-A+A_j$, $A-A_j$, $A-A_j$, $-A-A_j$, $A-A_j$,
 $A-A_j$, $A+A_j$, $-A+A_j$, $-A+A_j$, $-A-A_j$, $A+A_j$, $A+A_j$,
 $A+A_j$, $-A-A_j$, $-A+A_j$, $A-A_j$, $A+A_j$, $-A+A_j$, $-A-A_j$,
 $-A-A_j$, $A-A_j$, $A+A_j$, $A-A_j$, $A+A_j$, $A+A_j$, $-A+A_j$,
 $-A+A_j$, $-A-A_j$, $-A-A_j$, $-A-A_j$, $A+A_j$, $-A-A_j$, $A+A_j$,
 $-A-A_j$, $A-A_j$, $-A-A_j$, $A+A_j$, $-A-A_j$, $A-A_j$, $A+A_j$,
 $-A-A_j$, $-A-A_j$, $A-A_j$, $-A+A_j$, $A-A_j$, $-A+A_j$, $A+A_j$,
 $A-A_j$, $A-A_j$, $-A+A_j$, $A+A_j$, $-A-A_j$, $-A+A_j$, $-A-A_j$,
 $A+A_j$, $A+A_j$, $-A-A_j$, $A+A_j$, $-A+A_j$, $A-A_j$, $-A+A_j$,
 $A+A_j$, $-A+A_j$, $-A+A_j$, $-A+A_j$, $A+A_j$, $A-A_j$, $A+A_j$,
 $-A+A_j$, $A-A_j$, $-A-A_j$, $-A-A_j$, $A+A_j$, $-A+A_j$, $-A-A_j$,
 $A-A_j$, $A-A_j$, $-A+A_j$, $A-A_j$, and $-A-A_j$, and

wherein 96 pilot symbols in the second polarization direction are sequentially:

$A-A_j$, $A+A_j$, $A+A_j$, $-A-A_j$, $-A+A_j$, $-A+A_j$, $-A-A_j$,
 $-A+A_j$, $-A-A_j$, $A-A_j$, $-A-A_j$, $A-A_j$, $A+A_j$, $-A-A_j$,

$A+A_j$, $-A+A_j$, $-A+A_j$, $A-A_j$, $A+A_j$, $-A+A_j$, $A+A_j$,
 $A+A_j$, $A-A_j$, $A+A_j$, $-A-A_j$, $-A+A_j$, $A-A_j$, $-A-A_j$,
 $A-A_j$, $-A-A_j$, $-A-A_j$, $-A-A_j$, $-A+A_j$, $A+A_j$, $A-A_j$,
 $-A-A_j$, $-A+A_j$, $-A+A_j$, $-A-A_j$, $A-A_j$, $A+A_j$, $A-A_j$,
 $-A-A_j$, $-A-A_j$, $-A+A_j$, $-A+A_j$, $A-A_j$, $A-A_j$,
 $-A+A_j$, $A-A_j$, $A+A_j$, $A+A_j$, $-A-A_j$, $A-A_j$, $A-A_j$,
 $-A-A_j$, $-A+A_j$, $A-A_j$, $-A-A_j$, $-A+A_j$, $A+A_j$,
 $-A-A_j$, $-A-A_j$, $A+A_j$, $-A+A_j$, $A+A_j$, $-A+A_j$, $A-A_j$,
 $-A+A_j$, $-A-A_j$, $A-A_j$, $-A+A_j$, $-A+A_j$, $-A+A_j$,
 $-A-A_j$, $A-A_j$, $-A-A_j$, $-A+A_j$, $-A-A_j$, $-A-A_j$,
 $A+A_j$, $A+A_j$, $A+A_j$, $A+A_j$, $A-A_j$, $A-A_j$, $A+A_j$,
 $A+A_j$, $A-A_j$, $-A+A_j$, $A+A_j$, $-A-A_j$, $A-A_j$, $A+A_j$,
 $A+A_j$, and $A+A_j$.

51. A data transmission system, comprising:

a first data transmission apparatus comprising:

a first processor configured to generate data frames, wherein each data frame comprises N symbols in a polarization direction, wherein every M consecutive symbols in the N symbols comprise one of Q pilot symbols at a fixed location and $M-1$ payload symbols, wherein $N=M \times Q$, wherein Q is an even number, wherein M is an integer greater than or equal to 1, wherein the Q pilot symbols are a symbol sequence generated by a target polynomial and a seed, wherein each of the Q pilot symbols is one of four complex numbers: $-A-A_j$, $-A+A_j$, $A-A_j$, and $A+A_j$, wherein A is a real number, wherein a degree of the target polynomial is less than or equal to 10, and wherein a quantity of terms of the target polynomial is greater than or equal to 2 and less than or equal to 8; and

a transmitter coupled to the first processor and configured to send the data frames; and

a second data transmission apparatus connected to the first data transmission apparatus and comprising:

a receiver configured to receive the data frames; and

a second processor coupled to the receiver and configured to process the data frames.

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